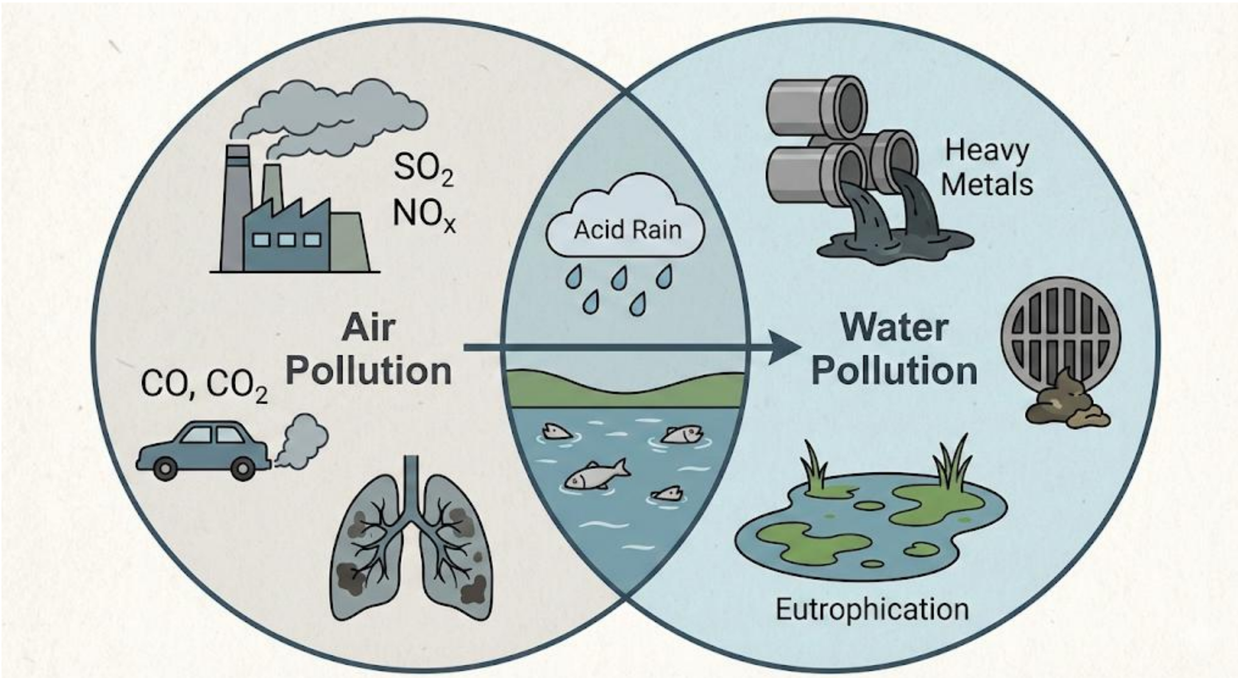
SS2 CHEMISTRY LESSON NOTES

FIRST TERM

1. Environmental Chemistry – Water and air pollution.
2. Electrolysis – Electrolytes, electrodes, Faraday's laws.
3. Electrochemical Cells – Galvanic and electrolytic cells.
4. Acid-Base Titration – Apparatus, procedures, calculations.
5. **Midterm Examination**
6. Solubility and Solubility Curves – Calculations and interpretation.
7. **Midterm Break**
8. Chemical Equilibrium – Equilibrium constant, calculations.
9. Chemical Kinetics – Rate laws, order of reaction.
10. Heat of Reaction – Enthalpy change, Hess's law.
11. Energetics of Reactions – Activation energy, diagrams.
12. Examination
13. Closing

WEEK 1: ENVIRONMENTAL CHEMISTRY – WATER AND AIR POLLUTION



INTRODUCTION

Environmental Chemistry is the branch of Chemistry that deals with **chemical processes occurring in the environment** and the **effects of human activities on the environment**. One major concern is **pollution**, which is the **introduction of harmful substances into the environment**, causing adverse effects on humans, animals, plants, and ecosystems.

Pollution is a serious global problem because it affects:

- **Health** (respiratory, skin, and waterborne diseases)
- **Agriculture** (reduced crop yield, soil degradation)
- **Climate** (global warming, acid rain)
- **Ecosystems** (loss of biodiversity, eutrophication)

The main types of environmental pollution are **water pollution** and **air pollution**, which are often chemically interrelated.

WATER POLLUTION

(a) Definition

Water pollution is the **contamination of water bodies** (rivers, lakes, oceans, underground aquifers) by **chemical, physical, or biological agents** that make the water harmful to organisms and humans.

(b) Causes of Water Pollution

1. **Industrial effluents**
    - Factories release chemicals such as **heavy metals (lead, mercury, cadmium)**, **acids, alkalis**, and **dyes** into water bodies.
2. **Domestic sewage**
    - Untreated sewage containing **organic waste, detergents, and pathogens** is discharged into rivers.
3. **Agricultural activities**
    - Runoff from farms carries **fertilizers, pesticides, and herbicides** into water bodies, leading to nutrient pollution.
4. **Oil spills**
    - Petroleum and oil products contaminate oceans and rivers, harming aquatic life.
5. **Mining activities**
    - Release of **toxic minerals and chemicals** into rivers and lakes.

(c) Effects of Water Pollution

1. **Health hazards**
    - Waterborne diseases such as **cholera, typhoid, and dysentery**.
    - Toxic metals can cause **kidney failure, neurological disorders, and cancer**.

2. **Eutrophication**
    - Excess nutrients (nitrates and phosphates) cause **algal blooms**, reducing oxygen in water and killing aquatic life.
3. **Bioaccumulation**
    - Toxic substances such as mercury accumulate in the food chain, affecting humans and wildlife.
4. **Aesthetic and economic impact**
    - Polluted water is unfit for recreation, fishing, and irrigation.

(d) Control Measures of Water Pollution

- Treating industrial and domestic effluents before discharge.
- Using **biodegradable detergents** instead of non-biodegradable ones.
- Proper **waste management and recycling**.
- Constructing **retention ponds** for runoff water.
- Implementing **laws and regulations** to control pollution.

AIR POLLUTION

(a) Definition

Air pollution is the **introduction of harmful substances** into the atmosphere, which can **adversely affect human health, animals, plants, and climate**.

(b) Causes of Air Pollution

1. **Industrial emissions**
    - Factories release **sulphur dioxide (SO_2), nitrogen oxides (NO_x), carbon monoxide (CO), and particulate matter**.
2. **Vehicular emissions**

- Combustion of petrol, diesel, and gas produces **CO, CO_2, NO_x, and hydrocarbons**.

3. **Burning of fossil fuels**
   - Power plants, refineries, and domestic burning produce **SO_2, CO_2, CO, and smoke**.
4. **Agricultural activities**
   - Use of fertilizers releases **ammonia**; burning of crop residues produces smoke and CO_2.
5. **Natural sources**
   - Volcanic eruptions, forest fires, and dust storms.

(c) Effects of Air Pollution

1. **Health effects**
   - Respiratory diseases: **asthma, bronchitis, lung cancer**.
   - Eye irritation and skin diseases.
2. **Environmental effects**
   - **Acid rain:** SO_2 and NO_x react with water in the atmosphere to form H_2SO_4 and HNO_3, damaging plants, soil, and water bodies.
   - **Global warming:** Increased CO_2 leads to the **greenhouse effect**, causing climate change.
   - **Ozone depletion:** Release of CFCs destroys stratospheric ozone, increasing UV radiation.
3. **Economic effects**
   - Damage to buildings, vehicles, and monuments due to corrosion by acidic pollutants.
   - Reduced agricultural yield due to acid rain and climate change.

(d) Control Measures of Air Pollution

- Use **clean fuels** such as LPG, natural gas, and renewable energy sources.
- Install **chimney filters and scrubbers** in industries.
- Afforestation and planting **green belts** around industrial areas.
- Enforce **vehicle emission standards**.
- Promote **public transportation** and reduce vehicular traffic.

INTERRELATIONSHIP BETWEEN WATER AND AIR POLLUTION

- Pollutants released into the air can be **washed into rivers and lakes** by rain (acid rain).
- Industrial smoke can contain particles that settle in water bodies, increasing chemical contamination.
- Climate change caused by air pollution can affect **water quality** and availability.

IMPORTANCE OF STUDYING ENVIRONMENTAL CHEMISTRY

1. To understand the **causes and effects of pollution**.
2. To **protect public health** and the ecosystem.
3. To **design chemical processes** that are environmentally friendly.
4. To ensure **sustainable use of natural resources**.
5. To comply with **environmental laws and regulations**.

EVALUATION

1. Define environmental chemistry and explain its significance.
2. What is water pollution? List four causes and four effects.

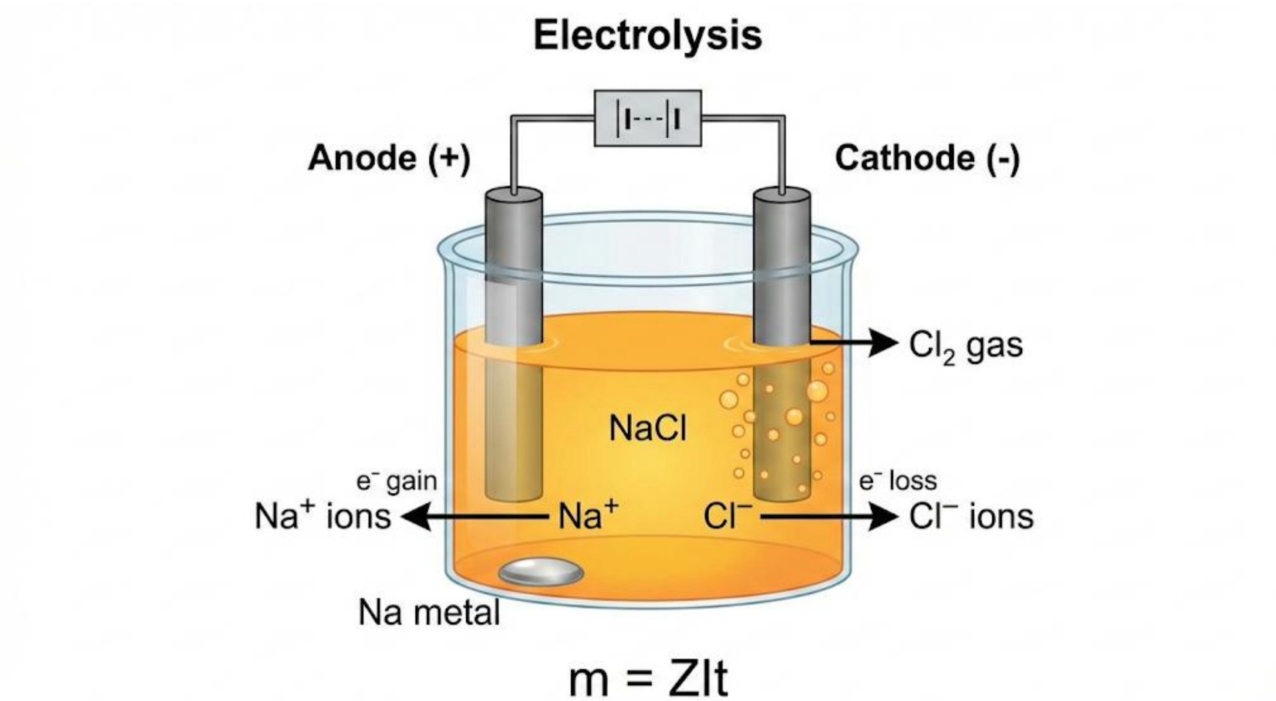
3. Define air pollution and mention five major sources.
4. Explain how SO_2 and NO_x contribute to acid rain.
5. Suggest five measures to reduce water and air pollution.
6. Explain the interrelationship between air and water pollution.

ASSIGNMENT

1. With examples, differentiate between **point source** and **non-point source water pollution**.
2. List ten pollutants (five water and five air) and describe their effects on humans, plants, and animals.
3. Prepare a brief report on **how industrial activities in your community contribute to water and air pollution**, and propose three chemical or practical solutions.

WEEK 2: ELECTROLYSIS

Electrolytes, electrodes, Faraday's laws



Introduction to Electrolysis

- **Definition:**
  Electrolysis is the process of using **direct electric current (DC)** to bring about a **chemical decomposition of an ionic compound (electrolyte)** in molten or aqueous solution.
- Electrolysis is an application of the movement of ions in solution.
- It is the basis of electroplating, extraction of metals, manufacture of chemicals, and electrorefining.

Basic Terms in Electrolysis

1. **Electrolyte**
   - A substance (usually ionic) that conducts electricity in the molten or aqueous state, undergoing chemical decomposition.

- Examples: NaCl (aq), H_2SO_4 (aq), molten NaCl, $CuSO_4$ (aq).
- Non-electrolytes: Do not conduct electricity or decompose (e.g., sugar, ethanol).

2. **Electrodes**
   - Conductors (usually metals or graphite) through which current enters or leaves the electrolyte.
   - **Anode (positive electrode):** Attracts anions (–) where oxidation occurs.
   - **Cathode (negative electrode):** Attracts cations (+) where reduction occurs.
3. **Electrochemical Cell**
   - Apparatus in which electrolysis occurs. It consists of:
     - **Power source (DC battery)**
     - **Electrolyte**
     - **Electrodes** connected to the power source.
4. **Ions**
   - **Cations (positive ions):** Migrate to cathode → gain electrons (reduction).
   - **Anions (negative ions):** Migrate to anode → lose electrons (oxidation).

Processes at Electrodes (Electrode Reactions)

- At **Cathode (reduction):**
  Cations gain electrons (electron consumption).
  Example:

$$Cu^{2+} + 2e^{-} \rightarrow Cu(s)$$

- At **Anode (oxidation):**
  Anions lose electrons (electron release).

Example:

$$2Cl^- \rightarrow Cl_2(g) + 2e^-$$

- General principle: **Cathode = Reduction; Anode = Oxidation (C.R.A.O.)**

Examples of Electrolysis

1. **Electrolysis of Acidified Water ($H_2O + H_2SO_4$)**
   - Cathode: H^+ ions reduced → H_2 gas.
   - Anode: OH^- ions oxidized → O_2 gas.

$$2H_2O(l) \xrightarrow{electricity} 2H_2(g) + O_2(g)$$

2. **Electrolysis of Molten NaCl**
   - Cathode: $Na^+ + e^- \rightarrow Na$
   - Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$
   - Products: Sodium metal and chlorine gas.
3. **Electrolysis of $CuSO_4$ (aq) using Carbon Electrodes**
   - Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$ (Cu deposited).
   - Anode: $2SO_4^{2-}$ and OH^- compete; oxygen gas liberated.

4. Electrolysis of $CuSO_4$ (aq) using Copper Electrodes

- Cathode: Copper deposited.
- Anode: Copper dissolves into solution.
- Used in **electroplating** and **refining of copper**.

Faraday's Laws of Electrolysis

1. **Faraday's First Law:**
    - *The mass of a substance deposited or liberated at an electrode is directly proportional to the quantity of electricity passed through the electrolyte.*

$m \propto Q$

$m = ZQ$

Where:

- m = mass of substance deposited (g)
- Q = total charge passed (Coulombs)
- Z = electrochemical equivalent (g/C)

Since $Q = It$,

$m = ZIt$

where I = current (A), t = time (s).

2. **Faraday's Second Law:**
    - *When the same quantity of electricity is passed through different electrolytes, the masses of substances deposited are proportional to their chemical equivalent weights.*

- Equivalent weight:

$$E = \frac{\text{Molar mass (M)}}{\text{Valency (n)}}$$

- So:

$$\frac{m_1}{m_2} = \frac{E_1}{E_2}$$

Applications of Electrolysis

- Extraction of metals (e.g., Al from bauxite via electrolysis of Al_2O_3 in cryolite).
- Electroplating (to prevent corrosion, improve appearance, or increase durability).
- Electrorefining of metals (purification of copper, zinc).
- Manufacture of chemicals (NaOH, Cl_2, H_2 by electrolysis of brine).
- Charging of rechargeable batteries.

Worked Examples

Example 1:
A current of 2.5 A is passed through $CuSO_4$ solution for 30 minutes. Calculate the mass of copper deposited. (Molar mass of Cu = 63.5 g/mol, valency = 2, F = 96500 C/mol e^-).

Solution:

- Time = 30 min = 1800 s
- Q = It = 2.5 × 1800 = 4500 C
- Equivalent weight = 63.5 ÷ 2 = 31.75 g
- Mass = (E × Q) / F = (31.75 × 4500) ÷ 96500 = **1.48 g**

Example 2:

During electrolysis of acidified water, 0.2 mol of O_2 is collected at the anode. Calculate the number of moles of H_2 formed at the cathode.

Solution:

Equation: $2H_2O \rightarrow 2H_2 + O_2$

Mole ratio: $H_2 : O_2 = 2 : 1$

So, if $O_2 = 0.2$ mol, $H_2 = 0.4$ mol.

Evaluation

1. Define electrolysis and give three examples of electrolytes.
2. Distinguish between anode and cathode.
3. Write electrode reactions for the electrolysis of:
   a) Molten NaCl
   b) Acidified water
4. State Faraday's First and Second Laws of Electrolysis.
5. Calculate the mass of silver deposited when a current of 5 A is passed through $AgNO_3$ solution for 1 hour (M = 108, valency = 1, F = 96500 C/mol e^-).
6. Mention four industrial applications of electrolysis.

Assignment

1. With equations, describe the electrolysis of $CuSO_4$ using copper electrodes.
2. A current of 4 A is passed through molten $PbBr_2$ for 30 minutes. Calculate:
   a) The quantity of electricity passed.
   b) The number of moles of electrons involved.
   c) The mass of lead deposited. (Molar mass of Pb = 207, valency = 2, F = 96500).
3. Research: Explain how aluminium is extracted from bauxite using electrolysis.

WEEK 3: ELECTROCHEMICAL CELLS

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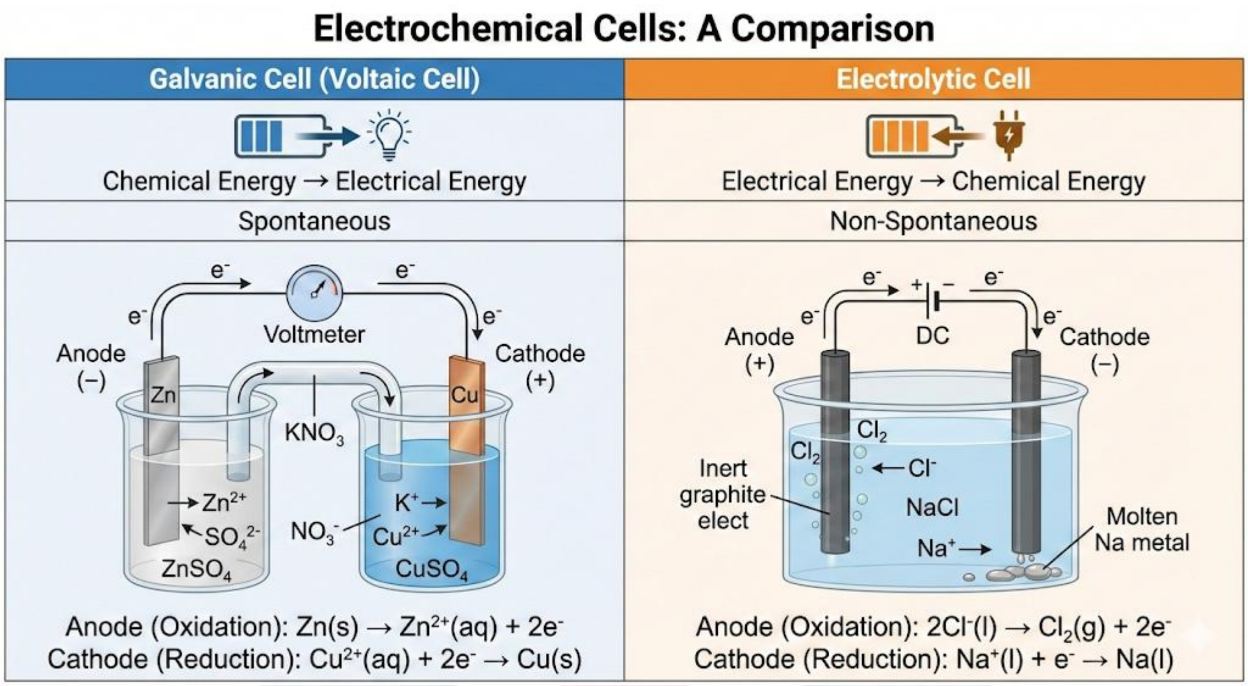
Galvanic and electrolytic cells



MEANING OF ELECTROCHEMICAL CELLS

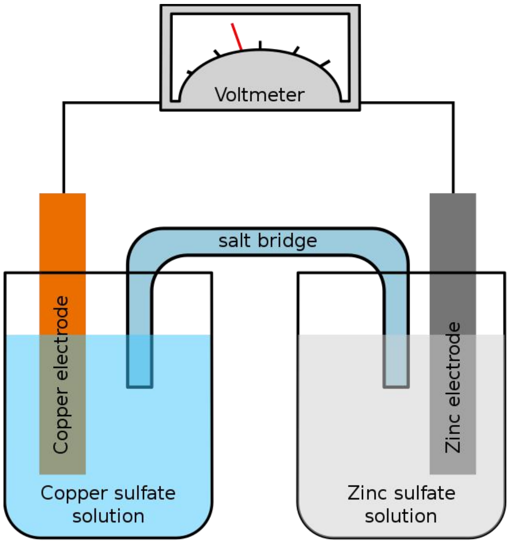
An **electrochemical cell** is a device that **converts chemical energy into electrical energy** or **electrical energy into chemical energy** through chemical reactions.

- **Key components:** two electrodes (conductors), an electrolyte (ionic conductor), and a medium to allow electron flow.
- Electrochemical cells are classified into **Galvanic (Voltaic) cells** and **Electrolytic cells** based on the direction of energy conversion.

GALVANIC (VOLTAIC) CELLS

Definition:

A **Galvanic cell** is an electrochemical cell in which a **spontaneous chemical reaction** produces electrical energy.



Features:

1. **Spontaneous reaction:** occurs naturally without external energy.
2. **Energy conversion:** chemical → electrical.
3. **Electrodes:** usually metals in contact with their ions.
4. **Electrolyte:** solution that conducts ions.
5. **Salt bridge:** maintains electrical neutrality by allowing ion flow.

Construction (Example – Daniell Cell):

- **Anode (negative electrode):** Zinc (Zn) → $Zn^{2+} + 2e^-$ (oxidation occurs)
- **Cathode (positive electrode):** Copper ($Cu^{2+} + 2e^- \rightarrow Cu$) (reduction occurs)
- **Salt bridge:** KNO_3 solution allows ions to flow and completes the circuit.
- **Electron flow:** from Zn to Cu through an external wire.

Cell notation:

$$Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$$

Working Principle:

1. Zinc loses electrons (oxidation) → Zn^{2+} goes into solution.

2. Electrons flow through the external wire to the copper electrode.
3. Copper ions gain electrons (reduction) → deposit on Cu electrode.
4. The flow of electrons constitutes an **electric current**.

Key Terms:

- **Anode:** electrode where oxidation occurs (Zn in Daniell cell, negative).
- **Cathode:** electrode where reduction occurs (Cu in Daniell cell, positive).
- **Electrolyte:** ionic conductor that allows ion movement.
- **Salt bridge:** neutralizes charge buildup.

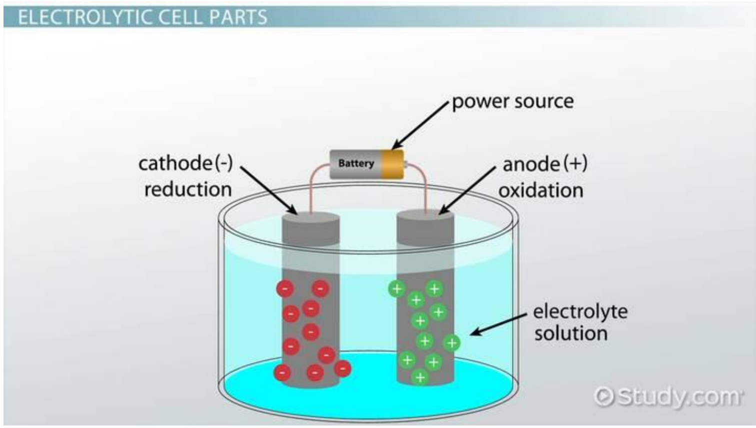
Applications of Galvanic Cells:

1. Batteries (dry cells, lead-acid, lithium-ion).
2. Powering small devices (torches, remote controls).
3. Demonstrating redox reactions in laboratories.

ELECTROLYTIC CELLS

Definition:

An **electrolytic cell** is an electrochemical cell in which **electrical energy is used to drive a non-spontaneous chemical reaction**.



Features:

1. **Non-spontaneous reaction:** requires external energy source (battery/power supply).
2. **Energy conversion:** electrical → chemical.
3. **Electrodes:** usually inert (graphite, platinum) or reactive depending on the reaction.
4. **Electrolyte:** ionic compound that melts or dissolves in water to allow ion conduction.

Construction & Working (Example – Electrolysis of Molten NaCl):

- **Electrodes:** graphite rods dipped in molten NaCl.
- **Power supply:** provides external electrical energy.
- **Electrolysis reactions:**
  - At **cathode (negative):** $Na^+ + e^- \rightarrow Na$ (reduction)
  - At **anode (positive):** $2Cl^- \rightarrow Cl_2 + 2e^-$ (oxidation)
- **Result:** Sodium metal deposits at cathode; chlorine gas released at anode.

Key Terms:

- **Cathode:** electrode connected to negative terminal; reduction occurs.

- **Anode:** electrode connected to positive terminal; oxidation occurs.
- **Electrolyte:** allows ion movement to complete the circuit.

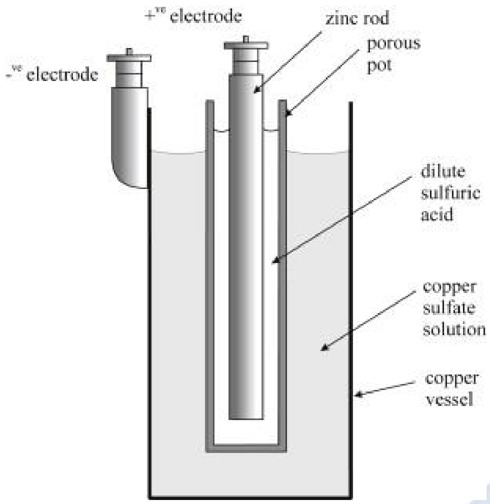
Applications of Electrolytic Cells:

1. Electroplating (gold, silver, copper).
2. Production of industrial chemicals ($NaOH$, Cl_2, H_2, Al).
3. Purification of metals (copper refining).
4. Electrolysis of water to produce hydrogen and oxygen.

DIFFERENCES BETWEEN GALVANIC AND ELECTROLYTIC CELLS

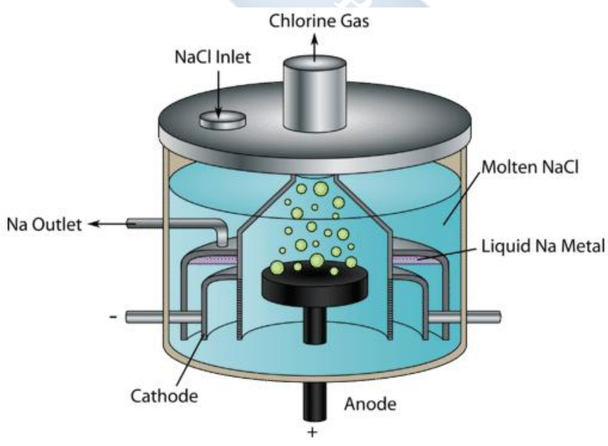
Feature	**Galvanic Cell**	**Electrolytic Cell**
Energy Conversion	Chemical → Electrical	Electrical → Chemical
Spontaneity	Spontaneous	Non-spontaneous
Electron Flow	Anode → Cathode	From negative terminal to positive terminal
Electrode Polarity	Anode negative, Cathode positive	Anode positive, Cathode negative
Examples	Daniell cell, Lead-acid battery	Electrolysis of molten NaCl, Electroplating

1. **Daniell Cell:**



- Two beakers: $ZnSO_4$ and $CuSO_4$ solutions
- Zn electrode in $ZnSO_4$, Cu electrode in $CuSO_4$
- Salt bridge (KNO_3 solution) connecting beakers
- External wire connecting Zn → Cu with voltmeter

2. **Electrolysis of Molten NaCl:**



- Container with molten NaCl
- Graphite electrodes connected to DC power supply
- Na^+ ions move to cathode, Cl^- ions move to anode

EVALUATION

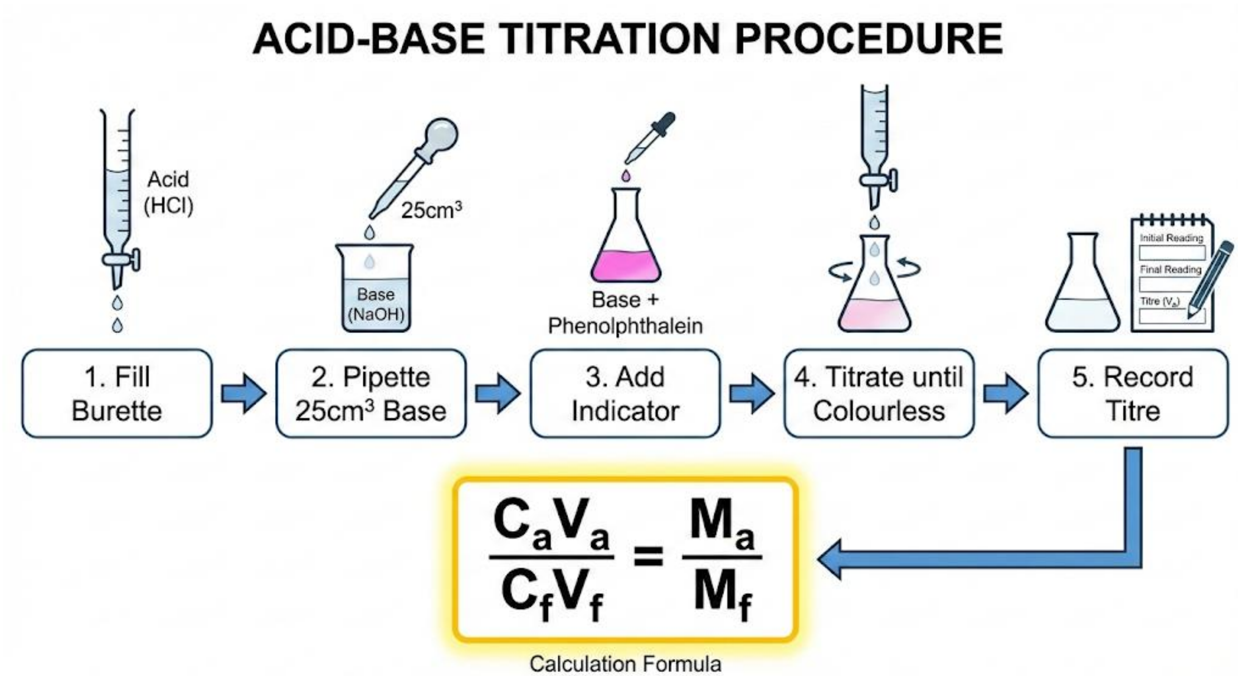
1. Define electrochemical cells and distinguish between galvanic and electrolytic cells.
2. Explain the working of a Daniell cell with equations.
3. State the roles of the salt bridge in a galvanic cell.
4. Describe the electrolysis of molten NaCl with reactions at each electrode.
5. Differentiate between anode and cathode in galvanic and electrolytic cells.

ASSIGNMENT

1. Draw and label a diagram of a Daniell cell and indicate the direction of electron flow.
2. Explain three industrial applications of electrolytic cells.
3. Compare galvanic and electrolytic cells in a tabular form, including energy conversion, spontaneity, and electrode polarity.

WEEK 4: ACID-BASE TITRATION

Apparatus, Procedures, Calculations



INTRODUCTION TO ACID-BASE TITRATION

- **Definition:**
  Acid-Base Titration is a **quantitative analytical technique** used to determine the **concentration of an acid or base solution** by reacting it with a solution of **known concentration**.
- It is a **neutralization reaction** between an acid and a base in the presence of an indicator to show the end point.
- Titrations are fundamental in **chemistry laboratories, food and beverage industries, pharmaceutical industries, and environmental studies**.

Concept of Titration

- Titration involves **accurately measuring a known volume of one solution** (usually using a **pipette**) and adding another solution of **known concentration** from a **burette** until **neutralization (end point)** is reached.

- The reaction is typically:

$$\text{Acid } (H^+) + \text{Base } (OH^-) \rightarrow \text{Salt} + \text{Water } (H_2O)$$

- The solution in the burette is called the **titrant**, and the solution in the conical flask is the **analyte**.

Acid-Base Theories (Recap)

1. **Arrhenius Theory:**
   - Acids produce **H^+ ions** in aqueous solutions, bases produce **OH^- ions**.
2. **Bronsted-Lowry Theory:**
   - Acid: Proton donor; Base: Proton acceptor.
3. **Lewis Theory:**
   - Acid: Electron-pair acceptor; Base: Electron-pair donor.

Apparatus and Materials Used in Titration

Apparatus/Material	**Description and Use**
Burette	Graduated glass tube with a tap at the bottom to deliver precise volumes of titrant.
Pipette	Used to accurately measure and transfer a fixed volume of solution (e.g., 25 cm^3).
Conical Flask	Holds the solution being analyzed (analyte).
Measuring Cylinder	For measuring approximate volumes of solutions.
White Tile	Placed under the flask to help observe color change clearly.

Funnels	Used to fill burettes safely without spillage.
Wash Bottle	Contains distilled water for rinsing equipment.
Indicators	Chemicals that change color to indicate end point.
Stand and Clamp	For holding burette vertically.

Indicators in Acid-Base Titration

- **Indicators** are substances that change color at or near the equivalence point.
- Examples:

Indicator	**Acid-Base Pair Example**	**Colour Change**	**pH Range**
Phenolphthalein	Strong acid vs strong base; weak acid vs strong base	Colourless → Pink	8.3 – 10.0
Methyl Orange	Strong acid vs strong base	Red → Orange → Yellow	3.1 – 4.4
Litmus	General indicator	Red (acidic) → Blue (basic)	~7.0

Procedure for Acid-Base Titration

Step-by-Step Process:

1. **Preparation of Apparatus:**
    - Rinse the burette with the **acid solution** to be used as titrant.
    - Rinse the pipette with the **alkali/base solution**.
    - Rinse the conical flask with **distilled water only**.
2. **Filling the Burette:**

- Fill the burette with the acid (titrant) and clamp it vertically. Record the initial reading (at the meniscus level) to **2 decimal places**.

3. **Measuring the Analyte:**
   - Use a pipette filler to draw exactly **25.0 cm³** of the base (e.g., NaOH) into the conical flask.
4. **Adding Indicator:**
   - Add 2–3 drops of the chosen indicator (e.g., phenolphthalein).
5. **Performing the Titration:**
   - Place the flask on a white tile.
   - Slowly run the acid from the burette into the base while swirling the flask.
   - Near the endpoint, add the titrant dropwise.
6. **Detecting Endpoint:**
   - The endpoint is reached when the indicator changes color (e.g., colorless to faint pink).
7. **Repeat for Accuracy:**
   - Perform 2–3 **concordant titres** (titres within 0.1 cm³ of each other).
   - Calculate the **average titre value**.

Calculations Involved in Titration

Formula:

$$C_1V_1/n_1 = C_2V_2/n_2$$

Where:

- C_1 = Concentration of acid (mol/dm³)
- V_1 = Volume of acid used (cm³)
- n_1 = Mole ratio of acid in balanced equation
- C_2 = Concentration of base (mol/dm³)
- V_2 = Volume of base used (cm³)
- n_2 = Mole ratio of base in balanced equation

Example:
If 25.0 cm^3 of NaOH solution requires 24.8 cm^3 of 0.10 mol/dm^3 HCl for neutralization, calculate the molarity of NaOH.

Reaction:

$$\mathrm{HCl} + \mathrm{NaOH} \rightarrow \mathrm{NaCl} + \mathrm{H_2O}$$

$$n_1 = 1, n_2 = 1$$

$$C_1V_1/n_1 = C_2V_2/n_2$$

$$0.10 \times 24.8/1000 = C_{\mathrm{NaOH}} \times 25.0/1000$$

$$C_{\mathrm{NaOH}} = \frac{0.10 \times 24.8}{25.0} = 0.0992\,\mathrm{mol/dm^3}$$

Practical Applications of Titration

1. Determining **acidity of vinegar and soft drinks**.
2. Quality control in **pharmaceutical industries**.
3. Estimating **alkalinity of water** in environmental monitoring.
4. **Food processing** (acidity and alkalinity regulation).
5. Used in **fertilizer and chemical manufacturing** to determine chemical purity.

Precautionary Measures

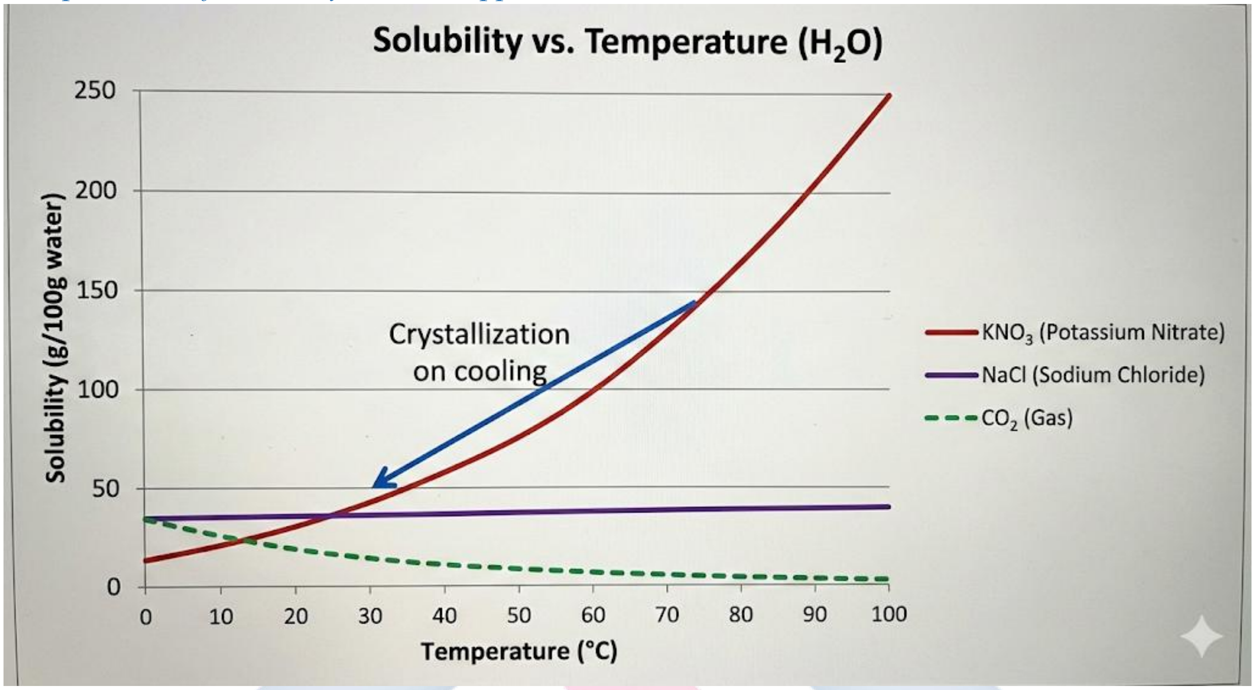
1. Rinse all apparatus with appropriate solutions before use.
2. Read burette values at **eye level** to avoid parallax error.
3. Add titrant slowly near endpoint.
4. Ensure concordant titres (repeat titration).
5. Use appropriate indicators for accurate detection.
6. Work in a well-lit lab with white tile to observe color change clearly.

Evaluation

1. Define titration and explain its importance.
2. List and describe **five apparatus** used in titration.
3. Outline the steps for carrying out an acid-base titration.
4. State three differences between **endpoint** and **equivalence point**.
5. Explain why a **pipette** is rinsed with base solution and a **burette** with acid.
6. A student used 0.10 M HCl to titrate 25.0 cm^3 of NaOH. If the average titre is 24.5 cm^3, calculate the concentration of NaOH in mol/dm^3.

WEEK 6: SOLUBILITY AND SOLUBILITY CURVES

Meaning of Solubility; Factors Affecting Solubility; Solubility Calculations; Construction and Interpretation of Solubility Curves; Applications



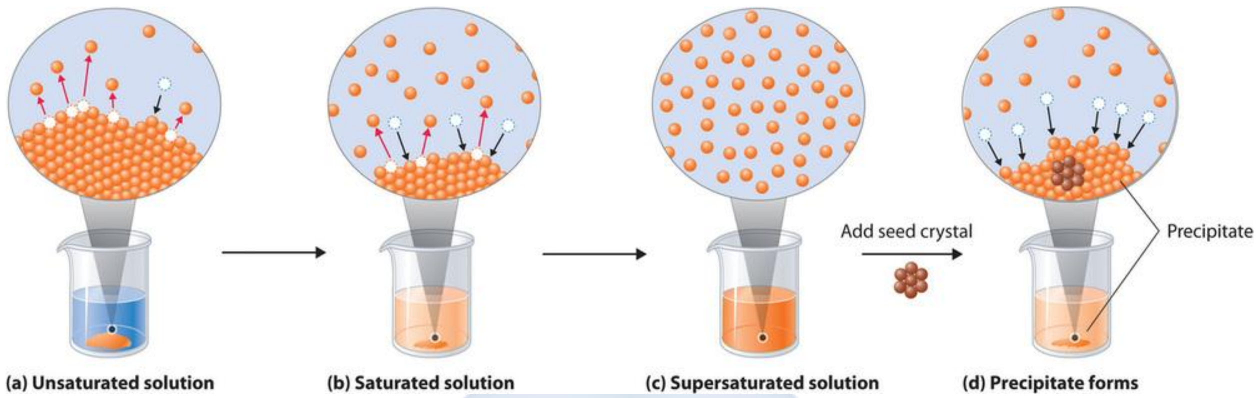
Meaning of Solubility

- **Solubility** is defined as the maximum amount of solute that can dissolve in 100 g of a solvent at a specific temperature to form a saturated solution.
- Units: usually expressed in **g of solute per 100 g of solvent** or **mol/dm³**.

Key Concepts:

- **Solute:** Substance being dissolved (e.g., NaCl, sugar).
- **Solvent:** Substance doing the dissolving (e.g., water, ethanol).
- **Saturated solution:** A solution that contains the maximum possible solute at a given temperature.
- **Unsaturated solution:** Contains less solute than the maximum.

- **Supersaturated solution:** Contains more solute than the normal maximum; achieved by heating and then slowly cooling. These are unstable.



Factors Affecting Solubility

1. **Nature of Solute and Solvent**
   - "Like dissolves like" principle:
     - Polar solutes dissolve in polar solvents (e.g., sugar in water).
     - Non-polar solutes dissolve in non-polar solvents (e.g., iodine in carbon tetrachloride).
2. **Temperature**
   - For most solids: solubility **increases** with temperature.
     - Example: KNO_3 dissolves much more at high temperature.
   - For gases: solubility **decreases** with temperature.
     - Example: Less oxygen dissolves in hot water → reason why fish struggle in warm water.
3. **Pressure**
   - Little effect on solids/liquids.

- For gases: solubility increases with pressure (Henry's Law).
  - Example: Carbon dioxide in soft drinks is kept under high pressure.

4. **Particle Size (Surface Area)**
   - Smaller particles dissolve faster because they expose a larger surface area.
5. **Agitation (Stirring/Shaking)**
   - Stirring increases the rate of dissolution but not the solubility limit.

Solubility Calculations

General Formula:

$$\text{Solubility} = \frac{\text{Mass of solute (g)}}{\text{Mass of solvent (g)}} \times 100$$

Worked Example 1:

20 g of KNO_3 dissolves in 50 g of water at 40°C. Find its solubility.

$$\text{Solubility} = \frac{20}{50} \times 100 = 40\, g/100g\, \text{H}_2\text{O}$$

Worked Example 2:

108 g of KCl dissolves in 200 g of water at 60°C. Calculate its solubility.

$$\text{Solubility} = \frac{108}{200} \times 100 = 54\, g/100g\, \text{H}_2\text{O}$$

Worked Example 3 (Crystallization):

At 70°C, 136 g of KNO_3 dissolves in 100 g of water. At 40°C, solubility = 64 g/100 g water.
How much KNO_3 will crystallize if a saturated solution at 70°C is cooled to 40°C?

$$136 - 64 = 72g \text{ will crystallize per 100 g water}$$

Worked Example 3 (Crystallization):

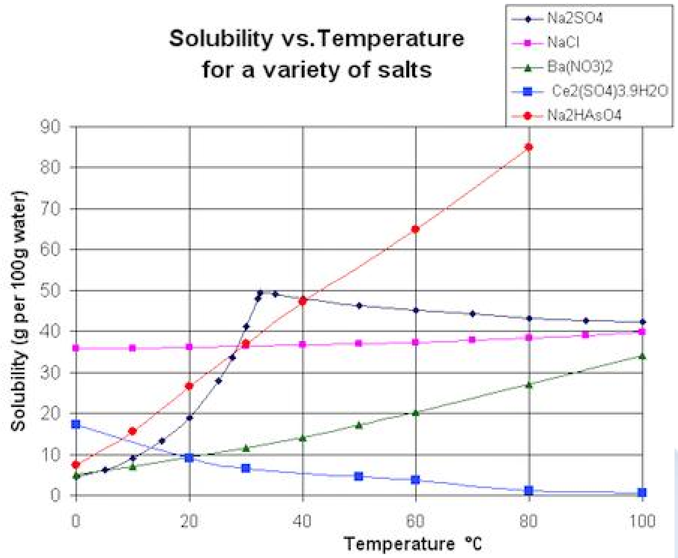
At 70°C, 136 g of KNO_3 dissolves in 100 g of water. At 40°C, solubility = 64 g/100 g water.

How much KNO_3 will crystallize if a saturated solution at 70°C is cooled to 40°C?

136 – 64 = 72g will crystallize per 100 g water

Solubility Curves

- A **solubility curve** is a graph showing the variation of solubility of a substance with temperature.



- **Construction:**
    - Measure solubility values at different temperatures.
    - Plot **temperature (x-axis)** vs **solubility (y-axis)**.
    - Join points smoothly to get the curve.

Interpretation:

1. A steep upward curve = solubility greatly increases with temperature (e.g., KNO_3).
2. A nearly flat curve = solubility little affected by temperature (e.g., NaCl).
3. A downward curve = gases, solubility decreases with temperature.
4. Intersection of curves = equal solubility of different salts at a certain temperature.

Case Studies:

- **Potassium Nitrate (KNO_3):** Highly soluble, steep curve → used in crystallization experiments.
- **Sodium Chloride (NaCl):** Almost flat curve → little effect of temperature.

- **Carbon dioxide (CO_2):** Decreasing curve → explains why warm soda loses fizz quickly.

Applications of Solubility and Solubility Curves

1. **Purification of substances (Crystallization):** Impurities remain in solution while pure crystals form.
2. **Pharmaceuticals:** Drug dissolution in body fluids is guided by solubility.
3. **Food industry:** Solubility of sugar and salts used in syrup production, preservation.
4. **Water bodies:** Oxygen solubility determines survival of aquatic organisms.
5. **Industrial design:** Boilers and cooling systems – solubility determines scale formation.
6. **Soft drinks industry:** Carbonation depends on solubility of CO_2 under pressure.

Evaluation

1. Define solubility and explain the difference between saturated, unsaturated, and supersaturated solutions.
2. State four factors that affect solubility, with examples.
3. A student dissolves 30 g of salt in 75 g of water at 40°C. Calculate the solubility in g/100 g of water.
4. Sketch a typical solubility curve for KNO_3 and describe its shape.
5. Why does CO_2 escape faster from a warm bottle of soda than from a cold one?
6. Explain why crystallization is possible when a hot saturated solution of KNO_3 is cooled.

Assignment

1. From the solubility curve of KNO_3:

- At 40°C → 64 g/100 g water
- At 70°C → 136 g/100 g water

  Calculate the mass of KNO_3 that will crystallize if 300 g of a hot saturated solution is cooled from 70°C to 40°C.

2. Sketch and label solubility curves for:

   a. A solid that increases in solubility with temperature (e.g., KNO_3).

   b. A solid whose solubility is almost independent of temperature (e.g., NaCl).

   c. A gas whose solubility decreases with temperature.

3. Research Questions:

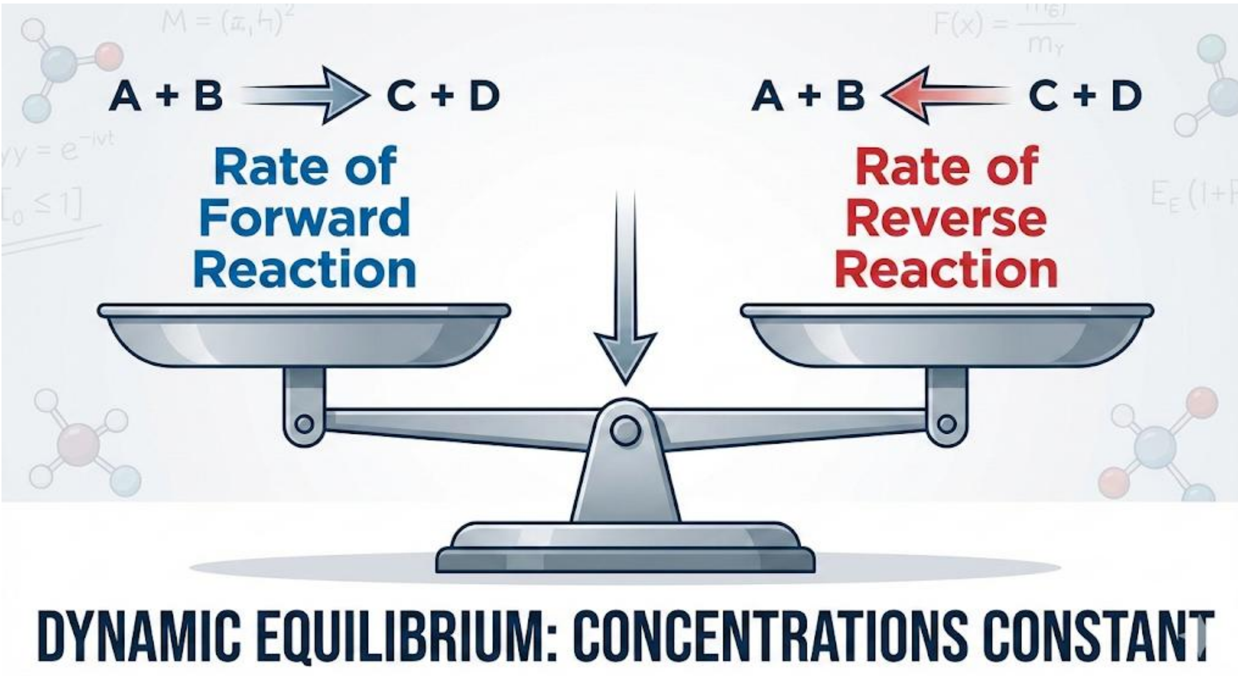
   a. Why do fish die in polluted or heated rivers? Relate your answer to solubility.

   b. Explain how crystallization is applied in the production of sugar from sugarcane juice.

   c. Investigate and report how the pharmaceutical industry uses solubility data in drug formulation.

WEEK 8: CHEMICAL EQUILIBRIUM

Equilibrium Constant and Calculations



In Chemistry, reactions may be classified as either *irreversible* or *reversible*.

- **Irreversible reactions** are those in which reactants are completely converted to products and cannot change back under the same conditions. For example:

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s)$$

Once magnesium burns in oxygen, magnesium oxide is formed and does not decompose under normal conditions.

- **Reversible reactions**, however, can proceed in both forward and backward directions depending on the conditions. This is represented with a double arrow (⇌). An example is:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Here, nitrogen and hydrogen combine to form ammonia, but ammonia can also decompose back into nitrogen and hydrogen.

In the early stages of a reversible reaction, the **forward reaction rate** is usually faster because there is an abundance of reactants. However, as the reaction proceeds, the concentration of reactants decreases while the concentration of products increases. This makes the **reverse reaction** begin to occur more significantly. Eventually, a stage is reached when the forward and backward reactions proceed at the same rate. At this point, the concentrations of reactants and products remain constant. This state is called **chemical equilibrium**.

It is important to note that equilibrium is **dynamic**, not static. Molecules are still colliding and reacting, but the overall concentrations remain unchanged because the rate of the forward reaction equals the rate of the backward reaction.

The Equilibrium Constant (Kc)

The concept of equilibrium can be expressed mathematically through the **equilibrium constant**.

For a general reversible reaction:

$$aA + bB \rightleftharpoons cC + dD$$

The equilibrium constant expression is given by:

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

where:

- [A], [B], [C], [D] are the equilibrium concentrations of the species, expressed in moles per cubic decimeter (mol dm^{-3}),
- a, b, c, d are the stoichiometric coefficients from the balanced chemical equation.

Characteristics of Kc:

1. The value of Kc depends **only on temperature** for a given reaction.
2. A large Kc (much greater than 1) indicates equilibrium strongly favors product formation.

3. A small Kc (much less than 1) indicates equilibrium strongly favors the reactants.
4. If Kc≈1, it means that neither side is favored – both reactants and products are present in comparable amounts.
5. Kc is dimensionless when the total number of moles of reactants and products are equal; otherwise, it has units depending on the reaction.

Worked Examples of Equilibrium Constant Calculations

Example 1

For the reaction:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

Suppose equilibrium concentrations are:

$[H_2] = 0.2\, mol\, dm^{-3}, [I_2] = 0.2\, mol\, dm^{-3}, [HI] = 0.6\, mol\, dm^{-3}$.

$$K_c = \frac{[HI]^2}{[H_2][I_2]}$$

$$K_c = \frac{(0.6)^2}{(0.2)(0.2)} = \frac{0.36}{0.04} = 9.0$$

This large value shows that equilibrium favors the product (HI).

Example 2

For the reaction:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

At equilibrium: $[N_2] = 0.1\, mol\, dm^{-3}$, $[H_2] = 0.3\, mol\, dm^{-3}$, $[NH_3] = 0.4\, mol\, dm^{-3}$.

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$K_c = \frac{(0.4)^2}{(0.1)(0.3)^3}$$

$$K_c = \frac{0.16}{0.0027} \approx 59.3$$

This very large Kc indicates that the forward reaction is highly favored, which is consistent with the industrial synthesis of ammonia (Haber process).

Factors Affecting Equilibrium Constant

- **Temperature:**
  - If the reaction is exothermic, increasing temperature shifts equilibrium backward, and Kc decreases.
  - If the reaction is endothermic, increasing temperature shifts equilibrium forward, and Kc increases.
- **Concentration and Pressure:**
  - Altering concentrations or pressure only shifts the **position of equilibrium** but does not affect the actual value of Kc.
- **Catalyst:**
  - A catalyst speeds up the rate of both forward and backward reactions equally, allowing equilibrium to be reached faster. However, it does not alter the equilibrium position or the value of Kc.

Real-Life Applications of Equilibrium

1. **Haber Process:** Production of ammonia from nitrogen and hydrogen is governed by equilibrium principles.
2. **Contact Process:** Manufacture of sulphuric acid involves equilibrium in the conversion of SO_2 to SO_3.
3. **Carbonated Drinks:** Equilibrium exists between dissolved carbon dioxide in liquid and gaseous carbon dioxide above the liquid in sealed bottles.
4. **Biological Systems:** Oxygen binding to hemoglobin in the blood is a reversible equilibrium reaction.

Evaluation:

1. Define chemical equilibrium and explain why it is described as *dynamic equilibrium*.
2. State two differences between irreversible and reversible reactions, giving one example of each.
3. Derive the equilibrium constant expression for the following reactions:
   a) $2SO_2(g)+O_2(g) \rightleftharpoons 2SO_3(g)$
   b) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
4. The equilibrium constant for the reaction $H_2+Cl_2 \rightleftharpoons 2HCl$ is very large at room temperature. What does this imply about the position of equilibrium?
5. List two factors that can change the **value** of Kc, and two factors that can change the equilibrium position but not the equilibrium constant.

Assignment:

1. A mixture of hydrogen and iodine gases is allowed to reach equilibrium at 500 K. At equilibrium, $[H_2] = 0.1\, mol\, dm^{-3}, [I_2] = 0.1\, mol\, dm^{-3}, [HI] = 0.8\, mol\, dm^{-3}$. Calculate the equilibrium constant, K_c.
2. At a certain temperature, the equilibrium constant for the reaction:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

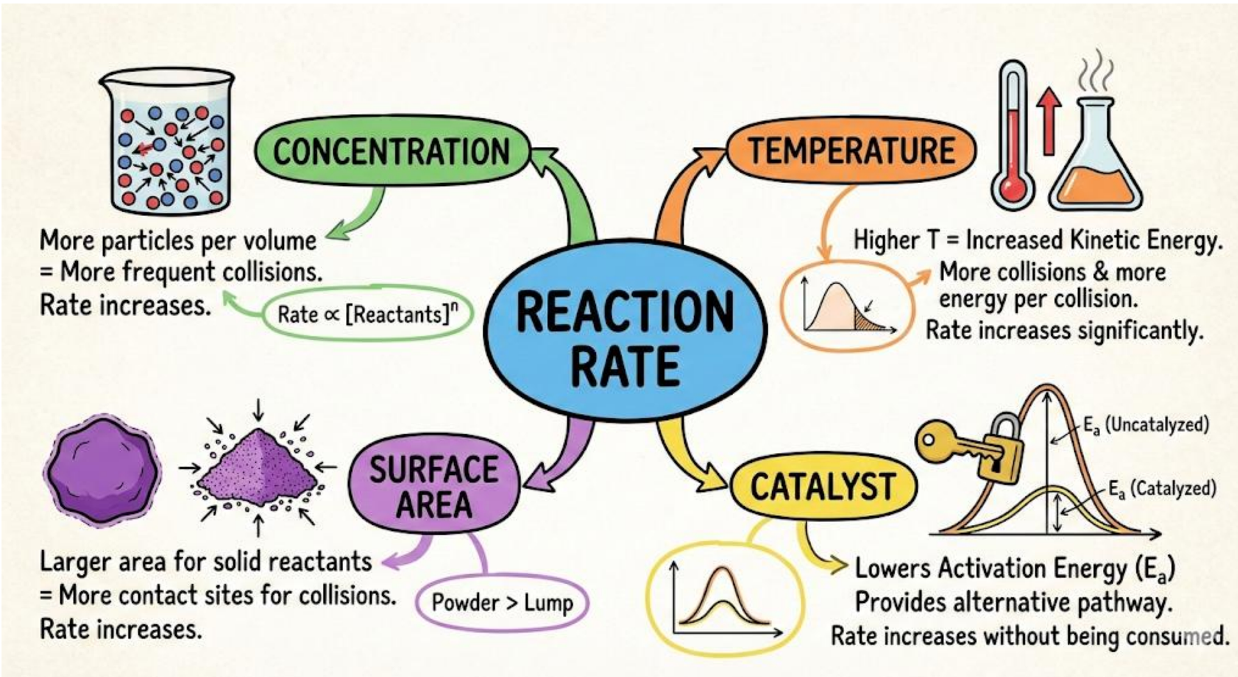
is 0.5. Does this mean that the reaction favors the reactants or the products? Explain your answer.

3. Using the equilibrium constant, explain why heating favors the decomposition of N_2O_4 into NO_2.

4. In your own words, explain why a catalyst cannot change the position of equilibrium but is very important in industrial chemical processes.

WEEK 9: CHEMICAL KINETICS

Rate Laws and Order of Reaction



Meaning of Chemical Kinetics

Chemical kinetics is the study of the *speed* at which chemical reactions occur and the *mechanisms* by which they proceed. It answers two key questions:

1. **How fast does a reaction proceed?**
2. **By what pathway does the reaction take place?**

While **thermodynamics** tells us whether a reaction is feasible (spontaneous or not), **kinetics** tells us how quickly that reaction will occur. For example:

- Rusting of iron is *thermodynamically favorable* but *kinetically slow*.
- Combustion of petrol is both thermodynamically favorable and *kinetically fast*.

Rate of Reaction

The rate of reaction measures the *change in concentration* of a reactant or product with time.

$$\text{Rate} = -\frac{d[A]}{dt} = \frac{d[B]}{dt}$$

- **Negative sign (–):** indicates that the reactant concentration decreases.
- **Positive rate for products:** concentration increases as reaction proceeds.

Example:

For the decomposition of hydrogen peroxide:

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$$

Rate can be expressed as:

$$\text{Rate} = -\frac{1}{2}\frac{d[H_2O_2]}{dt} = \frac{d[O_2]}{dt}$$

Types of Rates

- **Average Rate:** change in concentration over a time interval.

  $$\text{Average Rate} = \frac{\Delta[C]}{\Delta t}$$

- **Instantaneous Rate:** rate at a specific time (slope of tangent to the concentration vs. time graph).

Factors Affecting Rate of Reaction

1. **Concentration of Reactants** – higher concentration means more frequent collisions, hence faster reaction. (Collison theory).
   Example: A strong acid reacts faster with zinc than a dilute acid.
2. **Temperature** – increasing temperature gives molecules more kinetic energy; more particles have energy ≥ activation energy, so rate increases.
   Rule of thumb: Rate approximately doubles for every 10°C rise in temperature.

3. **Pressure** (for gaseous reactions) – increasing pressure increases the concentration of gas molecules, causing more collisions.

4. **Surface Area** – powdered solids react faster than large lumps because more surface is exposed.
   Example: Finely powdered coal ignites explosively while large lumps burn slowly.

5. **Catalyst** – a catalyst lowers the activation energy, providing an alternative pathway. It is not consumed in the reaction.
   Example: Manganese dioxide catalyzes the decomposition of hydrogen peroxide.

6. **Nature of Reactants** – ionic reactions (e.g., neutralization) are very fast, while covalent bond-breaking reactions are slower.

Rate Law (Rate Equation)

The rate law relates the reaction rate to the concentrations of reactants raised to certain powers (orders).

For a general reaction:

$$aA + bB \rightarrow cC + dD$$

Rate law:

$$\text{Rate} = k[A]^m[B]^n$$

where:

- k = rate constant (depends on temperature & catalyst, not concentration).
- m, n = order of reaction with respect to A and B.
- m + n = overall order of reaction.

Key Points:

- The rate law **must be determined experimentally**; it cannot be predicted from the stoichiometric equation.
- Units of the rate constant k depend on the order of reaction.

Example:

If $\text{Rate} = k[A][B]^2$, then:

- Order wrt A = 1
- Order wrt B = 2
- Overall order = 3

Order of Reaction

The order of a reaction is the sum of the powers of concentration terms in the rate law.

Types of Order of Reaction:

- **Zero-order:** Rate is independent of concentration.

  Rate = k

Example: Decomposition of NH_3 on a hot tungsten surface.

- **First-order:** Rate depends linearly on concentration.

  Rate = k[A]

Example: Radioactive decay, decomposition of N_2O_5.

- **Second-order:** Rate depends on the square of concentration (or product of two concentrations).

$$\text{Rate} = k[A]^2 \text{ or } k[A][B]$$

- **Fractional and Mixed Order:** Some reactions show fractional (e.g., 1.5) or different orders under different conditions.

Experimental Determination of Order of Reaction

1. **Method of Initial Rates** – measure the initial rate at varying concentrations of reactants. The rate dependence reveals the order.
2. **Integrated Rate Law Method** – use integrated rate equations to plot graphs:
    - Zero-order: [A] vs time → straight line (negative slope).
    - First-order: ln [A] vs time → straight line.
    - Second-order: 1/[A] vs time → straight line.
3. **Half-life Method** – half-life (time for concentration to reduce by half) is independent of concentration only in first-order reactions.

Significance and Applications of Rate Laws

- Helps in deducing **reaction mechanisms** (the stepwise process).
- Useful in **drug design** – understanding how medicines degrade in the body.
- Industrial applications:
    - **Haber process** (production of ammonia: controlling rate with pressure & catalyst).
    - **Contact process** (sulphuric acid manufacture).
    - **Petrochemical industry** – catalytic cracking of hydrocarbons.

Evaluation

1. Define chemical kinetics and explain how it differs from thermodynamics.
2. State and explain **four factors** that affect the rate of chemical reactions.
3. Write the rate law for the reaction:

$$2NO + O_2 \rightarrow 2NO_2$$

given that rate law is: $\mathrm{Rate} = k[NO]^2[O_2]$

- Find the order with respect to NO.
- Find the overall order of the reaction.

4. Explain why the rate law must be determined experimentally.
5. Differentiate between **zero-order** and **first-order** reactions with suitable examples.

Assignment

1. For the decomposition of N_2O_5:

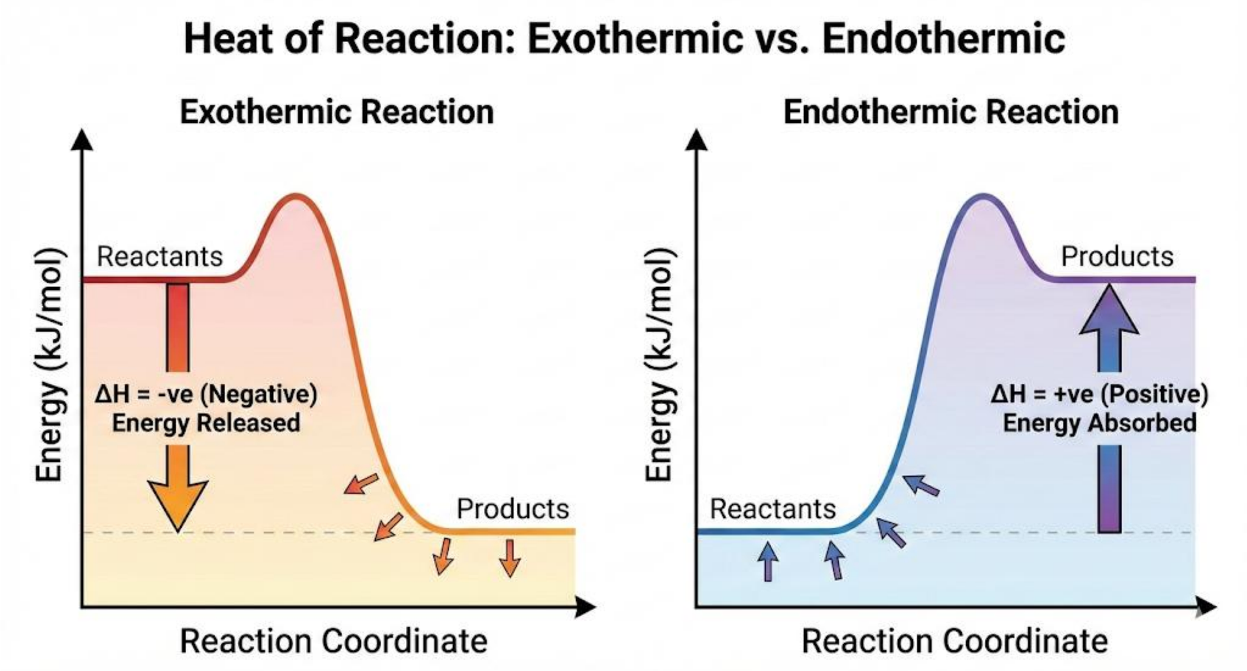
$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

The rate law is first-order: Rate = $k[N_2O_5]$
(a) Sketch the graph of ln $[N_2O_5]$ against time.
(b) Explain how this graph confirms the reaction order.

2. The half-life of a first-order reaction is **300 seconds**. If the initial concentration is 1.0 mol dm^{-3}, calculate:
   (a) The rate constant, k.
   (b) The concentration after 600 seconds.
3. Industrial Application:
   (a) Explain how catalysts are used in the Haber Process and their effect on the rate of reaction.
   (b) Why is high temperature not always the best condition for maximum yield in industrial reactions, even though it increases reaction rate?

WEEK 10: HEAT OF REACTION

Enthalpy Change, Hess's Law



INTRODUCTION

Chemical reactions are often accompanied by **energy changes**, usually in the form of **heat**. Some reactions release heat (exothermic), while others absorb heat (endothermic). The study of these heat changes is important because it helps chemists:

- Predict the feasibility of reactions,
- Design industrial processes,
- Calculate energy requirements for large-scale production,
- Understand environmental and biological processes.

This area of chemistry is known as **thermochemistry**.

HEAT OF REACTION

Definition

The heat of reaction (ΔH) is the amount of heat absorbed or released during a chemical reaction carried out under constant pressure. It is expressed in kilojoules per mole (kJ/mol).

Characteristics

1. It depends on the **nature of the reactants and products**.
2. It is affected by the **physical states** (solid, liquid, gas) of reactants and products.
3. It depends on **temperature and pressure**.
4. It is a **state function**, meaning it depends only on the initial and final states, not on the path of the reaction.

Types of Heat of Reaction

1. **Heat of Formation** – Heat change when **one mole of a compound is formed** from its constituent elements under standard conditions (25°C, 1 atm, 1M solution).
   - Example:
     $C (s) + O_2 (g) \rightarrow CO_2 (g), \Delta H = -394$ kJ/mol
2. **Heat of Combustion** – Heat change when **one mole of a substance is completely burnt in oxygen**.
   - Example:
     $CH_4 (g) + 2O_2 (g) \rightarrow CO_2 (g) + 2H_2O (l), \Delta H = -890$ kJ/mol
3. **Heat of Neutralization** – Heat change when **one mole of water is formed** from the neutralization of an acid and a base.
   - Example:
     $HCl (aq) + NaOH (aq) \rightarrow NaCl (aq) + H_2O (l), \Delta H = -57$ kJ/mol
4. **Heat of Solution** – Heat change when **one mole of a substance dissolves in excess solvent**.
5. **Heat of Atomization** – Heat required to **separate one mole of a compound into its atoms in gaseous state**.

ENTHALPY CHANGE (ΔH)

Definition

Enthalpy (H) is the total heat content of a system at constant pressure. The **enthalpy change (ΔH)** of a reaction is the difference between the enthalpy of products and reactants:

$\Delta H = H_{products} - H_{reactants}$

Classification Based on Sign of ΔH

1. **Exothermic Reactions ($\Delta H < 0$)**
   - Heat is **released** to the surroundings.
   - Example: Combustion of methane.
2. **Endothermic Reactions ($\Delta H > 0$)**
   - Heat is **absorbed** from the surroundings.
   - Example: Photosynthesis, decomposition of calcium carbonate.

HESS'S LAW OF CONSTANT HEAT SUMMATION

Statement

Hess's Law states that:

"The total enthalpy change of a chemical reaction is the same, regardless of the route by which the reaction occurs, provided the initial and final conditions are the same."

This is possible because **enthalpy is a state function**.

Importance

- It allows us to calculate ΔH for reactions that cannot be measured directly.
- It helps in determining enthalpies of formation, combustion, and atomization.

Illustration

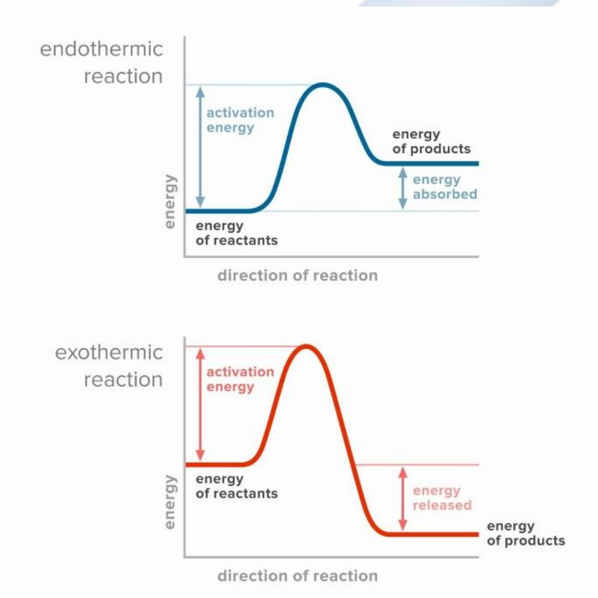
1. Direct reaction:
   $C\ (s) + O_2\ (g) \rightarrow CO_2\ (g) \quad \Delta H = -394$ kJ/mol

2. Indirect reaction:
   $C\ (s) + \frac{1}{2}O_2\ (g) \rightarrow CO\ (g) \quad \Delta H_1 = -110$ kJ/mol
   $CO\ (g) + \frac{1}{2}O_2\ (g) \rightarrow CO_2\ (g) \quad \Delta H_2 = -284$ kJ/mol

Total $\Delta H = \Delta H_1 + \Delta H_2 = -110 + (-284) = -394$ kJ/mol

This agrees with the direct method, confirming Hess's Law.

ENERGY LEVEL DIAGRAMS



- **Exothermic reaction**: Products have lower enthalpy than reactants. Heat is released to surroundings.
- **Endothermic reaction**: Products have higher enthalpy than reactants. Heat is absorbed from surroundings.

APPLICATIONS OF ENTHALPY AND HESS'S LAW

1. In calculating **standard enthalpy changes** for reactions.
2. In designing **energy-efficient industrial processes** (e.g., Haber process, contact process).
3. In determining **food calorific values**.
4. In understanding **biological energy changes** (respiration, photosynthesis).
5. In predicting **reaction feasibility**.

EVALUATION

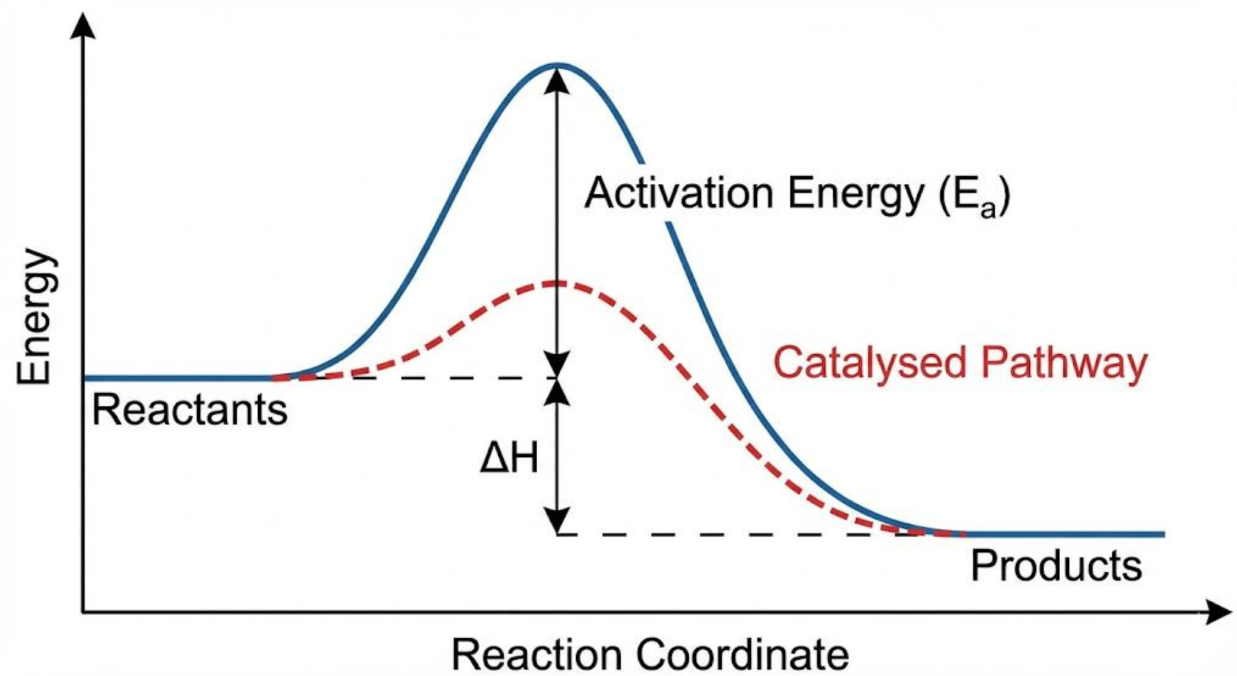
1. Define heat of reaction and list four types with examples.
2. Differentiate between **exothermic** and **endothermic** reactions with suitable equations.
3. State Hess's Law and explain why it is true.
4. Using Hess's Law, calculate ΔH for the combustion of carbon to CO_2 given:
   - $C + \frac{1}{2}O_2 \rightarrow CO$, $\Delta H = -110$ kJ/mol
   - $CO + \frac{1}{2}O_2 \rightarrow CO_2$, $\Delta H = -284$ kJ/mol
5. Explain the significance of enthalpy change in industrial processes.

ASSIGNMENT

1. Write short notes on **heat of neutralization, heat of combustion, and heat of formation**.
2. Draw and label energy level diagrams for (a) exothermic and (b) endothermic reactions.
3. State Hess's Law and use it to calculate the enthalpy change for the reaction:
   $N_2\,(g) + O_2\,(g) \rightarrow 2NO\,(g)$, given that:
   - $N_2\,(g) + 2O_2\,(g) \rightarrow 2NO_2\,(g)$, $\Delta H = +66$ kJ
   - $2NO\,(g) + O_2\,(g) \rightarrow 2NO_2\,(g)$, $\Delta H = -114$ kJ

WEEK 11: ENERGETICS OF REACTIONS

Activation Energy and Energy Profile Diagrams



Energetics in Chemistry

- All chemical reactions involve **energy changes** because bonds in the reactants must be broken and new bonds in the products formed.
- **Bond breaking** requires energy (endothermic).
- **Bond formation** releases energy (exothermic).
- The **overall energy change (ΔH)** of a reaction is the balance between energy absorbed (to break bonds) and energy released (to form bonds).
- Reactions are classified into two categories:
    - **Exothermic Reactions (ΔH negative):** Heat energy is released to the surroundings, temperature of the system increases.
        - Examples: combustion of fuels, neutralization (acid + base → salt + water), respiration.

- **Endothermic Reactions (ΔH positive):** Heat energy is absorbed from the surroundings, temperature of the system decreases.
    - Examples: thermal decomposition of calcium carbonate, photosynthesis, dissolving ammonium chloride in water.

Activation Energy (E_a)

- **Definition:** The minimum amount of energy that colliding particles must possess to initiate a chemical reaction.
- Think of activation energy as an **energy hill** that reactants must climb before forming products.
- Even **exothermic reactions**, though energy-releasing, cannot start spontaneously without the reactants reaching their activation energy.
- **Collision Theory:**
    - For a reaction to occur, particles must **collide**.
    - Not all collisions are effective.
    - For effective collisions:
        1. Particles must collide with energy **equal to or greater than E_a**.
        2. Particles must collide in the **proper orientation**.

Key Notes:

- High E_a → slow reaction (e.g., rusting of iron).
- Low E_a → fast reaction (e.g., neutralization).
- Catalysts lower E_a, speeding up reaction.

Energy Profile Diagrams

Energy profile diagrams show how the energy of the system changes during the course of a reaction.

a) Exothermic Reaction Diagram

- Reactants start at a **higher energy level**.
- Curve rises to a peak (activation energy).
- Then drops steeply because products are at **lower energy**.
- Difference in height between reactants and products = **ΔH (negative)**.

Examples:

- $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + energy
- Neutralization: $HCl + NaOH \rightarrow NaCl + H_2O$

b) Endothermic Reaction Diagram

- Reactants start at a **lower energy level**.
- Curve rises to a peak (activation energy).
- Products are at a **higher energy level** than reactants.
- ΔH = positive (heat absorbed).

Examples:

- $CaCO_3 \rightarrow CaO + CO_2$
- Photosynthesis: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + O_2$ (requires sunlight energy)

c) Effect of a Catalyst on the Energy Profile

- Catalyst provides an **alternative pathway** with lower activation energy.
- The diagram shows a **lower peak** compared to uncatalyzed reaction.
- ΔH remains **unchanged**, only the pathway differs.

Activation Energy in Real Life

- **Striking a matchstick:** The friction provides enough energy to overcome the activation energy barrier, after which combustion becomes self-sustaining (exothermic).
- **Respiration:** Glucose oxidation requires activation energy, which enzymes (biological catalysts) lower.
- **Cooking food:** Heat energy is supplied to overcome activation energies for breaking bonds in raw food substances.
- **Industrial processes:** Catalysts are used to lower activation energies and save cost:
  - Haber process: Fe catalyst for NH_3 production.
  - Contact process: V_2O_5 catalyst for SO_3 production.
  - Hydrogenation of vegetable oils: Ni catalyst.

Worked Examples

Example 1:
In a reaction, the activation energy is 150 kJ, and the enthalpy change is –75 kJ.

- Sketch the energy profile diagram.
- State whether the reaction is exothermic or endothermic.

Answer:

- ΔH is negative, hence the reaction is **exothermic**.
- Diagram: Reactants at higher energy, curve rises 150 kJ above them (E_a), then drops to a level 75 kJ lower than reactants (products).

Example 2:
Why does hydrogen peroxide (H_2O_2) decompose slowly on its own but rapidly in the presence of manganese(IV) oxide (MnO_2)?

Answer:

Because the decomposition has a **high activation energy**. MnO_2 acts as a catalyst to provide an alternative pathway with lower activation energy, thus speeding up the reaction.

Misconceptions to Address

- **Misconception 1:** Exothermic reactions start without any activation energy.
  - **Correction:** All reactions need some activation energy to begin.
- **Misconception 2:** A catalyst changes the enthalpy (ΔH) of a reaction.
  - **Correction:** A catalyst only lowers activation energy, it does not affect ΔH.
- **Misconception 3:** Faster reaction always means more heat is released.
  - **Correction:** Speed of reaction depends on E_a, while heat released depends on bond energies (ΔH).

Evaluation

1. Define activation energy.
2. Differentiate between exothermic and endothermic reactions with examples.
3. Why must even exothermic reactions require activation energy?
4. Sketch and label the energy profile of:
   a) An exothermic reaction
   b) An endothermic reaction
5. Explain how catalysts affect activation energy, using energy diagrams.
6. State two industrial applications of catalysts in lowering activation energy.
7. Why does rusting of iron occur slowly despite being an exothermic reaction?

Assignment

1. A reaction has $\Delta H = +120$ kJ/mol and an activation energy of 200 kJ/mol. Draw and explain the energy profile diagram.
2. Discuss the role of enzymes as biological catalysts in lowering activation energy, using respiration as an example.
3. Compare the energy changes in photosynthesis and respiration, highlighting their differences in terms of activation energy and ΔH.
4. Research: Write short notes on how activation energy is relevant in:
   a) The explosives industry (e.g., TNT detonation)
   b) Fuel combustion in car engines

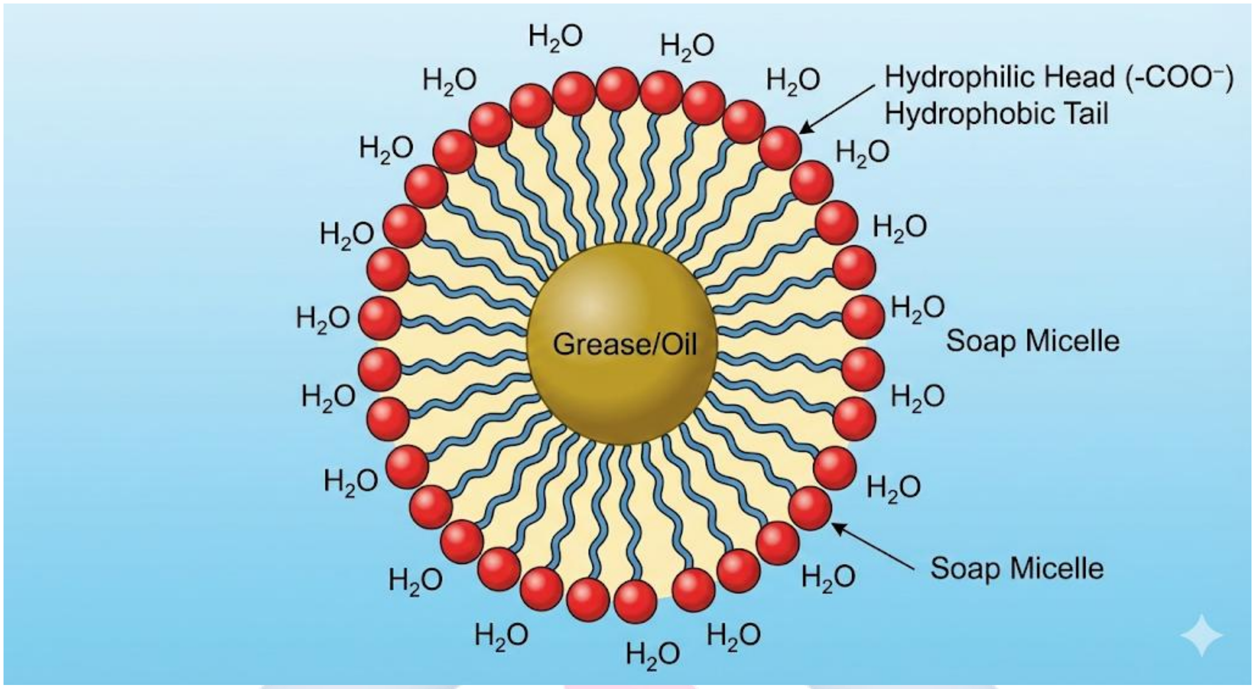
SS2 CHEMISTRY LESSON NOTES

SECOND TERM

1. Saponification and Detergents – Chemistry of soap making.
2. Introduction to Nuclear Chemistry – Radioactivity, applications.
3. Halogens – Properties, uses of fluorine, chlorine, etc.
4. Alkali and Alkaline Earth Metals – Characteristics and uses.
5. **Midterm Examination**
6. Transition Elements – Properties, complex compounds.
7. **Midterm Break**
8. General Properties of Metals – Reactivity, refining.
9. Non-Metals and Their Compounds – Hydrogen, oxygen, nitrogen.
10. Water – Hardness, purification, quality control.
11. Environmental Pollution – Causes, control, effects.
12. Examination
13. Closing

WEEK 1: SAPONIFICATION AND DETERGENTS

Chemistry of Soap Making



HISTORICAL BACKGROUND

- Soap has been used since **ancient Babylon (about 2800 BC)**, where people boiled fats with ashes to produce crude soap.
- Ancient Egyptians used a similar mixture for washing linen and treating skin diseases.
- By the 19th century, soap making became a **large-scale industry**, especially after the development of the chemical understanding of fats, oils, and alkalis.
- Detergents were first produced in the **20th century (1916, Germany)** during World War I, due to shortages of natural oils/fats for soap making.

NATURE OF FATS AND OILS

- Fats and oils are **triglycerides** – esters formed when glycerol ($C_3H_5(OH)_3$) combines with three fatty acids.
- **Fats** are solid at room temperature (rich in saturated fatty acids, e.g., tallow, butter).

- **Oils** are liquid at room temperature (rich in unsaturated fatty acids, e.g., palm oil, groundnut oil).
- Typical fatty acids: stearic acid ($C_{17}H_{35}COOH$), palmitic acid ($C_{15}H_{31}COOH$), oleic acid ($C_{17}H_{33}COOH$).

SAPONIFICATION

Definition

Saponification is the **hydrolysis of triglycerides (fats and oils)** with a strong alkali (NaOH or KOH), producing soap and glycerol.

Reaction Equation

$$C_3H_5(OOCR)_3 + 3NaOH \longrightarrow C_3H_5(OH)_3 + 3RCOONa$$

where:

- $C_3H_5(OOCR)_3$ = triglyceride (fat/oil)
- $C_3H_5(OH)_3$ = glycerol
- RCOONa = sodium salt of fatty acid (soap).

Laboratory Preparation of Soap

Materials: Palm oil, NaOH pellets, distilled water, ethanol, salt solution (NaCl).
Procedure:

1. Dissolve NaOH pellets in water to form caustic soda solution.
2. Mix palm oil with ethanol in a beaker.
3. Add the NaOH solution and heat gently for 30–40 minutes with stirring (avoid splashing).
4. A thick paste (soap) forms.
5. Add concentrated NaCl solution to precipitate the soap (salting out).

6. Filter and wash the soap with distilled water.
7. Dry and mold into desired shapes.

Observation: The soap separates from the mixture and solidifies, while glycerol remains dissolved in the liquid.

TYPES OF SOAP

1. **Hard Soap** – sodium salt of fatty acid; solid; prepared with NaOH; used for laundry.
2. **Soft Soap** – potassium salt of fatty acid; soft/semi-liquid; prepared with KOH; used in shampoos and shaving creams.
3. **Medicated Soap** – contains antiseptics like triclosan or sulfur; used for medical/dermatological purposes.
4. **Toilet Soap** – perfumed, often superfatted; mild on the skin.
5. **Transparent Soap** – made by adding excess glycerol and sugar to soap.

STRUCTURE AND CLEANING ACTION OF SOAP

Structure of Soap Molecule

- **Hydrophobic tail (R–)**: long hydrocarbon chain, repels water but dissolves in oil/grease.
- **Hydrophilic head (–COO^-Na^+)**: polar, soluble in water.

Cleaning Mechanism

- When soap is added to dirty fabric, hydrophobic tails penetrate greasy dirt.
- Hydrophilic heads remain in water.
- Soap molecules arrange themselves into **micelles** – spherical structures with grease trapped inside.
- Micelles remain suspended in water and are washed away during rinsing.

LIMITATIONS OF SOAP

- Ineffective in **hard water** (contains Ca^{2+}, Mg^{2+}, Fe^{2+}).
- Forms insoluble scum (calcium/magnesium stearates), reducing cleaning efficiency.
- Cannot be used effectively in acidic water, as H^+ ions liberate fatty acids from soap.

Equation:

$$2RCOONa + Ca^{2+} \longrightarrow (RCOO)_2Ca \downarrow + 2Na^+$$

DETERGENTS

Definition

Detergents are **synthetic cleaning agents** made from petroleum hydrocarbons. Unlike soaps, they do not form scum in hard water.

Structure

- **Hydrophobic tail**: hydrocarbon chain (similar to soap).
- **Hydrophilic head**: usually sulphonate ($–SO_3^-Na^+$), sulfate ($–OSO_3^-Na^+$), or ammonium ion.

Types of Detergents

1. **Anionic detergents**: hydrophilic head has negative charge (e.g., sodium alkylbenzene sulphonate).
2. **Cationic detergents**: hydrophilic head has positive charge (e.g., quaternary ammonium salts; used in hair conditioners).
3. **Non-ionic detergents**: contain polar groups (e.g., ethoxylated alcohols) but no ionic charge.

ADVANTAGES OF DETERGENTS OVER SOAPS

1. Work well in both soft and hard water.
2. Do not form scum with calcium or magnesium ions.
3. Can be formulated for special uses (industrial cleaners, shampoos, dishwashers).
4. Maintain effectiveness in acidic or neutral solutions.

ENVIRONMENTAL CONCERNS

- **Soaps**: Fully biodegradable; environmentally friendly.
- **Detergents**: Early detergents contained **branched hydrocarbons**, making them non-biodegradable → caused foaming in rivers and lakes, blocked oxygen exchange, and harmed aquatic life.
- **Modern detergents**: Made from **linear hydrocarbons**, which are biodegradable.

USES OF SOAP AND DETERGENTS

- **Soap**: Bathing, laundry, textile industries, emulsifying oils, shaving creams.
- **Detergents**: Laundry powders, dishwashing liquids, shampoos, industrial degreasers, emulsifiers in food and pharmaceuticals.

EVALUATION

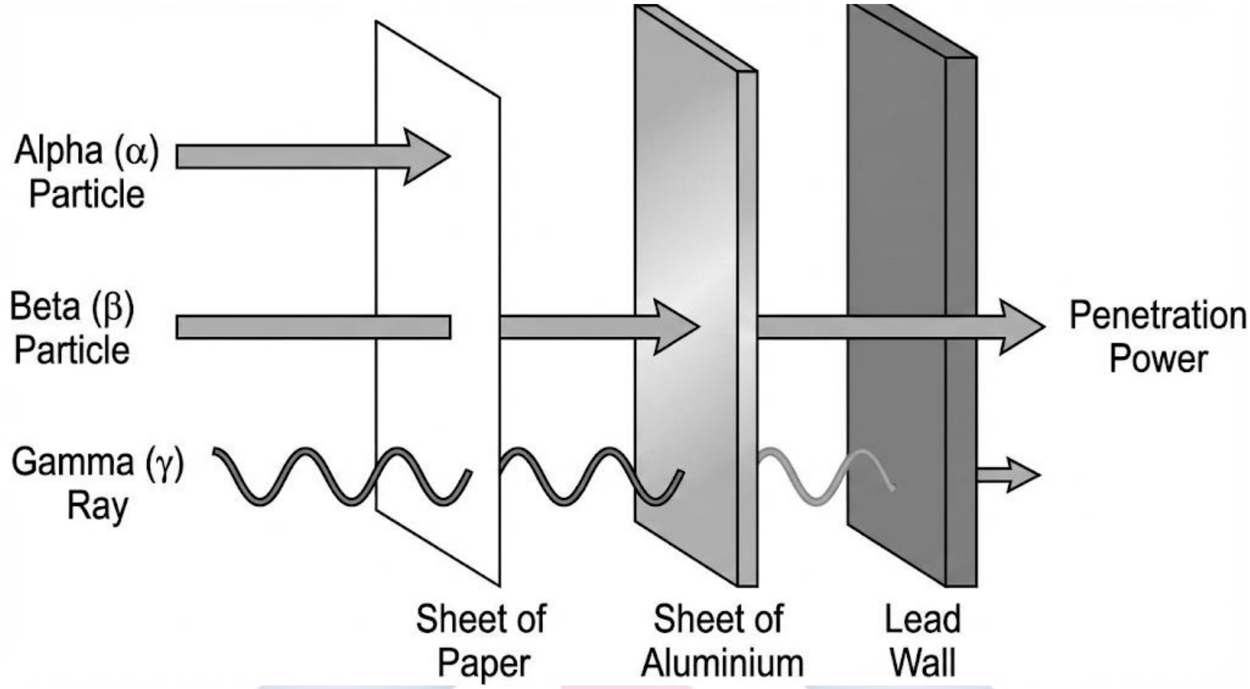
1. Define saponification and write a balanced equation for the process.
2. Explain the structural difference between soap and detergent molecules.
3. Describe the cleaning action of soap using the concept of micelles.
4. Why do soaps form scum in hard water but detergents do not?
5. Differentiate between hard soap and soft soap, giving one use of each.

ASSIGNMENT

1. Design a step-by-step **laboratory procedure** for the preparation of soap from palm oil, including the reagents, equipment, and observations.
2. With a diagram, explain the **micelle formation** process during cleaning with soap.
3. Compare and contrast the **environmental impact** of soaps and detergents, highlighting which is more eco-friendly and why.

WEEK 2: INTRODUCTION TO NUCLEAR CHEMISTRY

Radioactivity and Applications



Meaning of Nuclear Chemistry

- **Nuclear Chemistry** is the branch of chemistry concerned with the **structure of the nucleus of atoms** and the changes it undergoes.
- While **ordinary chemistry** focuses on chemical bonds and electron transfer/sharing, **nuclear chemistry** deals with processes that occur within the nucleus itself.
- These processes are accompanied by enormous energy changes, far greater than those in chemical reactions.

Nuclear vs Chemical Reactions

Feature	**Chemical Reaction**	**Nuclear Reaction**
Part involved	Electrons (outer shell)	Nucleus (protons & neutrons)
Energy change	Small (kJ/mol)	Very large (10^6 times more)
Effect of conditions	Affected by temperature, pressure	Not affected by ordinary conditions

Products	Same elements, new compounds	New elements/isotopes formed

Historical Background

- **1896 – Henri Becquerel** discovered radioactivity while working with uranium salts.
- **1898 – Marie and Pierre Curie** isolated radium and polonium, advancing research on radioactive elements.
- **Ernest Rutherford** later studied the nature of radioactive emissions and identified **α and β radiation**.
- **James Chadwick (1932)** discovered the neutron, which further explained nuclear structure and nuclear reactions.

Definition of Radioactivity

- **Radioactivity** is the **spontaneous emission** of particles (α, β) or electromagnetic radiation (γ) from the nucleus of an unstable atom until it transforms into a more stable form.
- The unstable atoms are called **radioisotopes** (radioactive isotopes).

Examples of Radioisotopes:

- Uranium-238, Thorium-232, Carbon-14, Radium-226, Iodine-131.

Types of Radioactive Emissions

a) Alpha (α) Particles

- **Nature/Structure:**
  An alpha particle consists of **2 protons and 2 neutrons**, making it identical to the nucleus of a helium atom $(^4_2He^{2+})$.
- **Charge and Mass:**

- Charge: **+2** (because of the two protons).
- Relative Mass: **4 atomic mass units (amu)**.

- **Speed:**
  They move at about **one-tenth the speed of light**, which is relatively slow compared to other radiations.
- **Penetrating Power:**
  Very low – easily stopped by a **sheet of paper, skin, or even a few centimeters of air**.
- **Ionizing Power:**
  Extremely high – because of their large mass and double positive charge, alpha particles can cause **intense ionization**, leading to serious damage if they come in direct contact with living tissues.
- **Example of Alpha Decay:**

  Uranium-238 undergoes alpha decay to form Thorium-234:

  $$^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He$$

b) Beta (β) Particles

- Structure: High-speed electron (β^-) or positron (β^+).
- Charge: –1 or +1.
- Mass: Negligible.
- Speed: Close to the speed of light.
- Penetrating power: Moderate (stopped by aluminium foil).
- Ionizing power: Medium.
- Example:

  $$^{14}_{6}C \longrightarrow {}^{14}_{7}N + \beta^-$$

c) Gamma (γ) Rays

- Structure: High-frequency electromagnetic waves.
- Charge: 0, Mass: 0.
- Speed: Speed of light.
- Penetrating power: Very high (needs thick lead/concrete to block).
- Ionizing power: Low.
- Usually emitted with α or β radiation.

Characteristics of Radioactive Decay

1. **Spontaneous** – cannot be influenced by heat, pressure, or catalysts.
2. Produces **new atoms/elements** (daughter nuclei).
3. Accompanied by **release of large energy**.
4. Follows a **definite rate**, independent of conditions.
5. Measured by **half-life (T½)** – the time taken for half the nuclei in a sample to decay.

Applications of Radioactivity

a) Medicine

- **Cancer treatment (Radiotherapy):** Cobalt-60 used to destroy cancerous tissues.
- **Diagnosis (Imaging):** Technetium-99m used as a tracer.
- **Treatment of goiter:** Iodine-131 concentrates in the thyroid gland and destroys abnormal cells.

b) Agriculture

- Radioisotopes used to trace absorption of fertilizers by plants.

- Radiation used to sterilize pests and prolong storage of foods (food irradiation).

c) Industry

- **Thickness gauge:** Beta radiation used to control the thickness of paper, plastic, and aluminium sheets.
- **Leak detection:** Radioactive tracers added to liquids or gases help detect leaks in pipelines.
- **Radiography:** Gamma rays used to detect cracks in metals and machine parts.

d) Archaeology & Geology

- **Carbon-14 dating:** Used to estimate the age of fossils, ancient bones, and artifacts up to 50,000 years.
- **Uranium-lead dating:** Used for dating rocks and minerals, often millions of years old.

e) Energy Production

- **Nuclear fission:** Uranium-235 and Plutonium-239 used as fuels in nuclear reactors to generate electricity.
- Nuclear energy is much more efficient than fossil fuels.

Dangers of Radioactivity

- Can cause **cancer, mutations, and genetic damage**.
- **Radiation burns** and sickness in cases of high exposure.
- **Environmental hazard** due to difficulty of disposing nuclear waste.
- **Nuclear accidents** (e.g., Chernobyl 1986, Fukushima 2011) highlight the risks of misuse.
- Potential use in **nuclear weapons**, leading to mass destruction.

Worked Examples

Example 1:

Differentiate between α, β, and γ radiation in terms of charge, mass, and penetrating power.

Answer:

- α: +2 charge, heavy mass, low penetration.
- β: –1 or +1 charge, negligible mass, moderate penetration.
- γ: Neutral, no mass, very high penetration.

Example 2:

Why is carbon-14 suitable for dating fossils?

Answer:

Because it is present in all living organisms, has a long half-life (5730 years), and begins to decay after death, allowing the age of remains to be determined.

Evaluation

1. Define nuclear chemistry and radioactivity.
2. State three differences between chemical and nuclear reactions.
3. List and describe the three types of radioactive emissions.
4. What is meant by the half-life of a radioactive element?
5. Mention four useful applications of radioactivity.
6. State three dangers of radioactivity.
7. Give one balanced nuclear equation for an alpha decay and one for beta decay.

Assignment

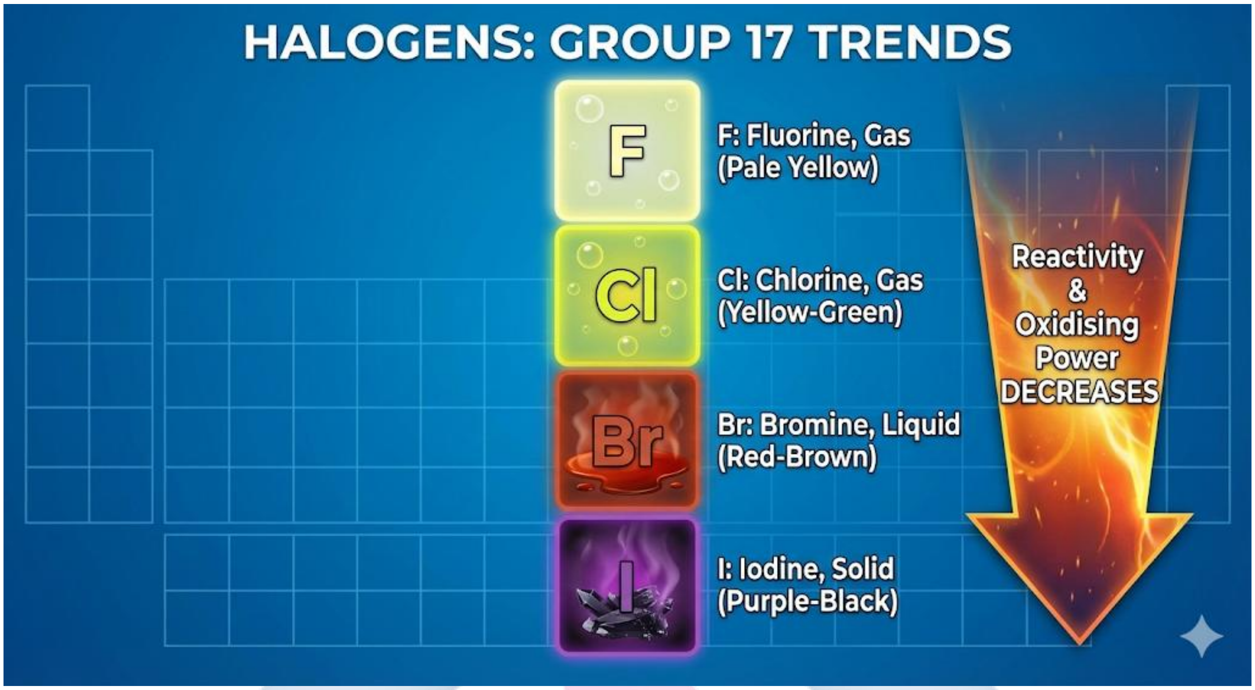
1. Complete the following nuclear equation:

$$^{210}_{84}Po \longrightarrow ? + ^{4}_{2}He$$

2. Research: Write a short note on **how nuclear power plants generate electricity**.
3. List two radioisotopes used in medicine and state their uses.
4. State two reasons why gamma rays are dangerous compared to alpha and beta particles.
5. Discuss two advantages and two disadvantages of using nuclear energy in society.

WEEK 3: HALOGENS

Properties and Uses of Fluorine, Chlorine, Bromine, and Iodine



INTRODUCTION TO HALOGENS

- The **halogens** are non-metals in **Group VIIA (Group 17)** of the Periodic Table.
- Members: **Fluorine (F), Chlorine (Cl), Bromine (Br), Iodine (I), and Astatine (At)**.
- The name *halogen* means "**salt former**" because they readily combine with metals to form salts such as NaCl (common salt).
- They all have **7 valence electrons** (outer electronic configuration ns^2np^5), which explains:
  - Their **high reactivity**.
  - Their tendency to **gain one electron** to form stable halide ions (X^-).
  - Their strong **oxidizing ability**.
- They exist naturally as **diatomic molecules (X_2)** and are rarely found free due to high reactivity.

GENERAL PROPERTIES OF HALOGENS

(A) Physical Properties

1. Exist as **diatomic molecules**: F_2, Cl_2, Br_2, I_2.
2. **Gradual change in physical state** down the group:
   - Fluorine → pale yellow gas.
   - Chlorine → greenish-yellow gas.
   - Bromine → reddish-brown liquid.
   - Iodine → grey crystalline solid.
3. **Color intensity increases** down the group (yellow → greenish → reddish-brown → purple-black).
4. **Melting and boiling points increase** down the group due to stronger intermolecular forces.
5. **Density increases** down the group.
6. Poor conductors of heat and electricity.
7. Solubility: Slightly soluble in water, more soluble in organic solvents (e.g., CCl_4, hexane) where they show distinct colors.

(B) Chemical Properties

1. **Reactivity:**
   - Reactivity decreases down the group: $F_2 > Cl_2 > Br_2 > I_2$.
   - Reason: atomic radius increases, shielding effect increases, electronegativity decreases.
2. **Oxidizing Power:**
   - Strong oxidizers, with ability decreasing down the group.
   - Example of displacement reactions:

$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$

Chlorine displaces bromine.

3. **Reaction with Hydrogen:**
   - Form hydrogen halides (HX), which are colorless acidic gases.
   - Example:

$H_2 + Cl_2 \rightarrow 2HCl$

(explosive in sunlight).

4. **Reaction with Metals:**
   - Form metal halides, usually ionic compounds.
   - Example:

$2Na + Cl_2 \rightarrow 2NaCl$

5. **Reaction with Water:**
   - Chlorine dissolves in water forming hydrochloric acid (HCl) and hypochlorous acid (HClO):

$Cl_2 + H_2O \rightleftharpoons HCl + HClO$

- HClO is responsible for chlorine's bleaching and disinfecting action.

6. **Bleaching Property:**
   - Chlorine and bromine act as bleaching agents due to liberation of oxygen from hypohalous acid:

$HClO \rightarrow HCl + [O]$

The nascent oxygen bleaches organic coloring matter.

DETAILED STUDY OF EACH HALOGEN

(A) Fluorine (F_2)

- **Physical Properties:**
  - Pale yellow gas, pungent smell, very toxic.
  - Most electronegative element.
- **Occurrence:** As fluorspar (CaF_2), cryolite (Na_3AlF_6), fluorapatite.
- **Chemical Properties:**
  - Very reactive, even reacts with noble gases under special conditions.
  - Reacts explosively with hydrogen even in the dark:

$H_2 + F_2 \rightarrow 2HF$

- **Uses:**

1. Manufacture of fluorocarbons like Teflon (PTFE) – non-stick coating.
2. Production of UF_6 in nuclear fuel enrichment.
3. Fluorides added to water and toothpaste to prevent tooth decay.

(B) Chlorine (Cl_2)

- **Physical Properties:** Greenish-yellow gas, choking smell, denser than air.
- **Occurrence:** Mainly in nature as sodium chloride (NaCl).
- **Chemical Properties:**
  - Reacts with hydrogen in sunlight explosively.
  - Dissolves in water giving HCl and HClO.
  - Displaces bromine and iodine from salts.
- **Uses:**

1. Sterilization of drinking water and swimming pools.

2. Manufacture of HCl, PVC, pesticides, and antiseptics.

3. Used as a bleaching agent in textile and paper industries.

4. Production of chloroform and other chlorinated hydrocarbons.

(C) Bromine (Br_2)

- **Physical Properties:** Reddish-brown volatile liquid, gives off suffocating fumes.
- **Occurrence:** Present in sea water as bromide salts (NaBr, KBr).
- **Chemical Properties:**
  - Reacts with hydrogen less vigorously:

$H_2 + Br_2 \rightarrow 2HBr$

  - Displaced from its salts by chlorine.
- **Uses:**

1. In production of flame retardants.

2. Manufacture of dyes and certain drugs.

3. Silver bromide (AgBr) in photography.

4. Water treatment (alternative to chlorine).

(D) Iodine (I_2)

- **Physical Properties:** Grey-black crystalline solid; sublimes to purple vapor.
- **Occurrence:** Found in seaweeds and iodine-containing salts.
- **Chemical Properties:**
  - Least reactive of common halogens.
  - Forms hydrogen iodide:

$H_2 + I_2 \rightarrow 2HI$

- HI is unstable and decomposes easily.

- **Uses:**

1. Tincture of iodine as antiseptic.
2. Iodized salt to prevent goiter.
3. Manufacture of dyes and pharmaceuticals.
4. Iodine solutions used in starch test (blue-black coloration).
5. Analytical chemistry (iodometry).

TRENDS IN PROPERTIES DOWN THE GROUP

1. **Atomic radius:** Increases down the group (more shells).
2. **Ionization energy:** Decreases down the group.
3. **Electronegativity:** Decreases down the group.
4. **Reactivity:** Decreases down the group.
5. **Melting & boiling points:** Increase down the group (stronger van der Waals forces).
6. **Oxidizing power:** Decreases from fluorine → iodine.
7. **Reducing power of halide ions (X^-):** Increases down the group (I^- is strongest reducing agent).

ENVIRONMENTAL AND HEALTH EFFECTS

- **Fluorine:**
  - Excess fluoride in water causes fluorosis (brown teeth, bone damage).
  - Controlled addition is beneficial to dental health.

- **Chlorine:**
  - Highly toxic; exposure can damage lungs.
  - Used as chemical weapon in World War I.
  - Controlled use disinfects water effectively.
- **Bromine:**
  - Vapors are irritating to eyes and skin.
  - Some brominated compounds are toxic and persistent pollutants.
- **Iodine:**
  - Essential trace element for thyroid hormone production.
  - Deficiency causes goiter; excess causes hyperthyroidism.

INDUSTRIAL AND DAILY IMPORTANCE OF HALOGENS

1. **Medicine:**
   - Iodine as antiseptic.
   - Halogenated drugs and anesthetics.
2. **Food and Nutrition:**
   - Fluorides in toothpaste.
   - Iodized salt.
3. **Water Treatment:**
   - Chlorine widely used.
4. **Plastics and Polymers:**
   - PVC (chlorine-based).
   - Teflon (fluorine-based).

5. **Photography:**
    - Silver bromide and silver iodide.
6. **Dyes and Bleaches:**
    - Chlorine and bromine for bleaching textiles and making organic dyes.

EXPERIMENTAL DEMONSTRATIONS

- **Displacement Reactions:**
    - Add chlorine water to potassium iodide solution: purple iodine appears.
    - Add chlorine water to potassium bromide solution: reddish-brown bromine appears.
- **Starch Test for Iodine:**
    - Iodine solution turns blue-black in starch solution.
- **Bleaching Action of Chlorine:**
    - A damp litmus paper decolorized when exposed to chlorine gas.

EVALUATION

1. Define halogens and state why they are called "salt formers."
2. List the physical states and colors of fluorine, chlorine, bromine, and iodine at room temperature.
3. Explain why reactivity decreases down the group.
4. Write balanced chemical equations for the following:
   a. Reaction of chlorine with hydrogen.
   b. Reaction of chlorine with sodium.
   c. Displacement of bromine by chlorine from KBr solution.

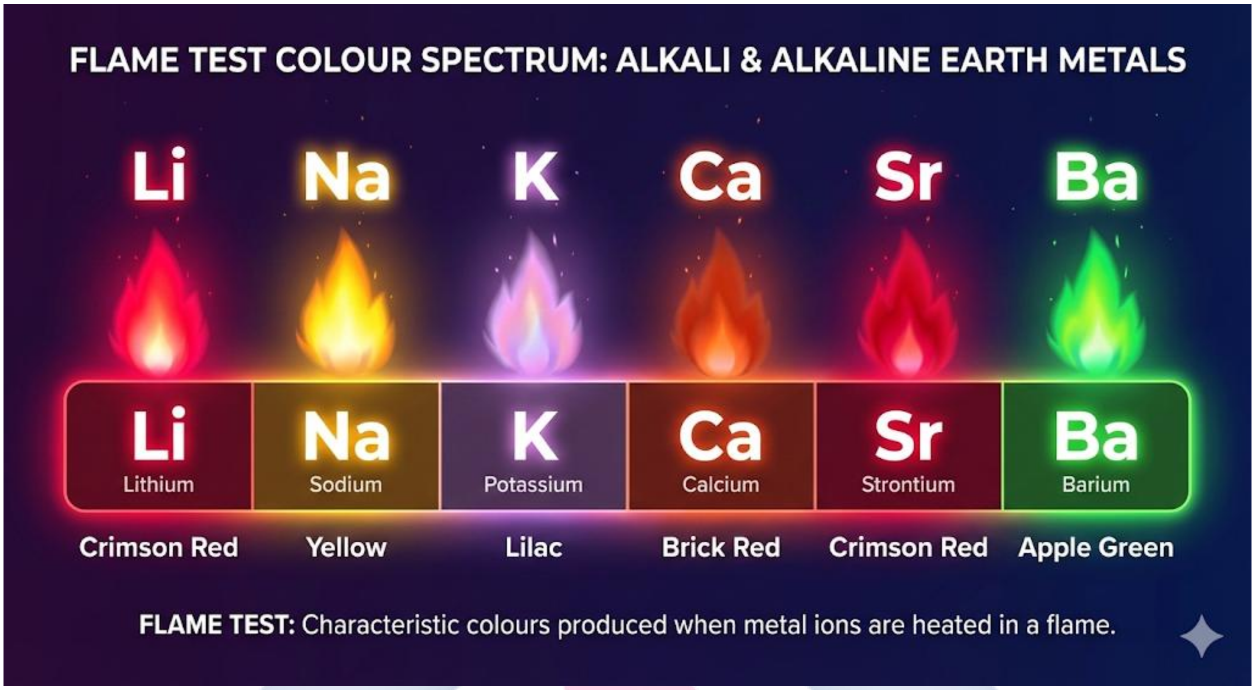
5. State four industrial uses each of chlorine and iodine.
6. Explain why fluorine is the strongest oxidizing agent among halogens.
7. Differentiate between oxidizing power of halogens and reducing power of halide ions.

ASSIGNMENT

1. Draw a comparative table showing the **appearance, physical state, common compounds, and uses** of fluorine, chlorine, bromine, and iodine.
2. Explain, with examples, why chlorine is used in water treatment despite being poisonous.
3. Discuss the environmental and health implications of **excessive exposure to halogens** in daily life.

WEEK 4: ALKALI AND ALKALINE EARTH METALS

Characteristics and Uses



Introduction to Alkali and Alkaline Earth Metals

- The **alkali metals** (Group 1 of the periodic table) and **alkaline earth metals** (Group 2) are **highly reactive metallic elements**.
- These metals are called **alkali metals** because their hydroxides are strongly **basic (alkaline)**, and **alkaline earth metals** because their oxides and hydroxides are also alkaline but less soluble in water, and they are found in the earth's crust.

Alkali Metals (Group 1)

- **Members:** Lithium (Li), Sodium (Na), Potassium (K), Rubidium (Rb), Cesium (Cs), and Francium (Fr).
- They are soft, silvery-white metals that react vigorously with water to form alkaline hydroxides and hydrogen gas.

Key General Properties:

Property	Description
Electronic Configuration	ns^1 (one electron in the outermost shell).
Metallic Character	Increases down the group due to increasing atomic size and decreasing ionization energy.
Softness	Soft metals that can be cut with a knife; softness increases down the group.
Melting/Boiling Point	Low and decreases down the group.
Density	Low; lithium, sodium, potassium are less dense than water.
Reactivity	Very reactive; increases down the group because the outer electron is easily lost.
Flame Colors	Lithium (crimson red), Sodium (yellow), Potassium (lilac), Rubidium (red-violet), Cesium (blue-violet).

Chemical Properties of Alkali Metals:

1. **Reaction with Air/Oxygen:**
   - Forms oxides, peroxides, or superoxides depending on size.
     Example: $4Na + O_2 \rightarrow 2Na_2O$.
2. **Reaction with Water:**
   - Reacts vigorously to form hydroxides and hydrogen gas.
     Example: $2Na + 2H_2O \rightarrow 2NaOH + H_2$.
3. **Reaction with Halogens:**
   - Forms ionic halides.
     Example: $2K + Cl_2 \rightarrow 2KCl$.
4. **Reaction with Acids:**
   - Highly exothermic; forms salts and hydrogen gas.

Uses of Alkali Metals:

- Sodium vapor lamps (street lighting).
- Lithium in rechargeable batteries.
- Potassium compounds (fertilizers, soap making).
- Sodium as a coolant in nuclear reactors.
- Sodium hydroxide in soap and detergent manufacturing.

Alkaline Earth Metals (Group 2)

- **Members:** Beryllium (Be), Magnesium (Mg), Calcium (Ca), Strontium (Sr), Barium (Ba), Radium (Ra).
- They are harder and denser than alkali metals but less reactive.

Key General Properties:

Property	**Description**
Electronic Configuration	ns^2 (two electrons in outer shell).
Metallic Character	Increases down the group.
Melting/Boiling Point	Higher than alkali metals, decreases slightly down the group.
Density	Higher than alkali metals.
Reactivity	Less reactive than alkali metals; reactivity increases down the group.
Flame Colors	Calcium (brick red), Strontium (crimson red), Barium (apple green).

Chemical Properties of Alkaline Earth Metals:

1. **Reaction with Oxygen:**

- Forms basic oxides.
  Example: $2Ca + O_2 \rightarrow 2CaO$.

2. **Reaction with Water:**
   - Reactivity increases down the group; calcium reacts slowly with cold water.
     Example: $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$.

3. **Reaction with Acids:**
   - Forms salts and releases hydrogen gas.
     Example: $Mg + 2HCl \rightarrow MgCl_2 + H_2$.

4. **Reaction with Halogens:**
   - Forms ionic halides.
     Example: $Sr + Cl_2 \rightarrow SrCl_2$.

Uses of Alkaline Earth Metals:

- **Magnesium:** Alloys (aircraft construction), fireworks, medicine (antacids).
- **Calcium:** Cement, plaster of Paris, building materials, fertilizers.
- **Strontium:** Fireworks (red color).
- **Barium:** X-ray shielding, paints, and ceramics.
- **Radium:** Formerly used in luminous paints, now for cancer treatment (radioactive).

Trends in the Two Groups

Property	**Group 1 (Alkali Metals)**	**Group 2 (Alkaline Earth Metals)**
Electronic configuration	ns^1	ns^2
Reactivity	Very high, increases down group	Moderate, increases down group
Hardness	Soft metals	Harder than alkali metals

Melting point	Very low	Relatively higher
Ionization energy	Lower	Higher
Flame colors	Distinctive colors	Distinctive colors
Compounds	Mostly soluble	Less soluble

Evaluation:

1. List **four general properties** of alkali metals and four of alkaline earth metals.
2. Explain why alkali metals are more reactive than alkaline earth metals.
3. Write balanced equations for:
   a) Sodium reacting with water.
   b) Magnesium reacting with hydrochloric acid.
   c) Calcium reacting with oxygen.
4. State **three uses** each of magnesium, calcium, and sodium.
5. Explain the trend of reactivity in:
   a) Group 1 metals
   b) Group 2 metals

Assignment:

1. Complete this table:

Element	Group	Electronic Configuration	Flame Color	One Major Use
Lithium				
Potassium				
Calcium				

Strontium				
Barium				

2. Research and describe the **biological roles** of magnesium and calcium in the human body.

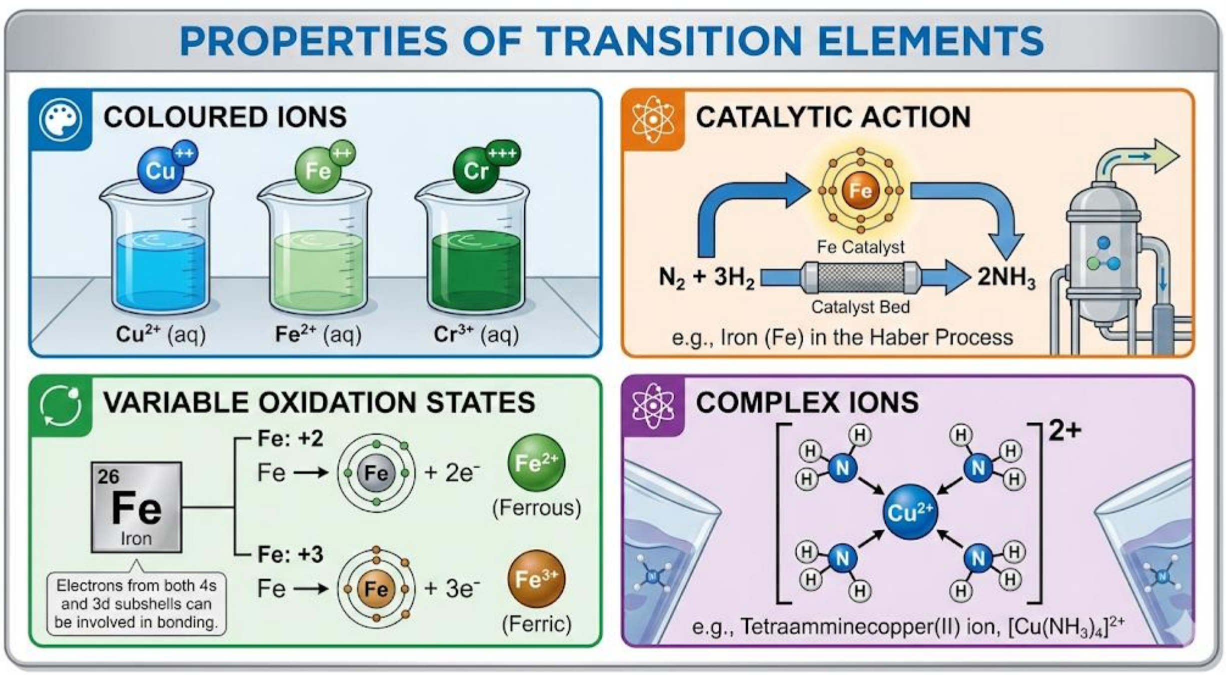
3. Explain why sodium and potassium are stored under oil.

4. Compare the solubility of hydroxides and sulfates of Group 1 and Group 2 metals with examples.

WEEK 6: TRANSITION ELEMENTS
Properties, complex compounds

Introduction to Transition Elements

- Transition elements are **d-block elements** found in Groups **3–12** of the Periodic Table.
- Defined as: *elements that form at least one stable ion with an incompletely filled d-orbital.*
- Examples: Ti, V, Cr, Mn, Fe, Co, Ni, Cu.
- Note: Zn is not a typical transition element because Zn^{2+} = [Ar] $3d^{10}$ (completely filled d-orbital).



Position in the Periodic Table & Electronic Configurations

- Located between **s-block** and **p-block** elements.
- General electronic configuration:

$$(n-1)d^{1-10}\, ns^{0-2}$$

Examples:

- Iron (Fe): [Ar] $3d^6\ 4s^2$

- Copper (Cu): [Ar] $3d^{10}$ $4s^1$
- Chromium (Cr): [Ar] $3d^5$ $4s^1$

General Properties of Transition Elements

Physical Properties

- High melting/boiling points (except Zn, Hg).
- Hard and dense metals.
- Good conductors of electricity and heat.
- Metallic lustre.
- Magnetic properties (Fe, Co, Ni are ferromagnetic).

Chemical Properties

- Show variable oxidation states.
- Form coloured ions and compounds.
- Form stable complex compounds.
- Act as catalysts.

Variable Oxidation States

- Transition metals can lose both s- and d-electrons → multiple oxidation states.

Examples in a Table:

Element	Oxidation States	Examples of Compounds
Fe	+2, +3	$FeCl_2$ (green), $FeCl_3$ (yellow-brown)
Mn	+2, +4, +6, +7	$MnCl_2$ (pink), MnO_2 (brown), $KMnO_4$ (purple)

Cr	+2, +3, +6	$CrCl_2$ (blue), $CrCl_3$ (green), $K_2Cr_2O_7$ (orange)
Cu	+1, +2	CuCl (white), $CuSO_4$ (blue)

Reason: Energy difference between (n-1)d and ns orbitals is small → easy electron loss.

Coloured Compounds

- Colours arise from **d–d transitions**: when light is absorbed, electrons jump between split d-orbitals.
- Different ligands and oxidation states cause different colours.

Examples:

- Fe^{2+} (aq): pale green
- Fe^{3+} (aq): yellow/brown
- Cu^{2+} (aq): blue
- $Cr_2O_7^{2-}$: orange
- MnO_4^-: purple

Catalytic Properties

Transition metals/compounds act as **heterogeneous or homogeneous catalysts**.

Examples:

- Fe in Haber process ($N_2 + 3H_2 \rightleftharpoons 2NH_3$)
- V_2O_5 in Contact process ($SO_2 + O_2 \rightleftharpoons SO_3$)
- Ni in hydrogenation of oils
- MnO_2 in $KClO_3$ decomposition ($2KClO_3 \rightarrow 2KCl + 3O_2$)

Reason:

- They provide alternative pathways with lower activation energy.
- Easily change oxidation states → help electron transfer.

Formation of Complex Compounds

- **Complex ion:** A central metal ion surrounded by ligands, held by coordinate covalent bonds.

Key Definitions:

- **Ligand:** Molecule/ion that donates a lone pair (H_2O, NH_3, Cl^-, CN^-).
- **Coordination number:** Number of ligand donor atoms attached to central ion.
- **Chelates:** Complexes with ligands that form ring structures (e.g., EDTA complexes).

Common Complex Examples:

- $[Cu(NH_3)_4]^{2+}$ → deep blue solution
- $[Fe(CN)_6]^{3-}$ → yellow complex
- $[Ag(NH_3)_2]^+$ → colourless (Tollens' reagent)

d-Orbital Splitting and Colour in Complexes

- In a free atom, all d-orbitals are degenerate (same energy).
- In complexes, ligands approach and cause **splitting of d-orbital energies**.
- **Octahedral complexes:** 6 ligands (e.g., $[Fe(H_2O)_6]^{2+}$).
- **Tetrahedral complexes:** 4 ligands (e.g., $[CuCl_4]^{2-}$).
- **Square planar complexes:** 4 ligands in a plane (e.g., $[Ni(CN)_4]^{2-}$).

Effect:

- Electrons absorb light energy equal to the energy gap.

- The complementary colour of absorbed light is seen.

Industrial, Biological and Environmental Applications

Industrial:

- Catalysts (Fe, Ni, V_2O_5).
- Electroplating (Ni, Cr) → rust prevention, decorative coatings.
- Pigments and dyes (chromates, permanganates).

Biological:

- Hemoglobin (Fe^{2+}): carries O_2.
- Vitamin B12 (Co): essential in metabolism.
- Cytochromes (Fe, Cu): electron transport in respiration.

Environmental:

- Transition metals in wastewater treatment (removal of toxins).
- Toxicity at high levels (Cr(VI), Cd, Hg are pollutants).

Evaluation

1. Define transition metals. Why is Zn not a typical transition metal?
2. Write electronic configurations for Fe, Cu, Cr, Zn. Identify which are typical transition metals.
3. List **three physical** and **three chemical** properties of transition elements.
4. State two reasons why transition metals act as good catalysts.
5. Give oxidation states and colours of:
   a. Fe in $FeCl_2$ and $FeCl_3$
   b. Mn in $KMnO_4$

6. Define ligand, coordination number, and give two examples of complexes.
7. Explain why transition metals form coloured compounds.
8. State one industrial, one biological, and one environmental importance of transition elements.

Assignment

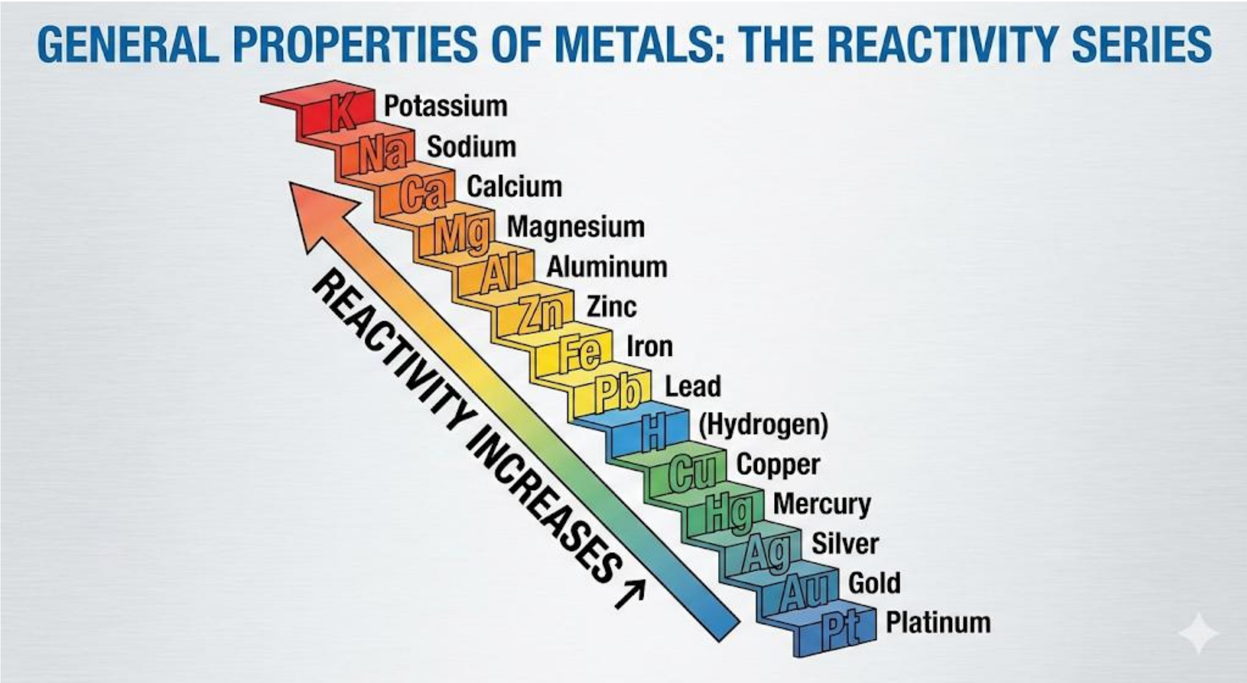
1. Complete the table below:

Element	Common Oxidation States	Colour of Compound/Ion	Example
Mn			
Cr			
Fe			
Cu			

2. Draw and describe (in words) the shapes of:
   a. Octahedral complex
   b. Tetrahedral complex
   c. Square planar complex
3. Research Task:
   a. Why do transition metals show magnetic properties?
   b. Discuss the importance of transition metals in the manufacture of stainless steel.
   c. Write a short note on the environmental hazard of Cr(VI) compounds.

WEEK 8: GENERAL PROPERTIES OF METALS

Reactivity and Refining



Metals are a group of elements that are widely used in our everyday lives because of their diverse physical and chemical properties. They constitute about three-quarters of all known elements in the periodic table and are essential in construction, transport, electricity, jewelry, communication, and many other aspects of modern living.

Their **general properties** can be considered under two main areas:

1. **Reactivity of Metals** (how easily they react with oxygen, water, acids, and salts, and how they are arranged in the reactivity series).
2. **Refining of Metals** (how pure metals are obtained after extraction from ores).

Reactivity of Metals

The Reactivity Series

Metals do not all react with the same vigor. Some, like potassium and sodium, are extremely reactive, while others, such as gold and platinum, are chemically inert. To compare their behaviors, chemists arrange them in a list called the **Reactivity Series of Metals**.

Order of the Reactivity Series:
K, Na, Ca, Mg, Al, Zn, Fe, Sn, Pb, H, Cu, Hg, Ag, Au, Pt.

- **Top of the series (K, Na, Ca, Mg, Al):** Very reactive, react readily with oxygen, water, and acids.
- **Middle (Zn, Fe, Sn, Pb):** Moderately reactive. They need stronger conditions (heating, steam, concentrated acids) to react.
- **Bottom (Cu, Hg, Ag, Au, Pt):** Least reactive, sometimes called **noble metals**. They resist corrosion and are found free in nature.

Explanation of Reactivity

- The reactivity of metals depends on how easily they lose electrons to form positive ions (cations).
- Metals with low ionization energies (like potassium and sodium) lose electrons easily and are highly reactive.
- Metals like gold and platinum have very high ionization energies and do not easily form ions, making them unreactive.

Reactions Showing Reactivity

1. **Reaction with Oxygen (Air):**
    - Potassium and sodium burn vigorously with air, producing bright flames.

      $$4Na + O_2 \rightarrow 2Na_2O$$

    - Magnesium burns with a dazzling white flame to form magnesium oxide.

      $$2Mg + O_2 \rightarrow 2MgO$$

- Aluminium forms an oxide coating (Al_2O_3) that protects it from further attack.
- Iron reacts slowly with oxygen and water to form **rust (hydrated iron(III) oxide)**.
- Copper only tarnishes after a long time; silver forms a black coating of Ag_2S.
- Gold and platinum do not react with oxygen at all.

2. **Reaction with Water:**
    - Potassium and sodium react explosively with cold water, producing hydrogen gas and a strong alkali (NaOH, KOH).

$$2Na + 2H_2O \rightarrow 2NaOH + H_2$$

    - Calcium reacts less violently, producing $Ca(OH)_2$ and H_2 gas.
    - Magnesium reacts slowly with hot water, while zinc and iron only react with **steam** to form oxides and hydrogen.

$$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$$

    - Copper, silver, and gold do not react with water.

3. **Reaction with Acids:**
    - Metals above hydrogen in the series (like Mg, Zn, Fe) displace hydrogen gas from dilute acids.

$$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$$

    - Metals below hydrogen (Cu, Ag, Au) do not react with dilute acids.

4. **Reaction with Other Metal Compounds (Displacement Reactions):**
   - A more reactive metal displaces a less reactive metal from its salt solution.
   - Example:

$$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$$

   - Iron nails placed in copper(II) sulphate solution get coated with a brown deposit of copper.

Applications of Reactivity

- Predicting which metals can be used to extract others from their compounds.
- Explaining corrosion resistance (why gold is used for jewelry).
- Designing sacrificial protection (zinc protects iron in galvanization).
- Industrial processes (Haber process, production of metals, etc.).

Refining of Metals

When metals are extracted from ores, they are not usually pure. They contain unwanted materials called **impurities** (such as sand, clay, or other metals). Refining is necessary to improve their strength, conductivity, and usefulness.

Methods of Refining Metals

1. **Electrolytic Refining (Most Common):**
   - Used for copper, silver, gold, and aluminium.
   - **Setup:**
     - Anode = impure metal.
     - Cathode = pure sheet of same metal.

- Electrolyte = solution of a salt of the metal.

- On passing electric current:
  - At the anode, the impure metal dissolves into ions.
  - At the cathode, pure metal is deposited.
  - Insoluble impurities settle as *anode mud*.

Example – Copper Refining:

- Anode: Impure copper rod
- Cathode: Pure copper plate
- Electrolyte: $CuSO_4$ solution + H_2SO_4
- Reactions:
  - Anode: $Cu \rightarrow Cu^{2+} + 2e^-$
  - Cathode: $Cu^{2+} + 2e^- \rightarrow Cu$

2. **Liquation:**
   - Used for metals with low melting points (e.g., tin, lead).
   - The impure metal is placed on a sloping surface and heated. The pure molten metal flows down, leaving impurities behind.

3. **Distillation:**
   - Used for volatile metals like zinc and mercury.
   - The impure metal is heated until it vaporizes, and the vapor is condensed to obtain pure metal.

4. **Zone Refining:**

- Used for highly pure metals required in electronics (e.g., silicon, germanium).
- A narrow region of the metal rod is melted by a moving heater. Impurities concentrate in the molten zone, which moves along the rod, leaving pure metal behind.

5. **Cupellation:**
   - Used for refining silver and other noble metals.
   - The impure metal is heated in a porous crucible (cupel). Base metals oxidize and are absorbed, leaving behind pure noble metal.

Importance of Refining Metals

- **Copper** must be pure for electrical wiring.
- **Aluminium** is refined for aircraft and construction use.
- **Gold and silver** require refining for jewelry and monetary reserves.
- **Silicon and Germanium** must be ultrapure for use in semiconductors, phones, and computers.

Evaluation

1. Define the term *reactivity series of metals* and explain its importance.
2. Arrange the following metals in order of decreasing reactivity: Zinc, Copper, Sodium, Iron.
3. Explain why aluminium, though reactive, does not corrode easily.
4. What is meant by displacement reaction? Give one balanced equation to illustrate.
5. State three observations when a piece of sodium is placed in cold water.

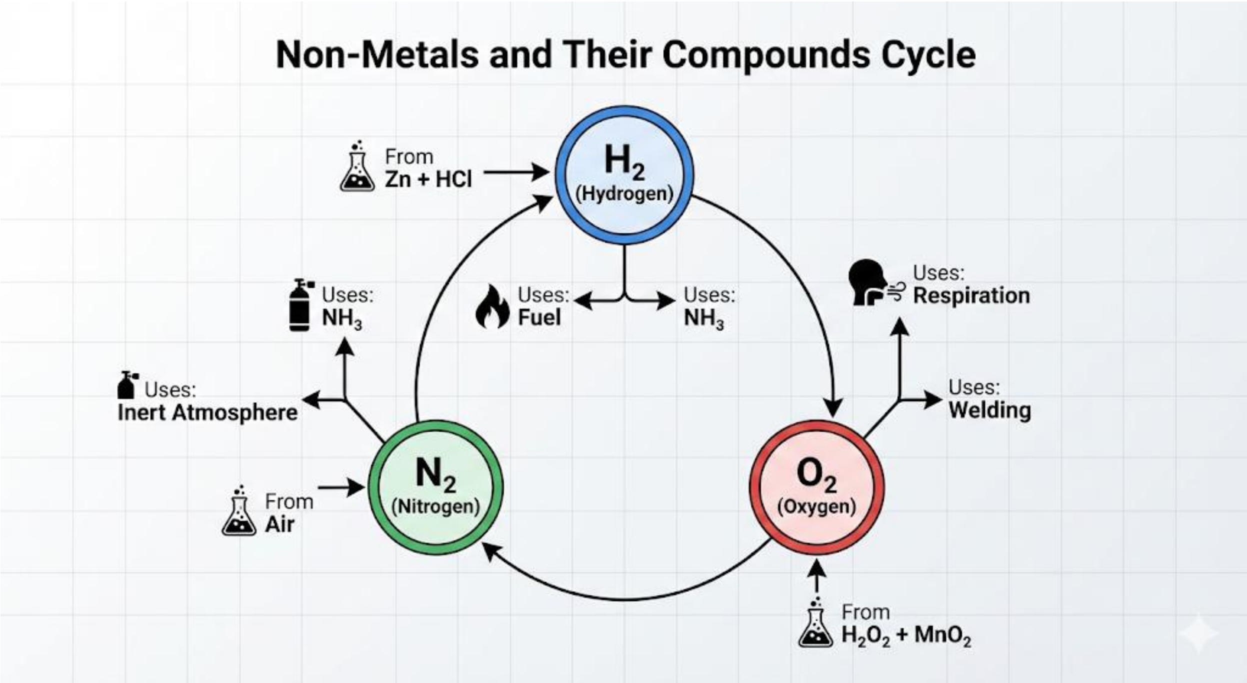
6. Explain why zinc can displace copper from copper (II) sulphate solution but copper cannot displace zinc from zinc sulphate solution.
7. List four methods of refining metals and mention one metal purified by each method.
8. In electrolytic refining of copper, what is the function of:
   a) the anode
   b) the cathode
   c) the electrolyte
9. Why is zone refining the most suitable method for silicon used in electronics?

Assignment

1. Compare the reactivity of calcium and magnesium with water, giving chemical equations.
2. During electrolytic refining of copper, impurities such as silver and gold are found in the anode mud. Explain why.
3. A student places an iron nail into copper (II) sulphate solution. State and explain the changes that occur.
4. A sample of impure zinc weighing 500 g contains 10% impurities. After refining, what mass of pure zinc will be obtained?
5. Why is gold considered a noble metal? Mention two uses of gold that arise from this property.

WEEK 9: NON-METALS AND THEIR COMPOUNDS

Hydrogen, Oxygen, and Nitrogen



General Properties of Non-Metals

- Non-metals are found on the **right side of the periodic table** (except hydrogen).
- They usually form **anions** (negative ions) by gaining electrons, or they share electrons to form **covalent bonds**.
- Non-metals are **electronegative** (tend to attract electrons).
- Unlike metals, they do not exhibit metallic luster, malleability, or ductility.
- They exist in various physical states:
  - Solids: sulphur (S), phosphorus (P), carbon (C).
  - Liquids: bromine (Br_2).
  - Gases: hydrogen (H_2), oxygen (O_2), nitrogen (N_2), chlorine (Cl_2).

General Characteristics:

1. Poor conductors of electricity (except graphite).

2. Low melting and boiling points (except diamond and graphite).
3. Form **acidic oxides** (CO_2, SO_2, P_2O_5).
4. React with hydrogen to form volatile covalent compounds (HCl, NH_3).
5. Have variable oxidation states.

Hydrogen (H_2)

Occurrence

- Most abundant element in the universe (~70% of universe's mass).
- Occurs combined in:
  - Water (H_2O)
  - Hydrocarbons (CH_4, C_2H_6, etc.)
  - Acids (HCl, H_2SO_4)
  - Organic compounds (proteins, carbohydrates, fats).

Laboratory Preparation

1. **Action of Dilute Acids on Reactive Metals** (Zn + HCl):

$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

  - Apparatus: flask, thistle funnel, delivery tube, gas jar over water.
  - Drying agent: calcium chloride.

2. **Action of Water on Alkali Metals** (e.g., Na + H_2O):

$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$

Physical Properties

- Colourless, odourless, tasteless gas.
- Lightest element (density = 0.069 relative to air).
- Slightly soluble in water (collected by downward displacement of water).
- Neutral to litmus paper.

Chemical Properties

1. **Combustion:** Burns in oxygen with a pale blue flame to form water.

$2H_2 + O_2 \rightarrow 2H_2O$

2. **Reaction with Halogens:** Explosively combines with chlorine in sunlight.

$H_2+Cl_2 \rightarrow 2HCl$

3. **Reducing Agent:** Reduces metal oxides to metals.

$CuO + H_2 \rightarrow Cu + H_2O$

4. **Synthesis Reactions:**
    - With nitrogen to form ammonia:

$N_2+3H_2 \rightarrow$Fe catalyst, 450∘C, 200atm

    - With unsaturated oils (hydrogenation):

$C_2H_4 + H_2 \rightarrow NiC_2H_6$

Uses of Hydrogen

- Production of **ammonia** (Haber process).
- Hydrogenation of vegetable oils into margarine.
- Used as **rocket fuel** (liquid H_2 + liquid O_2).
- Welding (oxy-hydrogen flame).

- As a reducing agent in metallurgy.

Oxygen (O_2)

Occurrence

- Constitutes 21% of atmospheric air.
- Present in water and oxides.
- Found in minerals (silicates, carbonates, sulphates).

Laboratory Preparation

1. **Decomposition of Hydrogen Peroxide:**

$$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2$$

2. **Thermal Decomposition of Potassium Chlorate(V):**

$$2KClO_3 \xrightarrow{MnO_2} 2KCl + 3O_2$$

Drying Agent: $CaCl_2$ or conc. H_2SO_4.

Physical Properties

- Colourless, odourless, tasteless gas.
- Slightly soluble in water.
- Denser than air.
- Neutral to litmus.

Chemical Properties

1. **Supports Combustion:** Candle burns brighter in oxygen.
2. **Reacts with Metals:**

$2Mg + O_2 \rightarrow 2MgO$

3. **Reacts with Non-Metals:**

   $C + O_2 \rightarrow CO_2$

   $S + O_2 \rightarrow SO_2$

4. **Formation of Oxides:** Basic oxides (Na_2O), acidic oxides (SO_2), amphoteric oxides (ZnO).

Uses of Oxygen

- Essential for **respiration** and **combustion**.
- Used in hospitals for patients.
- Oxy-acetylene flame for welding.
- Steel manufacturing (Bessemer process).
- Rocket fuel oxidant.

Nitrogen (N_2)

Occurrence

- Constitutes ~78% of atmospheric air.
- Found in nitrates (KNO_3, $NaNO_3$), proteins, fertilizers.

Laboratory Preparation

- By removing oxygen from air using alkaline pyrogallol.
- Industrially obtained by fractional distillation of liquid air.

Physical Properties

- Colourless, odourless, tasteless gas.
- Very inert at room temperature (strong triple bond, N≡N).
- Slightly soluble in water.
- Lighter than air.

Chemical Properties

1. **Inertness:** Does not react easily at room temperature.
2. **Reactions at high temperature/pressure:**
    - With hydrogen:

$$N_2 + 3H_2 \xrightarrow{Fe\ catalyst} 2NH_3$$

    - With oxygen:

$$N_2 + O_2 \rightarrow 2NO$$

    - With metals:

$$3Mg + N_2 \rightarrow Mg_3N_2$$

Uses of Nitrogen

- Production of **ammonia** and fertilizers.
- Manufacture of nitric acid (via Ostwald process).
- Liquid nitrogen used as a **coolant**.
- Provides inert atmosphere in food preservation and light bulbs.
- Manufacture of explosives (TNT, nitroglycerin).

Comparative Summary of Hydrogen, Oxygen, and Nitrogen

Property	Hydrogen (H_2)	Oxygen (O_2)	Nitrogen (N_2)
% in air	0.01%	21%	78%
Density	Lightest gas	Heavier than air	Slightly lighter than air
Reactivity	Moderate, reducing agent	Strong oxidizing agent	Very inert at room temp
Solubility in water	Slightly soluble	Slightly soluble	Slightly soluble
Industrial use	Haber process, welding	Respiration, welding	Fertilizers, preservation

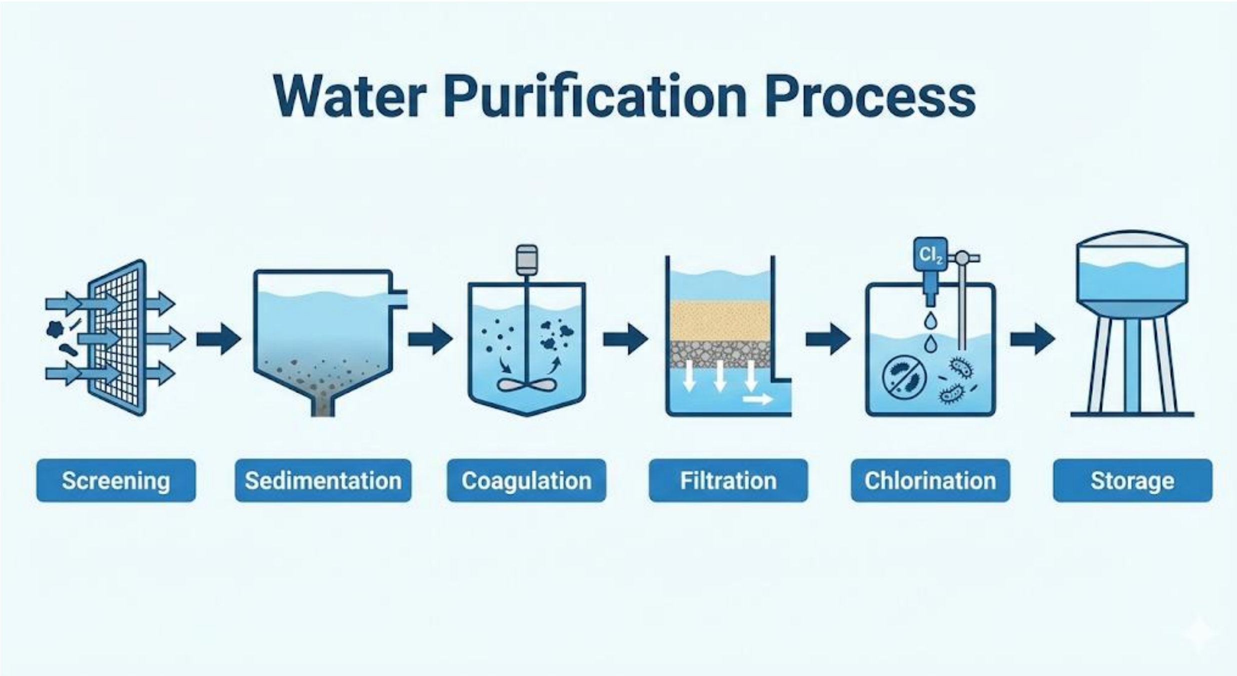
Evaluation

1. List four general properties of non-metals.
2. Write balanced equations for the laboratory preparation of:
   (a) hydrogen, (b) oxygen, (c) nitrogen.
3. Why is hydrogen described as a reducing agent?
4. State three uses of nitrogen in industries.
5. Differentiate between the chemical properties of oxygen and nitrogen.

Assignment

1. Compare the reactivity of hydrogen, oxygen, and nitrogen, explaining why nitrogen is inert at room temperature.
2. (a) Explain the role of iron catalyst in the Haber process.
   (b) Why are high pressure and moderate temperature used in the process?
3. Write balanced equations for the reactions of oxygen with:
   (i) magnesium, (ii) carbon, (iii) sulphur.
4. Describe one industrial use each of liquid hydrogen, liquid oxygen, and liquid nitrogen.

WEEK 10: WATER – HARDNESS, PURIFICATION, AND QUALITY CONTROL



HARDNESS OF WATER

Definition

Water is described as **hard** when it does not readily form lather with soap but instead produces a white scum. This is caused by the presence of **dissolved salts of calcium and magnesium** in the water.

Types of Hardness

1. **Temporary Hardness**
   - Caused by the presence of **calcium hydrogencarbonate ($Ca(HCO_3)_2$)** and **magnesium hydrogencarbonate ($Mg(HCO_3)_2$)**.
   - These salts are soluble in water and enter natural water sources when rainwater (which contains dissolved CO_2) reacts with limestone or chalk ($CaCO_3$) in the soil:

     $$CaCO_3(s) + CO_2(g) + H_2O(l) \longrightarrow Ca(HCO_3)_2(aq)$$
   - **Removal**: Temporary hardness can be removed by boiling or by adding slaked lime ($Ca(OH)_2$).

- On boiling, the hydrogencarbonates decompose to insoluble carbonates, CO_2, and water:

$$Ca(HCO_3)_2(aq) \xrightarrow{\Delta} CaCO_3(s) + CO_2(g) + H_2O(l)$$

$$Mg(HCO_3)_2(aq) \xrightarrow{\Delta} MgCO_3(s) + CO_2(g) + H_2O(l)$$

- The insoluble $CaCO_3$ and $MgCO_3$ precipitate out, leaving the water soft.

2. **Permanent Hardness**
   - Caused by the presence of **calcium sulfate ($CaSO_4$)**, **magnesium sulfate ($MgSO_4$)**, and chlorides or nitrates of calcium and magnesium.
   - These salts are highly soluble in water and **cannot be removed by boiling**.
   - **Removal methods**:
     - **Addition of washing soda (Na_2CO_3):**
       Sodium carbonate reacts with calcium or magnesium ions to form insoluble carbonates.

$$CaSO_4(aq) + Na_2CO_3(aq) \longrightarrow CaCO_3(s) + Na_2SO_4(aq)$$

     - **Ion exchange/zeolite method (permutit process):**
       - Zeolites are complex sodium aluminosilicates.
       - In the process, Ca^{2+} and Mg^{2+} ions are exchanged for Na^+ ions:

$$Na_2Zeolite + Ca^{2+} \longrightarrow CaZeolite + 2Na^+$$

       - Eventually, the zeolite becomes "exhausted" and must be regenerated by washing with brine (NaCl solution).
     - **Use of complexing agents (Calgon method):**
       Sodium hexametaphosphate $(NaPO_3)_6$, commonly called calgon, binds Ca^{2+}

and Mg^{2+} ions to form soluble complexes, preventing them from interfering with soap.

Purification of Water

Importance

Water purification is the process of removing undesirable chemical, biological, and physical contaminants to make water **safe for human consumption, domestic use, and industrial purposes**.

Steps in Water Purification (as used in municipal waterworks)

1. **Screening**
    - Large debris such as leaves, sticks, and plastics are removed using wire meshes or metal screens.
2. **Sedimentation**
    - Water is allowed to stand in large tanks to let suspended particles settle by gravity.
3. **Coagulation/Flocculation**
    - A coagulant such as **alum ($KAl(SO_4)_2 \cdot 12H_2O$)** is added.
    - Alum hydrolyzes to form gelatinous $Al(OH)_3$ which traps fine suspended particles, bacteria, and colloids into larger flocs that settle out.
4. **Filtration**
    - Water passes through beds of sand, gravel, and sometimes activated charcoal to remove remaining solid particles and improve clarity.
    - Activated charcoal also removes bad taste and odour.
5. **Chlorination (Disinfection)**
    - A small amount of chlorine (Cl_2) or compounds such as sodium hypochlorite ($NaOCl$) is added to kill harmful microorganisms (bacteria, viruses, protozoa).

- Reaction:

$$Cl_2 + H_2O \longrightarrow HCl + HOCl$$

Hypochlorous acid (HOCl) is the active germicidal agent.

6. **Aeration**
   - Air is bubbled through water to remove dissolved gases such as hydrogen sulfide (H_2S) and carbon dioxide, and to increase oxygen content which improves taste.
7. **Softening (if required)**
   - Industrial and domestic water may be softened using lime-soda or ion-exchange methods.
8. **Distillation (for laboratory purposes only)**
   - Distillation produces pure water free from dissolved salts and microorganisms.
   - Distilled water is used in chemical analysis, preparation of solutions, and in car batteries.

3. Water Quality Control

Meaning

Water quality control refers to the **monitoring, testing, and regulation** of water to ensure it meets the standards for safe use.

Parameters of Water Quality

1. **Physical parameters**
   - Colour: Pure water should be colourless.
   - Taste: Water should be tasteless.
   - Odour: Water should be odourless.
   - Turbidity: Water should be clear and free from suspended particles.

2. **Chemical parameters**
    - **pH**: Acceptable range is 6.5 – 8.5.
    - **Hardness**: Should not exceed recommended limits (about 150 ppm).
    - **Chloride content**: Excess chloride makes water salty.
    - **Heavy metals**: Lead (Pb^{2+}), cadmium (Cd^{2+}), mercury (Hg^{2+}), and arsenic (As^{3+}) must be absent or below safe limits.
3. **Biological parameters**
    - Drinking water must be free from harmful microorganisms.
    - Presence of **coliform bacteria** is an indication of faecal contamination.

Importance of Water Quality Control

- Prevents outbreaks of waterborne diseases such as **cholera, dysentery, and typhoid**.
- Ensures public health and safety.
- Prevents scaling and corrosion in boilers, turbines, and pipelines.
- Guarantees water suitability for sensitive industries (e.g., pharmaceuticals, beverages, textiles, electronics).
- Helps meet international water standards set by organizations such as WHO and NAFDAC.

Evaluation

1. Define hardness of water and explain why hard water forms scum with soap.
2. Differentiate between temporary and permanent hardness, giving their causes.
3. State, with equations, two methods of removing:
    - (a) Temporary hardness

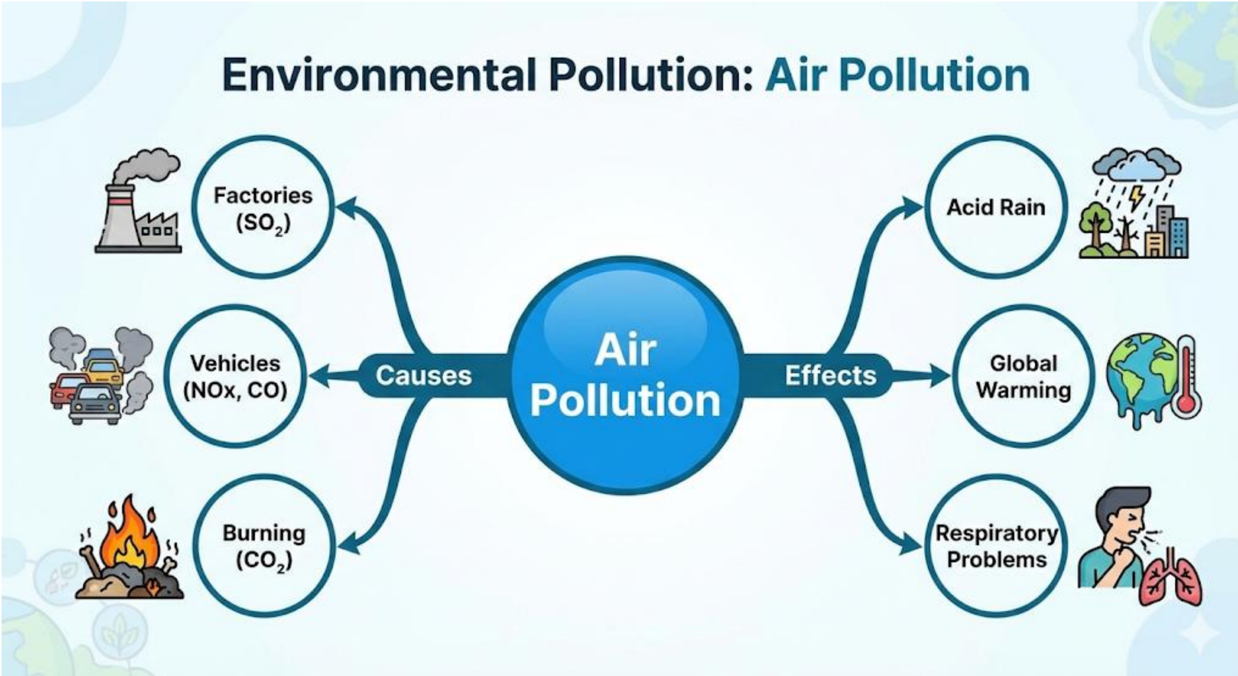
- (b) Permanent hardness

4. List six steps involved in the purification of municipal water.
5. Mention three physical, three chemical, and two biological parameters of water quality.
6. Why is water quality control important for industries and households?

Assignment

1. Explain the chemistry of how boiling removes temporary hardness of water.
2. With balanced chemical equations, show how washing soda removes permanent hardness.
3. Write short notes on the following methods of water softening:
   - Ion exchange method
   - Calgon method
4. Distinguish between distilled water and potable water, stating one use of each.
5. Suggest **five measures** that governments can take to ensure quality control of drinking water in cities.

WEEK 11: ENVIRONMENTAL POLLUTION – CAUSES, CONTROL, AND EFFECTS



MEANING OF ENVIRONMENTAL POLLUTION

- The **environment** is the sum of all external conditions (air, water, soil, plants, animals, etc.) that affect living organisms.
- **Pollution** is the **undesirable alteration** in the physical, chemical, or biological characteristics of the environment, making it harmful to life or unsuitable for normal use.
- Pollution may be **natural** (e.g., volcanic eruptions, dust storms) or **man-made/anthropogenic** (industrialization, deforestation, excessive burning of fuels).
- A **pollutant** is any material or factor that causes pollution. Pollutants can be:
  - **Biodegradable pollutants** → easily decomposed by microorganisms (e.g., food waste, sewage, paper).
  - **Non-biodegradable pollutants** → persist in the environment and accumulate in the food chain (e.g., plastics, heavy metals, DDT, radioactive isotopes).

MAJOR TYPES OF POLLUTION, CAUSES AND EXAMPLES

(a) Air Pollution

- **Definition:** Contamination of the atmosphere by harmful substances in excessive concentrations.
- **Major Causes and Pollutants:**
    - **Carbon monoxide (CO):** From incomplete combustion of petrol in vehicles → poisonous, binds with haemoglobin.
    - **Carbon dioxide (CO_2):** From burning fossil fuels, deforestation → greenhouse effect and global warming.
    - **Sulphur dioxide (SO_2):** From burning coal, oil refineries, volcanic eruptions → forms acid rain.
    - **Nitrogen oxides (NO, NO_2):** From vehicles, industries → contribute to photochemical smog and acid rain.
    - **Particulates (dust, smoke, soot, asbestos):** From factories, cement industries → cause lung diseases.
    - **Chlorofluorocarbons (CFCs):** From refrigerants and sprays → deplete ozone layer.
- **Real-Life Example:** Severe smog in industrial cities like Beijing and Lagos reduces visibility and causes respiratory illnesses.

(b) Water Pollution

- **Definition:** Contamination of rivers, lakes, seas, and underground water by harmful substances.
- **Causes and Pollutants:**
    - **Sewage and domestic waste:** Introduces pathogens → cholera, typhoid, dysentery.

- **Industrial effluents:** Contain toxic metals like mercury (Hg), cadmium (Cd), lead (Pb). Example: *Minamata disease* in Japan was caused by mercury poisoning from industrial discharge.
- **Agricultural runoff:** Fertilizers add nitrates/phosphates → eutrophication (excessive algae growth → oxygen depletion → fish kill).
- **Oil spills:** Coats the surface of water, preventing oxygen dissolution and killing marine life.

- **Equation (Eutrophication):**
  Nitrate/Phosphate → Algal Bloom → Oxygen Depletion → Death of Aquatic Life

(c) Soil Pollution

- **Definition:** Degradation of land by harmful chemicals, waste, or poor land use.
- **Causes:**
  - Use of excessive pesticides (e.g., DDT) and fertilizers.
  - Dumping of plastics, metals, glass, and non-biodegradable materials.
  - Crude oil spillage in the Niger Delta → loss of farmlands.
  - Acid rain → reduces soil pH, making it infertile.
- **Consequence:** Contaminated food crops, poor agricultural productivity, entry of toxic chemicals into food chains.

(d) Noise Pollution

- **Definition:** Excessive, unpleasant sound that disturbs health and well-being.
- **Causes:** Traffic, loudspeakers, industries, aircraft, blasting in mining.
- **Effects:** Hearing loss, hypertension, lack of concentration, insomnia.

(e) Thermal Pollution

- **Definition:** Rise in temperature of water bodies due to discharge of heated effluents.
- **Sources:** Power plants, steel factories, petroleum refineries.
- **Effects:**
    - Reduces dissolved oxygen in water.
    - Disrupts breeding and survival of fish.

(f) Radioactive Pollution

- **Definition:** Release of radioactive substances into the environment.
- **Sources:** Nuclear power plants, uranium mining, nuclear weapons testing, improper disposal of radioactive waste.
- **Effects:**
    - Genetic mutations, cancer, birth defects.
    - Long-term contamination of soil and water (lasting thousands of years).
- **Example:** The **Chernobyl nuclear disaster (1986)** in Ukraine and **Fukushima (2011)** in Japan released dangerous radiation affecting generations.

EFFECTS OF ENVIRONMENTAL POLLUTION

On Human Health

- Respiratory diseases (asthma, bronchitis, lung cancer) from air pollutants.
- Neurological damage from lead and mercury poisoning.
- Gastrointestinal diseases from polluted water.

- Psychological stress and hearing impairment from noise pollution.

On Plants and Agriculture

- Acid rain destroys crops and reduces soil nutrients.
- Heavy metals accumulate in plants → enter food chain.
- Oil spills destroy farmlands and reduce crop yield.

On Aquatic Life

- Fish kills due to oxygen depletion from eutrophication.
- Bioaccumulation of mercury and cadmium in fish → harmful when consumed by humans.
- Coral reef destruction from oil spills and acidification.

On Climate and Ecosystem

- Global warming from greenhouse gases (CO_2, CH_4).
- Ozone layer depletion → increased UV radiation → skin cancer and cataracts.
- Extinction of species and loss of biodiversity.

CONTROL MEASURES

Air Pollution

- Use of alternative fuels (biofuel, LPG, solar energy).
- Installation of catalytic converters in vehicles.
- Industrial treatment plants (scrubbers, filters).
- Afforestation to absorb CO_2.

Water Pollution

- Proper sewage treatment before discharge.
- Use of biodegradable detergents.

- Quick clean-up of oil spills using dispersants and absorbents.
- Regulation of industrial effluents.

Soil Pollution

- Recycling and reuse of non-biodegradable materials.
- Organic farming to reduce fertilizer use.
- Proper waste disposal and controlled landfills.

Noise Pollution

- Use of silencers in machinery and vehicles.
- Zoning laws to separate residential and industrial areas.
- Enforcement of noise control regulations.

Thermal Pollution

- Cooling of hot effluents before discharge.
- Use of cooling towers in industries.

Radioactive Pollution

- Secure underground storage of nuclear waste.
- International treaties to regulate nuclear tests.
- Use of protective shielding and monitoring systems in nuclear plants.

IMPORTANCE OF POLLUTION CONTROL

- Safeguards human and animal health.
- Protects biodiversity and ecosystems.
- Prevents climate change and natural disasters.
- Ensures sustainable use of resources.

- Promotes cleaner, safer, and healthier environments for future generations.

EVALUATION

1. Define pollution and differentiate between biodegradable and non-biodegradable pollutants with examples.
2. List four causes of air pollution and explain how each affects human health.
3. Describe eutrophication and explain how it affects aquatic life.
4. Explain how acid rain is formed and state three of its harmful effects.
5. Mention two historical examples of radioactive pollution and their consequences.

ASSIGNMENT

1. With suitable examples, explain five major types of pollution and their specific effects.
2. Discuss five practical methods by which air pollution can be controlled in Nigeria.
3. Suggest a community-based waste management strategy that can reduce soil and water pollution in your locality.

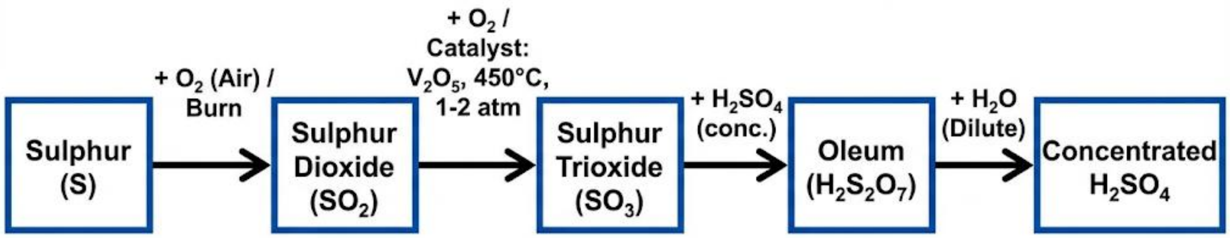
SS2 CHEMISTRY LESSON NOTES

THIRD TERM

1. Sulphur and Its Compounds – Sulphur dioxide, sulphuric acid.
2. Carbon and Its Compounds – Oxides, carbonates, hydrocarbons.
3. Industrial Preparation of Chemicals – HCl, NH_3, H_2SO_4.
4. Organic Chemistry (Revision) – Functional groups, IUPAC naming.
5. **Midterm Examination**
6. Hydrocarbons – Properties, tests.
7. **Midterm break**
8. Alcohols, Ethers, and Phenols – Uses, reactions.
9. Aldehydes and Ketones – Preparation, identification.
10. Carboxylic Acids and Esters – Structure and reactions.
11. Polymers and Plastics – Types, uses, dangers.
12. Examination
13. Closing

WEEK 1: SULPHUR AND ITS COMPOUNDS

Sulphur Dioxide and Sulphuric Acid



INTRODUCTION TO SULPHUR

- **Sulphur (S)** is a **yellow, brittle, non-metallic element** with atomic number 16 and atomic mass 32.06 g/mol.
- **Occurrence:**
  - **Native sulphur deposits:** Found in volcanic regions (e.g., Sicily, USA).
  - **Combined forms:** Present in minerals such as:
    - Iron pyrites (FeS_2)
    - Galena (PbS)
    - Zinc blende (ZnS)
    - Copper pyrites ($CuFeS_2$)
    - Gypsum ($CaSO_4 \cdot 2H_2O$)
- **Allotropes of Sulphur:**

1. **Rhombic sulphur:** Stable at room temperature; yellow crystals; melts at 114 °C; density = 2.06 g/cm³.

2. **Monoclinic sulphur:** Stable above 96 °C; needle-like crystals; melts at 119 °C; less dense than rhombic sulphur.

- **General properties:**
  - Yellow solid, brittle, and odourless.
  - Insoluble in water; soluble in carbon disulphide (CS_2).
  - Burns with a blue flame to produce sulphur dioxide (SO_2).

SULPHUR DIOXIDE (SO_2)

(a) Preparation

Laboratory method:

$$Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + SO_2$$

Industrial method:

- Roasting sulphide ores:

$$2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$$

$$4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$$

(b) Physical properties

- Colourless gas with a **pungent, choking odour**.
- Denser than air (density ~2.26 kg/m³).
- Soluble in water to form **sulphurous acid (H_2SO_3)**:

$$SO_2 + H_2O \rightarrow H_2SO_3$$

(c) Chemical properties

1. **As an acidic oxide:**

    - Reacts with bases to form salts (sulphites and bisulphites):

$$SO_2 + 2NaOH \rightarrow Na_2SO_3 + H_2O$$

$$SO_2 + NaOH \rightarrow NaHSO_3$$

2. **As a reducing agent:**

- Reduces ferric ions to ferrous ions:

$$2Fe^{3+} + SO_2 + 2H_2O \rightarrow 2Fe^{2+} + H_2SO_4$$

- Reduces iodine to hydrogen iodide:

$$I_2 + SO_2 + 2H_2O \rightarrow 2HI + H_2SO_4$$

3. **Bleaching action:**

- Bleaches wool, silk, and paper temporarily due to reduction of coloured compounds.

(d) Uses of Sulphur Dioxide

- Bleaching agent in paper and textile industries.
- Preservative for dried fruits and wine.
- Raw material in the manufacture of **sulphuric acid** (Contact Process).
- Fumigant and disinfectant in food and agriculture.

3. SULPHURIC ACID (H_2SO_4)

(a) Industrial preparation – Contact Process

1. **Burning sulphur:**

$$S + O_2 \rightarrow SO_2$$

2. **Catalytic oxidation:**

$$2SO_2 + O_2 \xrightarrow{V_2O_5, 450^\circ C, 2atm} 2SO_3 \quad \text{(exothermic)}$$

3. **Absorption in concentrated H_2SO_4 (oleum formation):**

$$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$$

4. **Dilution to form concentrated H_2SO_4:**

$$H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$$

Note: Sulphuric acid is **highly exothermic** when mixed with water; always add acid to water, not water to acid.

(b) Physical properties

- Colourless, viscous, oily liquid.
- Density: 1.84 g/cm^3; boiling point: 338 °C.
- Miscible with water; very hygroscopic.
- Highly corrosive to skin and metals.

(c) Chemical properties

1. **As a strong acid:**

$$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$$

2. **As a dehydrating agent:**

- Dehydrates carbohydrates to carbon and water:

$$C_{12}H_{22}O_{11} \xrightarrow{H_2SO_4} 12C + 11H_2O$$

3. **As an oxidizing agent (concentrated H_2SO_4):**

$$Cu + 2H_2SO_4(conc.) \rightarrow CuSO_4 + SO_2 + 2H_2O$$

4. **Formation of salts:**

- Reacts with metals, bases, and metal oxides to form **sulphates**.

(d) Uses of Sulphuric Acid

- Manufacture of **fertilizers**: ammonium sulphate, superphosphate.
- **Petroleum refining** and manufacture of detergents.
- **Electrolyte in lead-acid batteries.**
- Production of dyes, paints, drugs, and explosives.
- Laboratory reagent for dehydration, oxidation, and sulphonation reactions.

(e) Environmental and Safety Considerations

- **Sulphur dioxide** is a major **air pollutant**, causing acid rain:

$SO_2 + H_2O + 12O_2 \rightarrow H_2SO_4$

- **Sulphuric acid** is **corrosive**; can cause burns, blindness, and environmental hazards if spilled.

- Adequate **ventilation**, **protective clothing**, and **neutralization procedures** are necessary when handling these compounds.

COMPARATIVE OVERVIEW

Property / Feature	Sulphur Dioxide (SO_2)	Sulphuric Acid (H_2SO_4)
Physical State	Colourless gas	Colourless, viscous liquid
Odour	Pungent, choking	Odourless (fuming in concentrated form)
Solubility	Soluble in water → H_2SO_3	Highly soluble, hygroscopic
Chemical Nature	Acidic oxide, reducing agent	Strong acid, oxidizing & dehydrating agent
Uses	Bleaching, preservative, acid manufacture	Fertilizers, batteries, industrial reagent
Hazards	Irritates respiratory tract	Corrosive, causes burns, environmental hazard

EVALUATION

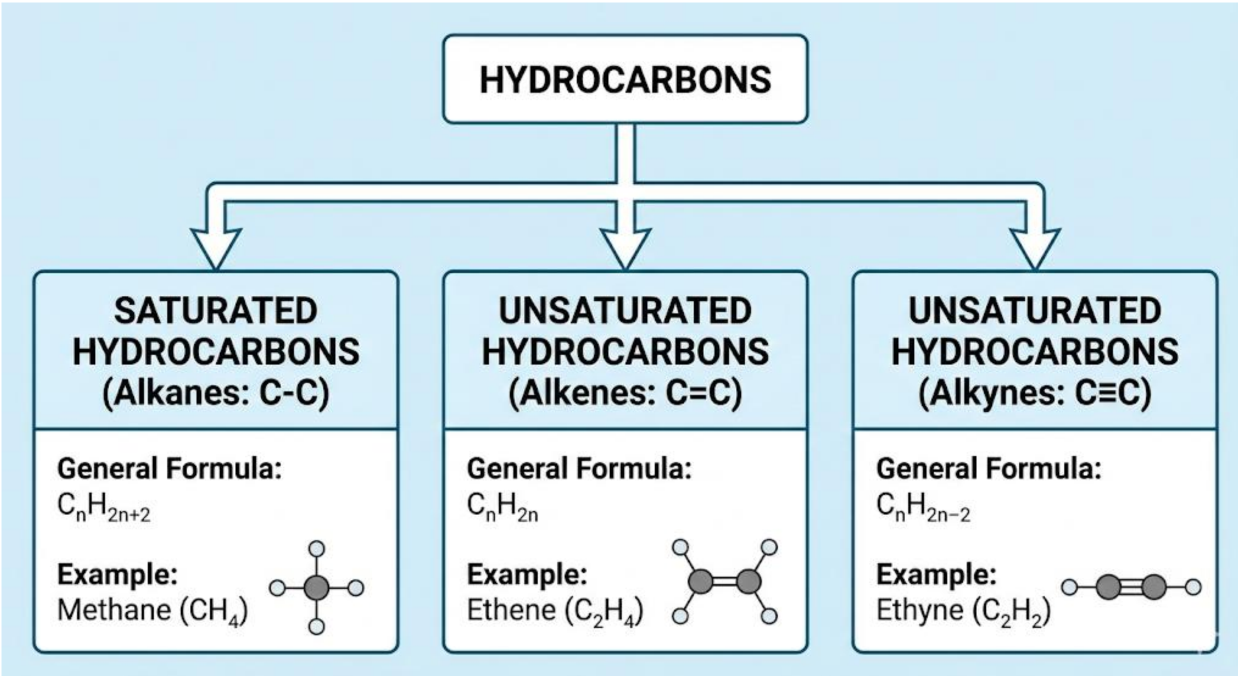
1. List two natural sources of sulphur.
2. Write balanced equations for the laboratory preparation of SO_2 and its reaction with water.
3. Explain why SO_2 is considered a reducing agent.
4. Outline the major steps of the Contact Process for H_2SO_4 production.
5. Give three industrial uses of concentrated sulphuric acid.

ASSIGNMENT

1. Explain the chemical properties of SO_2 and H_2SO_4, highlighting their similarities and differences.
2. Discuss environmental hazards associated with SO_2 and sulphuric acid, and suggest preventive measures.
3. Draw a labeled diagram of the Contact Process, including the roles of catalyst, temperature, and pressure.

WEEK 2: CARBON AND ITS COMPOUNDS

Oxides, Carbonates, Hydrocarbons



INTRODUCTION TO CARBON

- **Atomic Number:** 6
- **Electronic Configuration:** $1s^2\ 2s^2\ 2p^2$
- **Valency:** 4 (tetravalent – can form 4 covalent bonds).
- **Allotropes of Carbon:**
  - **Crystalline** – Diamond, Graphite.
  - **Amorphous** – Coal, Charcoal, Coke.
- **Unique Properties:**
  - **Catenation:** Ability to form long chains and rings with itself.
  - **Tetravalency:** Bonds strongly with H, O, N, S, etc.
  - **Formation of multiple bonds:** (single, double, triple).
  - **Isomerism:** Existence of compounds with same formula but different structures.

These properties make carbon the **backbone of organic chemistry**.

OXIDES OF CARBON

a) Carbon (II) oxide (CO)

- **Nature:** Colourless, odourless, toxic gas.
- **Preparation:**
    - By incomplete combustion of carbon:

      $2C + O_2 \longrightarrow 2CO$
    - By dehydration of formic acid with conc. H_2SO_4:

$$HCOOH \xrightarrow{conc.H_2SO_4} CO + H_2O$$

- **Properties:**
    - Combustible, burns with blue flame:

    $2CO + O_2 \longrightarrow 2CO_2$
    - Powerful reducing agent:

    $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$
- **Uses:**
    - As a reducing agent in metallurgy (extraction of iron).
    - Manufacture of methanol and synthetic fuels.
- **Hazard:** Toxic – forms **carboxyhaemoglobin** with blood, blocking oxygen transport → suffocation.

b) Carbon(IV) oxide (CO_2)

- **Nature:** Colourless, odourless, slightly acidic, heavier than air.
- **Preparation:**
  - By decomposition of carbonates:

    $CaCO_3 \rightarrow \Delta\ CaO + CO_2$

  - By acid + carbonate reaction:

    $CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$

- **Properties:**
  - Non-flammable, extinguishes fire.
  - Slightly soluble in water → carbonic acid (H_2CO_3).
  - Turns limewater milky:

    $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$

- **Uses:**
  - Fire extinguishers.
  - Carbonated drinks.
  - Refrigerant (dry ice).
  - Photosynthesis by plants.

c) Comparative Table: CO vs CO_2

Feature	**CO**	**CO_2**
Colour & Smell	Colourless, odourless	Colourless, odourless
Combustibility	Combustible	Non-combustible
Toxicity	Highly poisonous	Non-toxic (but suffocating in excess)
Reaction with Limewater	No effect	Turns limewater milky
Main Uses	Metallurgy, methanol	Fire extinguishers, drinks

CARBONATES

- **Definition:** Salts of carbonic acid (H2CO3)(H_2CO_3).
- **Examples:**
  - Sodium carbonate (Na_2CO_3).
  - Calcium carbonate ($CaCO_3$).
  - Potassium carbonate (K_2CO_3).
- **Properties:**
  - Solubility: Na_2CO_3 and K_2CO_3 are soluble; most others insoluble.
  - Thermal decomposition:

    $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
  - Reaction with acids:

  $Na_2CO_3 + 2HCl \longrightarrow 2NaCl + H_2O + CO_2$
- **Uses:**
  - Na_2CO_3 → glass, soap, detergent manufacture.
  - $CaCO_3$ → cement, chalk, marble.
  - K_2CO_3 → fertilizer production.

Hydrocarbons

Classification

1. **Alkanes** (Saturated, single bonds)
   - Formula: C_nH_{2n+2}.
   - Example: Methane (CH_4), Ethane (C_2H_6).

- Properties:
    - Relatively unreactive (due to strong C–C and C–H bonds).
    - Undergo **substitution** reactions with halogens in UV light.
    - Burn with clean flame:

  $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

2. **Alkenes** (Unsaturated, double bonds)
    - Formula: C_nH_{2n}.
    - Example: Ethene (C_2H_4).
    - Properties:
        - Very reactive.
        - Undergo **addition reactions** (e.g., hydrogenation, halogenation).
        - Burn with smoky flame.
    - Test: Decolourizes bromine water.
3. **Alkynes** (Unsaturated, triple bonds)
    - Formula: C_nH_{2n-2}
    - Example: Ethyne (C_2H_2).
    - Properties:
        - Very reactive.
        - Undergo addition reactions.
        - Burns with very sooty flame.

Isomerism

- Hydrocarbons show **structural isomerism**: same molecular formula but different arrangement.
    - Example: C_4H_{10} → n-butane and isobutane.

Preparation of Hydrocarbons

- **Methane:**
  From heating sodium acetate with soda lime:

$$CH_3COONa + NaOH \xrightarrow{CaO,\Delta} CH_4 + Na_2CO_3$$

- **Ethyne (acetylene):**
  From calcium carbide with water:

$CaC_2 + 2H_2O \longrightarrow C_2H_2 + Ca(OH)_2$

Uses of Hydrocarbons

- Fuels (LPG, petrol, kerosene, diesel).
- Raw materials for plastics, alcohols, detergents.
- Source of synthetic fibres and drugs.

Importance of Carbon Compounds

- Fuels (coal, petroleum, gas).
- Industrial raw materials (plastics, rubbers, detergents, cement, glass).
- Essential to life (proteins, fats, carbohydrates are carbon compounds).

Environmental Effects

- **CO_2:** Greenhouse gas → global warming, climate change.
- **CO:** Toxic to humans.
- **Hydrocarbon combustion:** Produces **soot, CO, NO_2, SO_2** → air pollution, acid rain.

Worked Example

Question: Differentiate between alkanes, alkenes, and alkynes with examples and reactions.

Answer:

- Alkanes: Saturated, substitution, $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$.
- Alkenes: Unsaturated, addition, $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$.
- Alkynes: Unsaturated, triple bond, $C_2H_2 + H_2 \rightarrow C_2H_4$.

Evaluation

1. Write two differences between CO and CO_2.
2. Write a balanced equation for:
   a) Decomposition of $CaCO_3$.
   b) Reaction of Na_2CO_3 with HCl.
3. State three properties of alkenes that distinguish them from alkanes.
4. Why do alkynes burn with sooty flames?
5. Mention three industrial uses of calcium carbonate.

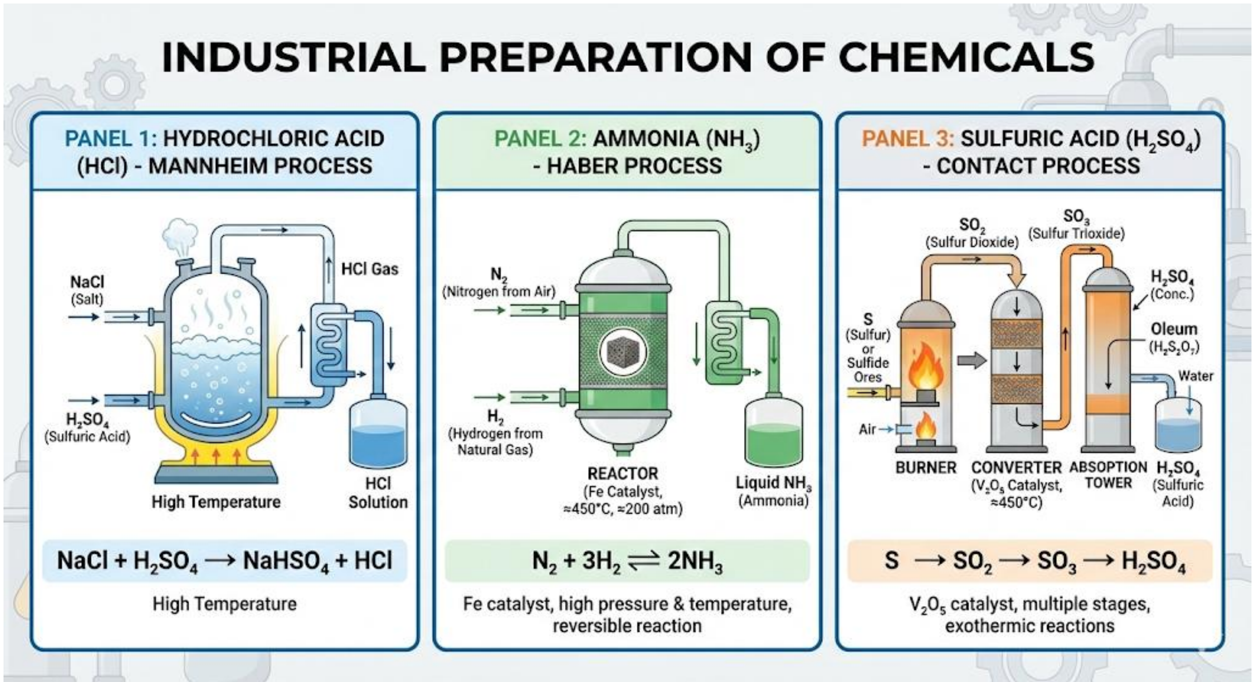
Assignment

1. Research the role of CO_2 in climate change and global warming.
2. With balanced equations, describe the laboratory preparation of methane and ethyne.

3. Compare the ionization effect of CO and CO_2 on human health.
4. State three uses of hydrocarbons in the petrochemical industry.
5. Differentiate between **structural isomerism** and **functional group isomerism** with examples.

WEEK 3: INDUSTRIAL PREPARATION OF CHEMICALS

Hydrochloric Acid (HCl), Ammonia (NH_3), and Sulphuric Acid (H_2SO_4)



INTRODUCTION

- **Industrial chemicals** such as HCl, NH_3, and H_2SO_4 are fundamental to modern chemical industries.
- They serve as **raw materials for fertilizers, dyes, pharmaceuticals, plastics, explosives, and cleaning agents**.
- The production of these chemicals is a key indicator of a country's **industrial development**.
- Knowledge of their preparation involves understanding **chemical principles, reaction conditions, apparatus design, and safety measures**.

INDUSTRIAL PREPARATION OF HYDROCHLORIC ACID (HCl)

(A) Historical Background

- Hydrochloric acid has been known since the **13th century**, originally produced by reacting **salt with vitriol (sulphuric acid)**.
- Today, it is produced industrially in **large-scale plants** as a by-product or by direct synthesis.

(B) Raw Materials

- Sodium chloride ($NaCl$)
- Concentrated sulphuric acid (H_2SO_4)

(C) Principle of Preparation

- **Mannheim Process**: Involves reaction of $NaCl$ with H_2SO_4 at elevated temperatures (200–250 °C) to produce HCl gas and sodium bisulphate ($NaHSO_4$).

  $NaCl + H_2SO_4 \rightarrow \Delta NaHSO_4 + HCl\uparrow$

- On further heating, $NaHSO_4$ reacts with more $NaCl$ to form sodium sulphate:

  $NaHSO_4 + NaCl \rightarrow 400°C\ Na_2SO_4 + HCl\uparrow$

(D) Collection

- HCl gas is **absorbed in water** to form concentrated hydrochloric acid.
- **Apparatus**: Lead chamber or absorption tower; water must be free of impurities.

(E) Properties

- Colorless, pungent gas.
- Highly soluble in water.
- Forms hydrochloric acid in solution.

(F) Uses

1. Pickling of metals (removing rust).
2. Manufacture of PVC and other chlorides.

3. Laboratory reagent in chemical analysis.
4. pH adjustment in water treatment and food industry.
5. Regeneration of ion-exchange resins.

(G) Safety and Hazards

- Corrosive to metals and skin.
- Fumes are toxic; must be handled in fume chambers with protective gear.

INDUSTRIAL PREPARATION OF AMMONIA (NH_3)

(A) Historical Background

- Discovered in **Ancient times** as a pungent gas from decomposing organic matter.
- Modern industrial preparation is via the **Haber–Bosch process**, developed in the early 20th century.

(B) Raw Materials

- Nitrogen (from air via fractional distillation).
- Hydrogen (from natural gas, water gas, or electrolysis).

(C) Principle

- Ammonia is formed by **direct combination of nitrogen and hydrogen** under high pressure and moderate temperature.

  $N_2 + 3H_2 \rightleftharpoons 2NH_3$ $\Delta H = -92$ kJ/mol

(D) Conditions

1. **Temperature:** 400–500 °C (to balance rate and yield).
2. **Pressure:** 200–250 atm (high pressure favors formation of NH_3).
3. **Catalyst:** Iron (Fe) with promoters such as potassium and aluminum oxides.

4. **Recycling:** Unreacted gases are recycled to increase yield.

(E) Collection and Purification

- Ammonia gas is **cooled under high pressure** to liquefy.
- Stored in **pressurized tanks** for industrial use.

(F) Properties

- Colorless gas with pungent odor.
- Highly soluble in water forming ammonium hydroxide:

  $NH_3 + H_2O \rightarrow NH_4OH$

- Weakly basic in aqueous solution.

(G) Uses

1. Manufacture of nitrogenous fertilizers: urea, ammonium nitrate, ammonium sulphate.
2. Production of nitric acid via **Ostwald process**.
3. Refrigerant in cooling systems.
4. Cleaning agents (ammonium hydroxide).
5. Production of explosives and pharmaceuticals.

(H) Safety Considerations

- NH_3 is **toxic and irritating**; exposure can damage eyes, lungs, and skin.
- High pressure storage requires careful monitoring.

INDUSTRIAL PREPARATION OF SULPHURIC ACID (H_2SO_4)

(A) Historical Background

- Known as **oil of vitriol** since the Middle Ages.
- Today, produced via **Contact Process**, replacing the older lead chamber process.

(B) Raw Materials

- Sulphur or sulphide ores (e.g., pyrite FeS_2).
- Oxygen (from air).
- Water (for absorption).

(C) Principle

- Oxidation of sulphur to SO_2, conversion to SO_3 using V_2O_5 catalyst, then absorption in H_2SO_4 to form oleum ($H_2S_2O_7$).
- Dilution of oleum yields concentrated sulphuric acid.

(D) Steps

1. **Burning of Sulphur:**

   $S + O_2 \rightarrow SO_2$

2. **Conversion of SO_2 to SO_3:**

   $2SO_2 + O_2 \rightleftharpoons 2SO_3 \Delta H = -197$ kJ/mol[2]

- Catalyst: Vanadium(V) oxide.
- Temperature: 450 °C, Pressure: 2 atm.

3. **Absorption of SO_3 in H_2SO_4 (Oleum formation):**

   $SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$

4. **Dilution of Oleum:**

   $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$

(E) Properties

- Dense, oily, colorless liquid (darkens on exposure to air due to impurities).
- Strongly acidic and hygroscopic.
- Powerful dehydrating and oxidizing agent.

(F) Uses

1. Manufacture of fertilizers (superphosphate, ammonium sulphate).
2. Production of detergents, dyes, and pigments.
3. Petroleum refining and processing.
4. Electrolyte in lead-acid batteries.
5. Laboratory reagent for dehydration and oxidation.

(G) Safety and Hazards

- Corrosive; causes severe burns.
- Strong oxidizer; reacts violently with organics.
- Must be stored in corrosion-resistant containers (e.g., glass, lead-lined steel).

COMPARATIVE IMPORTANCE

Chemical	Raw Material	Industrial Process	Main Use
HCl	$NaCl + H_2SO_4$	Mannheim Process	PVC, pickling steel
NH_3	$N_2 + H_2$	Haber Process	Fertilizers, nitric acid
H_2SO_4	$S + O_2 + H_2O$	Contact Process	Fertilizers, detergents, batteries

- H_2SO_4 is called **"king of chemicals"** due to its central role in industry.
- NH_3 is critical for agriculture as a **fertilizer precursor**.
- HCl is used widely in **metal processing and chemical synthesis**.

ENVIRONMENTAL AND SAFETY CONSIDERATIONS

- HCl: Acidic fumes can cause air pollution; must be scrubbed before release.
- NH_3: Toxic gas; can cause burns and respiratory problems.

- H_2SO_4: Spillage causes soil and water acidification; strong oxidizer; handled with PPE.

EVALUATION

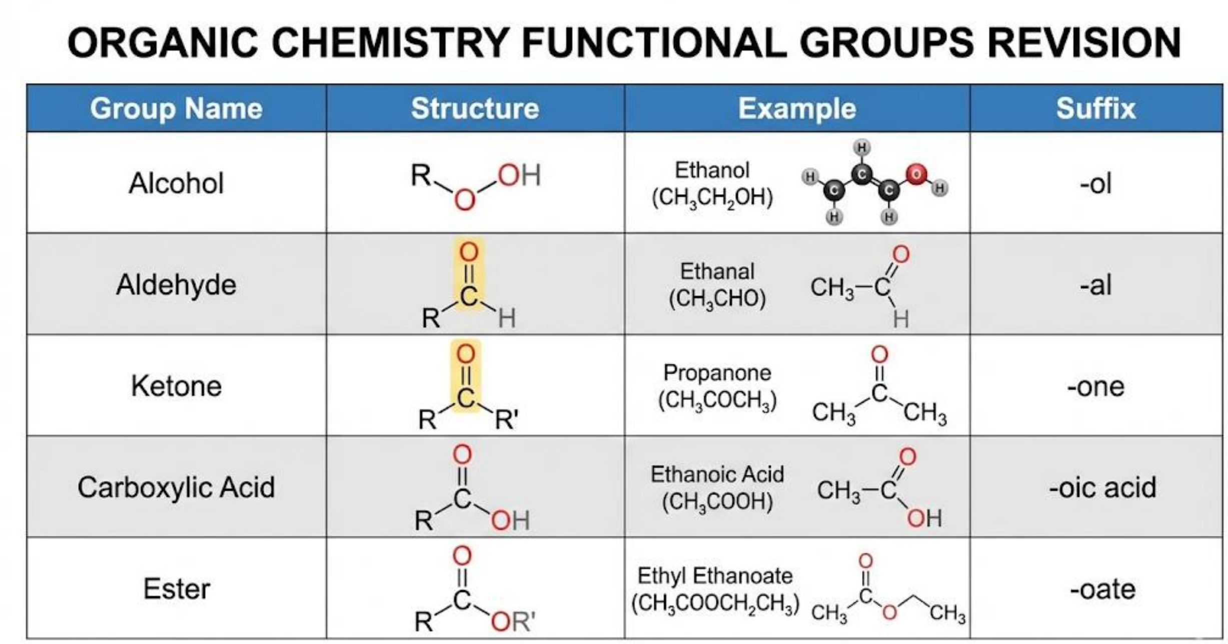
1. State the raw materials for HCl, NH_3, and H_2SO_4.
2. Write balanced chemical equations for:
   - Manufacture of HCl by Mannheim process.
   - Formation of NH_3 in Haber process.
   - Contact process for H_2SO_4.
3. Explain the role of catalyst, temperature, and pressure in the Haber and Contact processes.
4. Discuss two industrial uses for each chemical.
5. Explain why H_2SO_4 is called the "king of chemicals."

ASSIGNMENT

1. Draw a detailed **flow diagram** showing the industrial preparation of H_2SO_4 by Contact process.
2. Compare and contrast the **production conditions** of HCl, NH_3, and H_2SO_4.
3. Discuss environmental hazards and safety precautions associated with the industrial production of these chemicals.
4. Explain how recycling unreacted gases increases efficiency in the Haber process.
5. State why sulphur trioxide is absorbed in concentrated H_2SO_4 instead of water during production.

WEEK 4: ORGANIC CHEMISTRY (REVISION)

Functional Groups and IUPAC Naming



ORGANIC CHEMISTRY FUNCTIONAL GROUPS REVISION

Group Name	Structure	Example	Suffix
Alcohol	R–O–OH	Ethanol (CH_3CH_2OH)	-ol
Aldehyde	R–C(=O)–H	Ethanal (CH_3CHO) CH_3–C(=O)H	-al
Ketone	R–C(=O)–R'	Propanone (CH_3COCH_3) CH_3–C(=O)–CH_3	-one
Carboxylic Acid	R–C(=O)–OH	Ethanoic Acid (CH_3COOH) CH_3–C(=O)OH	-oic acid
Ester	R–C(=O)–OR'	Ethyl Ethanoate ($CH_3COOCH_2CH_3$) CH_3–C(=O)–O–CH_3	-oate

INTRODUCTION TO ORGANIC CHEMISTRY

- Organic Chemistry is the branch of chemistry concerned with **carbon-containing compounds** (except simple oxides of carbon like CO, CO_2, carbonates, and carbides).
- Organic compounds are **covalent in nature** and often exist as **molecules** rather than ions.
- The chemistry of carbon is unique because of:
  - **Tetravalency**: Carbon forms four covalent bonds.
  - **Catenation**: Ability to form long chains and rings with itself.
  - **Hybridization**: sp^3 (single bonds), sp^2 (double bonds), sp (triple bonds).
- Organic chemistry is essential in **petrochemicals, polymers, pharmaceuticals, cosmetics, agrochemicals, dyes, detergents, and bio-organic systems**.

SOURCES OF ORGANIC COMPOUNDS

1. **Natural Sources:**
    - **Petroleum**: Main source of hydrocarbons and petrochemicals.
    - **Coal**: Source of aromatic hydrocarbons and industrial chemicals.
    - **Natural gas**: Provides methane and other alkanes.
    - **Plants and animals**: Sources of fats, oils, carbohydrates, proteins, alkaloids, etc.
2. **Synthetic Sources:**
    - Laboratory and industrial synthesis of compounds (plastics, synthetic drugs, fertilizers, etc.).

GENERAL CHARACTERISTICS OF ORGANIC COMPOUNDS

- **Low melting and boiling points** (due to covalent bonding).
- **Poor electrical conductivity** (non-electrolytes).
- **Solubility:** Generally insoluble in water but soluble in organic solvents (benzene, ether, etc.).
- **Combustibility:** Most organic compounds are combustible.
- **Slow chemical reactions** compared to inorganic compounds.

CLASSIFICATION OF ORGANIC COMPOUNDS

Organic compounds are classified as follows:

1. **Based on Structure:**
    - **Acyclic (aliphatic):** Open-chain compounds (alkanes, alkenes, alkynes).
    - **Cyclic (alicyclic):** Ring compounds that are non-aromatic.
    - **Aromatic:** Benzene and its derivatives.

2. **Based on Saturation:**
    - **Saturated hydrocarbons:** Only single bonds (alkanes).
    - **Unsaturated hydrocarbons:** Double or triple bonds (alkenes, alkynes).
3. **Based on Functional Groups:**
    - Compounds containing characteristic reactive groups (alcohols, aldehydes, ketones, etc.).

HOMOLOGOUS SERIES

- A **homologous series** is a family of organic compounds having:
    - The same **general formula**.
    - A **common functional group**.
    - Members differ by a **CH_2 unit**.
    - Similar chemical properties but a gradual change in physical properties.

Series	**General Formula**	**Example**	**Suffix**
Alkanes	C_nH_{2n+2}	Methane (CH_4)	-ane
Alkenes	C_nH_{2n}	Ethene (C_2H_4)	-ene
Alkynes	C_nH_{2n-2}	Ethyne (C_2H_2)	-yne
Alcohols	$C_nH_{2n+1}OH$	Methanol (CH_3OH)	-ol
Carboxylic Acids	$C_nH_{2n+1}COOH$	Ethanoic acid (CH_3COOH)	-oic acid

FUNCTIONAL GROUPS IN DETAIL

A **functional group** is a specific atom or group of atoms that gives a compound its characteristic properties.

Functional Group	Structure	Compound Class	Example	Uses
Hydroxyl (–OH)	R–OH	Alcohols	Ethanol	Fuel, solvent
Carbonyl (–CHO)	R–CHO	Aldehydes	Ethanal	Perfumes
Carbonyl (>C=O)	R–CO–R	Ketones	Propanone	Solvent
Carboxyl (–COOH)	R–COOH	Carboxylic acids	Ethanoic acid	Vinegar
Ester (–COOR)	R–COOR	Esters	Methyl ethanoate	Flavorings
Amino (–NH_2)	R–NH_2	Amines	Methylamine	Fertilizers
Halogen (–X)	R–X	Haloalkanes	Chloromethane	Refrigerants
Alkene (C=C)	C=C	Alkenes	Ethene	Plastics
Alkyne (C≡C)	C≡C	Alkynes	Ethyne	Welding

RULES OF IUPAC NOMENCLATURE (STEP-BY-STEP)

1. **Find the longest carbon chain** (this is the parent hydrocarbon).
2. **Number the chain** from the end nearer to the principal functional group or substituent.
3. **Identify and name substituents** (alkyl or halogen groups).
4. **Position numbers** are assigned to show where substituents or functional groups are located.
5. **Combine names**: Prefix (substituents) + Parent name + Suffix (functional group).
6. For **multiple identical substituents**, use di-, tri-, tetra-.
7. For **double or triple bonds**, use numbers to indicate their position.
8. Functional groups take priority in numbering over double/triple bonds and substituents.

WORKED EXAMPLES

Compound	Structure	Name	Reason
$CH_3–CH_2–CH_3$	C_3H_8	Propane	3-carbon alkane
$CH_3–CH_2–OH$	C_2H_5OH	Ethanol	–OH group on C1
$CH_3–CH_2–CHO$	C_3H_6O	Propanal	Aldehyde group on C1
$CH_3–CO–CH_3$	C_3H_6O	Propanone	Ketone group on C2
$CH_3–CH=CH–CH_3$	C_4H_8	But-2-ene	Double bond at C2
$CH≡C–CH_3$	C_3H_4	Prop-1-yne	Triple bond at C1

INDUSTRIAL AND EVERYDAY IMPORTANCE OF ORGANIC COMPOUNDS

- **Petrochemicals:** Fuels (petrol, diesel), lubricants, plastics.
- **Pharmaceuticals:** Antibiotics, analgesics, antiseptics.
- **Agriculture:** Fertilizers, pesticides, herbicides.
- **Food Industry:** Preservatives, flavorings, sweeteners.
- **Daily life:** Detergents, cosmetics, synthetic fibers, paints, and dyes.

EVALUATION:

1. Define a **functional group** and explain its importance in organic chemistry.
2. Draw and name all isomers of **C_4H_{10}**.
3. Name these compounds:
   a) $CH_3–CH_2–COOH$
   b) $CH_3–CHCl–CH_3$
4. What is the **difference between aldehydes and ketones**?
5. List **four physical properties** that change gradually in a homologous series.

ASSIGNMENT:

1. Name the following systematically:

   a) $CH_3–CH_2–CH=CH–CH_3$

   b) $CH_3–C≡C–CH_3$

   c) $CH_3–CH_2–CH_2–OH$

2. Draw structures for:

   - Pent-2-ene
   - But-2-yne

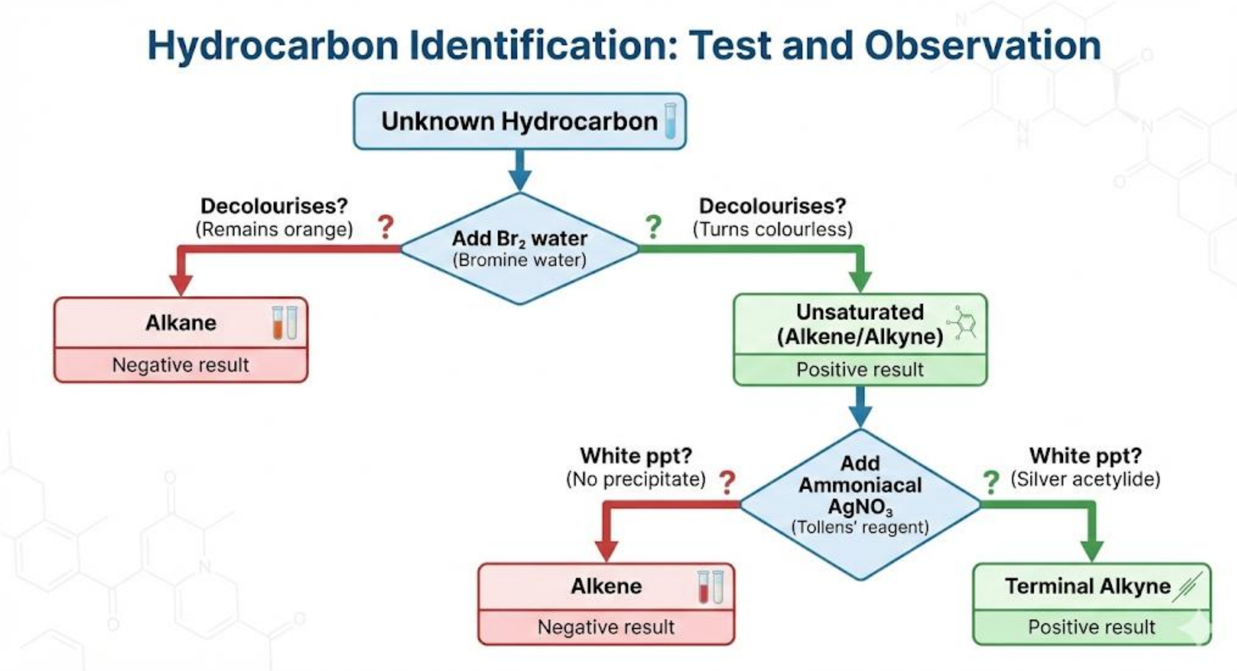
3. List **five industrial uses** of alkanes and alkenes.

4. Differentiate between **saturated and unsaturated hydrocarbons**.

5. Write the first three members of the alkyne series with names and molecular formulae.

WEEK 6: HYDROCARBONS

Properties, Tests



MEANING AND IMPORTANCE OF HYDROCARBONS

- **Definition:** Hydrocarbons are **organic compounds composed only of carbon and hydrogen atoms**.
- They form the **backbone of organic chemistry** because most other organic compounds are derived from them.
- Found in **crude oil, natural gas, coal, and plant materials**.
- Serve as **fuels, solvents, raw materials for plastics, lubricants, and explosives**.

CLASSIFICATION OF HYDROCARBONS

Hydrocarbons are broadly classified into:

1. **Alkanes (Saturated Hydrocarbons):**
    - General formula: C_nH_{2n+2}

- Only **single bonds (C–C, C–H)**
- Example: Methane (CH_4), Propane (C_3H_8).

2. **Alkenes (Unsaturated Hydrocarbons with Double Bond):**
    - General formula: C_nH_{2n}
    - Contain at least **one double bond (C=C)**
    - Example: Ethene (C_2H_4), Butene (C_4H_8).
3. **Alkynes (Unsaturated Hydrocarbons with Triple Bond):**
    - General formula: C_nH_{2n-2}
    - Contain at least **one triple bond (C≡C)**
    - Example: Ethyne (C_2H_2), Propyne (C_3H_4).

GENERAL CHARACTERISTICS OF HYDROCARBONS

- Non-polar compounds → **insoluble in water** but soluble in organic solvents (benzene, ether, chloroform).
- Highly **combustible** → major fuels.
- Lower members are **gases** (C_1–C_4), middle are **liquids** (C_5–C_{16}), higher members are **waxes/solids**.
- Exhibit **homologous series**: each successive member differs by a $–CH_2–$ group.

PHYSICAL PROPERTIES

Property	**Alkanes**	**Alkenes**	**Alkynes**
State at Room Temp	C_1–C_4 gases, C_5–C_{16} liquids, >C_{16} solids	Similar trend as alkanes	Similar trend as alkanes

Solubility	Insoluble in water, soluble in organic solvents	Same	Same
Density	Less dense than water	Less dense than water	Less dense than water
Boiling Point	Increases with molecular mass	Slightly lower than alkanes	Slightly higher than alkenes
Flame type	Blue, non-sooty	Smoky (due to higher C:H ratio)	Very smoky, luminous flame

CHEMICAL PROPERTIES OF HYDROCARBONS

(a) Combustion Reactions

- **Alkanes:**

  $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

(Complete combustion)

  $2CH_4 + 3O_2 \rightarrow 2CO + 4H_2O$

(Incomplete combustion → carbon monoxide)

- **Alkenes/Alkynes:** burn with **smoky flame** due to high carbon content.
  Example:

$$C_2H_2 + \frac{5}{2}O_2 \rightarrow 2CO_2 + H_2O$$

(b) Substitution Reactions (Alkanes Only)

- In presence of **UV light or heat**, halogens replace H in alkanes.

$$CH_4 + Cl_2 \xrightarrow{UV} CH_3Cl + HCl$$

$$CH_3Cl + Cl_2 \xrightarrow{UV} CH_2Cl_2 + HCl$$

→ leads to multiple substitution products.

(c) Addition Reactions (Alkenes and Alkynes)

- **Hydrogenation (with Ni catalyst):**

$$C_2H_4 + H_2 \xrightarrow{Ni} C_2H_6$$

- **Halogenation:**

$$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$$

(Decolourization of bromine water → test for unsaturation)

- **Hydration:**

$$C_2H_4 + H_2O \xrightarrow{H_3PO_4, 300^\circ C} C_2H_5OH$$

(Industrial ethanol production).

- **Alkynes** undergo **addition twice** due to triple bond.

$$C_2H_2 + 2Br_2 \rightarrow C_2H_2Br_4$$

(d) Oxidation Reactions

- **Combustion** already covered.
- **Mild oxidation of alkenes:** with **acidified $KMnO_4$ (Baeyer's Test)** → gives diols.

$$C_2H_4 + [O] \xrightarrow{KMnO_4} CH_2OH\text{–}CH_2OH$$

- **Strong oxidation** breaks C=C bond into carboxylic acids or ketones.

Laboratory Tests for Hydrocarbons (Step-by-Step)

1. **Bromine Water Test**
    - Add bromine water (orange) to sample hydrocarbon.
    - Shake.
    - **Observation:**
        - Alkanes: no change (unless with UV).
        - Alkenes/Alkynes: colour disappears → indicates unsaturation.
2. **Baeyer's Test (Acidified $KMnO_4$ Solution)**
    - Add purple $KMnO_4$ solution to hydrocarbon.
    - **Observation:**
        - Alkanes: no change.
        - Alkenes/Alkynes: solution decolourizes, brown MnO_2 precipitate forms.
3. **Ignition Test**
    - Burn the hydrocarbon in a Bunsen flame.
    - **Observation:**
        - Alkanes: blue, clean flame.
        - Alkenes: smoky flame.
        - Alkynes: very sooty, luminous flame.
4. **Ammoniacal Silver Nitrate Test (for terminal alkynes)**
    - Add sample hydrocarbon to ammoniacal $AgNO_3$ solution.
    - **Observation:** White precipitate of **silver acetylide** forms.

$$HC \equiv CH + AgNO_3 \rightarrow AgC \equiv CAg(ppt) + HNO_3$$

Comparison Table of Hydrocarbons

Feature	Alkanes	Alkenes	Alkynes
Bond type	Single (C–C)	Double (C=C)	Triple (C≡C)
Formula	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}
Reactivity	Least reactive	Reactive	Most reactive
Flame type	Clean, non-sooty	Smoky	Very smoky, luminous
Example	CH_4 (Methane)	C_2H_4 (Ethene)	C_2H_2 (Ethyne)
Characteristic Test	Substitution with halogen (UV)	Decolourizes Br_2, $KMnO_4$	Same as alkene + $AgNO_3$ ppt for terminal

Industrial and Domestic Uses of Hydrocarbons

- **Alkanes:** cooking gas (butane, propane), petrol (octane), lubricants, wax.
- **Alkenes:** production of plastics (polyethene, PVC), alcohols (ethanol), synthetic rubber.
- **Alkynes:** oxy-acetylene welding, artificial rubber, synthesis of plastics and solvents.

Environmental Effects of Hydrocarbons

- **Air Pollution:** Incomplete combustion produces carbon monoxide and soot.
- **Greenhouse Effect:** CO_2 emissions from burning fuels → global warming.
- **Oil Spills:** Hydrocarbons pollute water, harming aquatic life.
- **Smog Formation:** Unburnt hydrocarbons + NOx gases produce photochemical smog.

Evaluation

1. Define hydrocarbons and classify them with examples.
2. State two physical properties each of alkanes, alkenes, and alkynes.

3. Differentiate between substitution and addition reactions with balanced equations.

4. Write chemical equations for the following:

   a. Combustion of propane.

   b. Addition of hydrogen bromide to ethene.

   c. Reaction of ethyne with ammoniacal silver nitrate.

5. Explain two laboratory tests that distinguish between alkanes, alkenes, and alkynes.

6. List three environmental problems caused by hydrocarbons.

Assignment

1. Fill in the blanks in the table below:

Hydrocarbon	**Formula**	**Type of Reaction**	**Equation**	**Observation**
Methane		Substitution		
Ethene		Addition		
Ethyne		Silver Nitrate Test		

2. Why do alkenes decolourize bromine water, while alkanes do not?

3. Write short notes on three uses of hydrocarbons in:

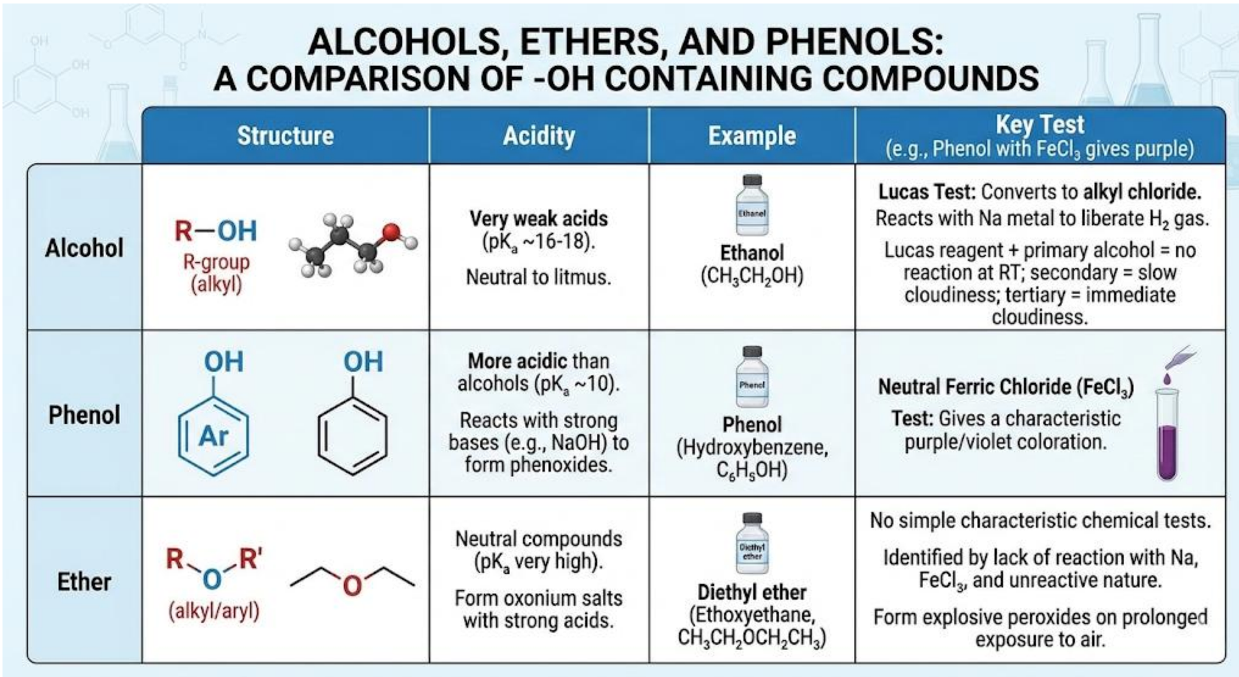
   a. Industry

   b. Household applications

   c. Energy production

WEEK 8: ALCOHOLS, ETHERS, AND PHENOLS

Uses and Reactions



ALCOHOLS, ETHERS, AND PHENOLS: A COMPARISON OF -OH CONTAINING COMPOUNDS

	Structure	Acidity	Example	Key Test (e.g., Phenol with $FeCl_3$ gives purple)
Alcohol	R–OH R-group (alkyl)	**Very weak acids** (pK_a ~16-18). Neutral to litmus.	**Ethanol** (CH_3CH_2OH)	**Lucas Test:** Converts to **alkyl chloride.** Reacts with Na metal to liberate H_2 gas. Lucas reagent + primary alcohol = no reaction at RT; secondary = slow cloudiness; tertiary = immediate cloudiness.
Phenol	OH Ar; OH	**More acidic** than alcohols (pK_a ~10). Reacts with strong bases (e.g., NaOH) to form phenoxides.	**Phenol** (Hydroxybenzene, C_6H_5OH)	**Neutral Ferric Chloride ($FeCl_3$) Test:** Gives a characteristic purple/violet coloration.
Ether	R–O–R' (alkyl/aryl)	Neutral compounds (pK_a very high). Form oxonium salts with strong acids.	**Diethyl ether** (Ethoxyethane, $CH_3CH_2OCH_2CH_3$)	No simple characteristic chemical tests. Identified by lack of reaction with Na, $FeCl_3$, and unreactive nature. Form explosive peroxides on prolonged exposure to air.

Introduction

Organic compounds containing oxygen play an essential role in industrial, pharmaceutical, domestic, and laboratory applications. Three of the most significant oxygenated organic compounds are **alcohols, ethers, and phenols**. Despite all containing oxygen, they differ in their **functional groups, physical properties, chemical reactivity, and industrial importance**. A clear understanding of their properties and reactions forms a foundation for advanced studies in organic chemistry and industrial processes.

ALCOHOLS

Definition and Structure

Alcohols are **hydroxy derivatives of alkanes** in which one or more hydrogen atoms are replaced by hydroxyl (–OH) groups.

General formula: $C_nH_{2n+1}OH$ (for aliphatic alcohols).

Functional group: Hydroxyl group (–OH).

Classification of Alcohols

1. **Based on number of –OH groups**:
    - **Monohydric alcohols:** contain one –OH group. Example: Ethanol, CH_3CH_2OH
    - **Dihydric alcohols:** contain two –OH groups. Example: Ethane-1,2-diol (glycol).
    - **Trihydric alcohols:** contain three –OH groups. Example: Glycerol (propane-1,2,3-triol).
    - **Polyhydric alcohols:** contain more than three –OH groups. Example: Sugars like glucose.
2. **Based on position of –OH group (Primary, Secondary, Tertiary):**
    - **Primary (1°):** –OH attached to a carbon bonded to one other carbon.
      Example: CH_3CH_2OH.
    - **Secondary (2°):** –OH attached to a carbon bonded to two other carbons.
      Example: $CH_3CHOHCH_3$.
    - **Tertiary (3°):** –OH attached to a carbon bonded to three other carbons.
      Example: 2-methylpropan-2-ol.

Physical Properties of Alcohols

- **Boiling point:** Higher than alkanes due to hydrogen bonding.
- **Solubility:** Lower alcohols (methanol, ethanol, propanol) are soluble in water due to hydrogen bonding. Solubility decreases with increase in molecular weight.
- **Odour:** Characteristic sweet odour (especially ethanol).

- **Conductivity:** Poor conductors of electricity because they do not ionize significantly.

Preparation of Alcohols

1. **Fermentation of carbohydrates (ethanol):**
   - Sugars like glucose undergo fermentation in the presence of enzymes (zymase from yeast).

$$C_2H_4 + H_2O \xrightarrow{H_3PO_4} C_2H_5OH$$

2. **Hydration of alkenes:**
   - Ethene reacts with steam in the presence of concentrated phosphoric acid catalyst at high temperature and pressure to give ethanol.

$$C_2H_4 + H_2O \xrightarrow{H_3PO_4} C_2H_5OH$$

3. **Reduction of aldehydes and ketones:**
   - Aldehyde + H_2 → Primary alcohol.
   - Ketone + H_2 → Secondary alcohol.

Chemical Reactions of Alcohols

1. **Combustion:**
   Alcohols burn in oxygen with a clean flame.

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$$

2. **Reaction with Sodium:**

$$2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$$

3. **Oxidation:**
    - Primary alcohol → Aldehyde → Carboxylic acid.
    - Secondary alcohol → Ketone.
    - Tertiary alcohols are resistant.
4. **Esterification:**
   Reaction with carboxylic acid in the presence of concentrated H_2SO_4.

$$C_2H_5OH + CH_3COOH \xrightarrow{H_2SO_4} CH_3COOC_2H_5 + H_2O$$

Uses of Alcohols

- Ethanol: Alcoholic beverages, solvent, antiseptic, motor fuel, manufacture of esters, perfumes, and medicines.
- Methanol: Manufacture of formaldehyde, solvent, antifreeze, alternative fuel.
- Glycerol: Used in cosmetics, explosives (nitroglycerin), and food industry.
- Ethylene glycol: Used in antifreeze and polyester production.

ETHERS

Definition and Structure

Ethers are organic compounds in which two alkyl or aryl groups are bonded to the same oxygen atom.
General formula: R–O–R.

Examples:

- Symmetrical ether: Diethyl ether $(C_2H_5–O–C_2H_5)$.
- Unsymmetrical ether: Methyl ethyl ether $(CH_3–O–C_2H_5)$.

Physical Properties of Ethers

- Volatile and highly flammable.
- Slightly soluble in water (less hydrogen bonding).
- Pleasant smell but dangerous due to anaesthetic effects.
- Lower boiling points compared to alcohols of similar molecular weight.

Preparation of Ethers

1. **Dehydration of alcohols:**

$$2C_2H_5OH \xrightarrow{H_2SO_4, 140^\circ C} C_2H_5–O–C_2H_5 + H_2O$$

2. **Williamson Ether Synthesis:**

$$C_2H_5ONa + CH_3I \rightarrow CH_3–O–C_2H_5 + NaI$$

Reaction of sodium alkoxide with alkyl halide.

Chemical Reactions of Ethers

1. **Combustion:** Produces CO_2 and H_2O.
2. **Reaction with acids (HI, HBr):**

$$C_2H_5–O–C_2H_5 + HI \rightarrow C_2H_5I + C_2H_5OH$$

3. **Auto-oxidation:** In air, ethers form unstable and explosive peroxides.

Uses of Ethers

- Diethyl ether: Anaesthetic (though largely replaced due to flammability).
- Solvents in laboratories and industries.
- Starting materials in synthesis of plastics, perfumes, and drugs.

PHENOLS

Definition and Structure

Phenols are aromatic compounds in which a hydroxyl group (–OH) is directly bonded to a benzene ring.
Example: Phenol, C_6H_5OH.

Physical Properties of Phenols

- Solid with characteristic smell.
- Soluble in water due to hydrogen bonding.
- Weakly acidic.

Preparation of Phenols

1. From **benzene sulfonic acid** by fusion with sodium hydroxide.

$$C_6H_5SO_3Na + 2NaOH \xrightarrow{\Delta} C_6H_5ONa + Na_2SO_3$$

Acidification gives phenol.

2. From **chlorobenzene** by hydrolysis with NaOH at high temperature and pressure.

Chemical Reactions of Phenols

1. **Acidic Reaction with Sodium:**

$$C_6H_5OH + Na \rightarrow C_6H_5ONa + \frac{1}{2}H_2$$

2. **Reaction with NaOH:**

$$C_6H_5OH + NaOH \rightarrow C_6H_5ONa + H_2O$$

3. **Electrophilic Substitution in Benzene Ring:**
   - **Nitration:** Phenol + dilute HNO_3 → 2-nitrophenol + 4-nitrophenol.
   - **Halogenation:** Phenol + Br_2 water → 2,4,6-tribromophenol (white precipitate).
4. **Test with Iron (III) Chloride:** Gives violet coloration, a confirmatory test for phenols.

Uses of Phenols

- Used as antiseptics and disinfectants.
- Starting material for aspirin and other drugs.
- Manufacture of plastics, dyes, Bakelite, and explosives.

Comparison of Alcohols, Ethers, and Phenols

Property	Alcohols	Ethers	Phenols
Functional group	–OH on alkyl	–O– between alkyl groups	–OH on benzene ring
Acidity	Neutral/weak	Neutral	Weakly acidic
Solubility	Soluble (low MW)	Slightly soluble	Soluble
Boiling point	High (H-bonding)	Lower (no H-bonding)	High due to H-bonding

Typical reaction	Oxidation, esterification	Cleavage by HI/auto-oxid.	Electrophilic substitution, $FeCl_3$ test

Evaluation

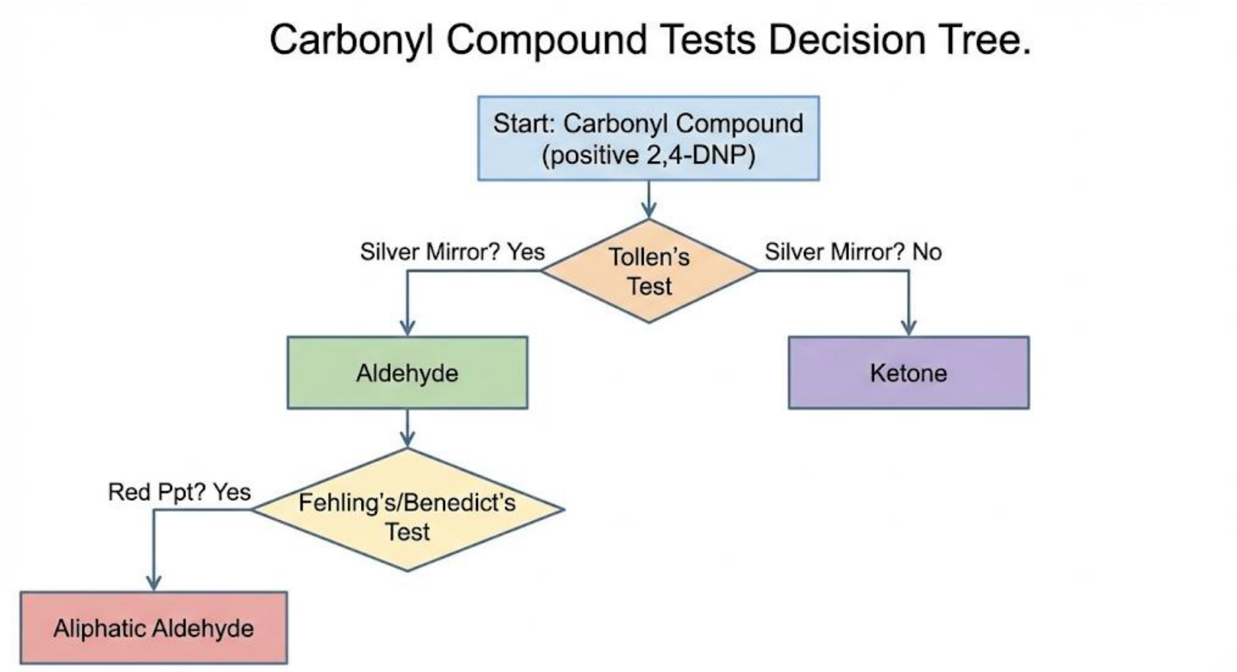
1. Differentiate between alcohols, ethers, and phenols, giving one example each.
2. State three laboratory tests that can distinguish alcohols from phenols.
3. Write equations for:
   a) Combustion of ethanol.
   b) Esterification of ethanol with ethanoic acid.
   c) Reaction of phenol with bromine water.
4. Explain why phenol is more acidic than ethanol.
5. Why are ethers more volatile than alcohols of similar molecular mass?

Assignment

1. Discuss the **industrial preparation of ethanol by fermentation** and **manufacture of phenol from benzene**.
2. State **three major uses each** of ethanol, diethyl ether, and phenol in everyday life.
3. Write balanced equations for the laboratory preparation of:
   a) Diethyl ether from ethanol.
   b) Phenol from sodium benzene sulfonate.
4. Compare the reactions of **ethanol and phenol** with sodium hydroxide solution.
5. Explain the difference between **primary, secondary, and tertiary alcohols** with clear examples.

WEEK 9: ALDEHYDES AND KETONES

Preparation and Identification



1. Introduction to Aldehydes and Ketones

- **Carbonyl Compounds:** Aldehydes and ketones are two important homologous series of **carbonyl compounds**, containing the **carbonyl functional group (C=O)**.
- The **carbonyl carbon** is **sp^2 hybridized** and planar. The oxygen atom is more electronegative than carbon, making the **C=O group highly polar**.
- This polarity makes the carbon atom **electrophilic** (electron-deficient) and susceptible to attack by nucleophiles, while the oxygen atom is **nucleophilic** (electron-rich).

General Formula

- **Aldehydes:** R–CHO (carbonyl group attached to at least one hydrogen atom).
- **Ketones:** R–CO–R′ (carbonyl group bonded to two alkyl or aryl groups).

Structures

Aldehyde:

O

||

R–C–H

- Ketone:

O

||

R–C–R′

Examples

- Aldehydes:
    - Methanal ($HCHO$)
    - Ethanal (CH_3CHO)
    - Benzaldehyde (C_6H_5CHO)
- Ketones:
    - Propanone (CH_3COCH_3)
    - Butanone ($CH_3COC_2H_5$)
    - Phenylacetone ($C_6H_5CH_2COCH_3$)

2. Preparation of Aldehydes and Ketones

A. Preparation of Aldehydes

1. **Oxidation of Primary Alcohols**
    - Primary alcohols are oxidized under controlled conditions (mild oxidizing agents).

- Reagents: **PCC (Pyridinium chlorochromate)** or **acidified $K_2Cr_2O_7$ under distillation**.
- Example:

$$CH_3CH_2OH \xrightarrow{[O]} CH_3CHO + H_2O$$

2. **Controlled Oxidation of Methane**
   - Industrial production of methanal (formaldehyde) by catalytic oxidation of methane in the presence of molybdenum or copper catalysts.
   - Reaction:

$$CH_4 + O_2 \xrightarrow{Mo/Cu, 400^\circ C} HCHO + H_2O$$

3. **Hydroformylation (Oxo process)**
   - Alkenes react with carbon monoxide and hydrogen (syngas) in the presence of a cobalt or rhodium catalyst.
   - Example:

$$CH_2 = CH_2 + CO + H_2 \longrightarrow CH_3CH_2CHO$$

4. **Partial Oxidation of Methanol (Industrial)**
   - Oxidation of methanol with air using a silver catalyst produces methanal:

$$CH_3CH(OH)CH_3 \xrightarrow{[O]} CH_3COCH_3 + H_2O$$

B. Preparation of Ketones

1. **Oxidation of Secondary Alcohols**
   - Secondary alcohols are oxidized to ketones using acidified potassium dichromate or permanganate.

- Example:

$$CH_3CH(OH)CH_3 \xrightarrow{[O]} CH_3COCH_3 + H_2O$$

2. **Friedel-Crafts Acylation (Industrial)**
   - Aromatic ketones are prepared by reacting benzene with an acyl chloride in the presence of **$AlCl_3$ catalyst**.
   - Example:

$$C_6H_6 + CH_3COCl \xrightarrow{AlCl_3} C_6H_5COCH_3 + HCl$$

3. **Dry Distillation of Calcium Salts**
   - Calcium salts of carboxylic acids when heated produce ketones.
   - Example:

$$C_6H_6 + CH_3COCl \xrightarrow{AlCl_3} C_6H_5COCH_3 + HCl$$

4. **Hydration of Alkynes (Markovnikov's rule)**
   - Addition of water to alkynes in the presence of dilute H_2SO_4 and Hg^{2+} catalyst yields ketones.
   - Example:

$$(CH_3COO)_2Ca \xrightarrow{\Delta} CH_3COCH_3 + CaCO_3$$

3. Identification of Aldehydes and Ketones

Because both aldehydes and ketones contain the **C=O group**, they respond to general tests for carbonyl compounds, but specific reagents can distinguish them.

(a) General Test for Carbonyl Group

- **2,4-Dinitrophenylhydrazine (2,4-DNP) Test**

- Both aldehydes and ketones react with 2,4-DNP to form a **yellow/orange crystalline precipitate** of hydrazone derivatives.
- Confirms the presence of a carbonyl group.

(b) Distinguishing Tests

1. **Tollen's Test (Ammoniacal Silver Nitrate)**
    - Aldehydes reduce Tollen's reagent to metallic silver, producing a **silver mirror**.
    - Reaction:

$$RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \longrightarrow RCOO^- + 2Ag(s) + 4NH_3 + 2H_2O$$

    - Ketones do not react.
2. **Fehling's Test (Fehling's Solution A + B)**
    - Aldehydes reduce the deep-blue Cu^{2+} complex to a **brick-red precipitate** of Cu_2O.
    - Ketones do not react (except α-hydroxy ketones).
3. **Benedict's Test**
    - Similar to Fehling's test, aldehydes reduce Benedict's solution to a red precipitate of Cu_2O.
4. **Schiff's Test**
    - Aldehydes restore the pink/magenta colour of Schiff's reagent.
    - Ketones do not react.

Mechanism of Oxidation of Alcohols

- **Primary alcohol → Aldehyde → Carboxylic acid**
    - With mild oxidation → aldehyde

- With strong oxidation → carboxylic acid

- **Secondary alcohol → Ketone**
  - With further oxidation, ketones generally resist oxidation, except under strong conditions, when they break into smaller acids.

Comparison of Aldehydes and Ketones

Feature	Aldehydes (R–CHO)	Ketones (R–CO–R′)
Position of carbonyl group	Terminal (end of chain)	Internal (middle of chain)
Reactivity	More reactive (due to –H attached to carbonyl carbon)	Less reactive
Oxidation	Easily oxidized to carboxylic acids	Resistant to oxidation
Tollen's/Fehling's test	Positive (silver mirror/brick red ppt)	Negative
Examples	Ethanal, Benzaldehyde	Propanone, Butanone

Uses of Aldehydes and Ketones

- **Aldehydes**
  - Formaldehyde: manufacture of plastics, resins, disinfectants, preservatives.
  - Benzaldehyde: flavouring agent in food and perfumes.
  - Ethanal: raw material for acetic acid production.
- **Ketones**
  - Acetone (propanone): solvent for varnishes, paints, plastics, pharmaceuticals.
  - Used in nail polish removers.

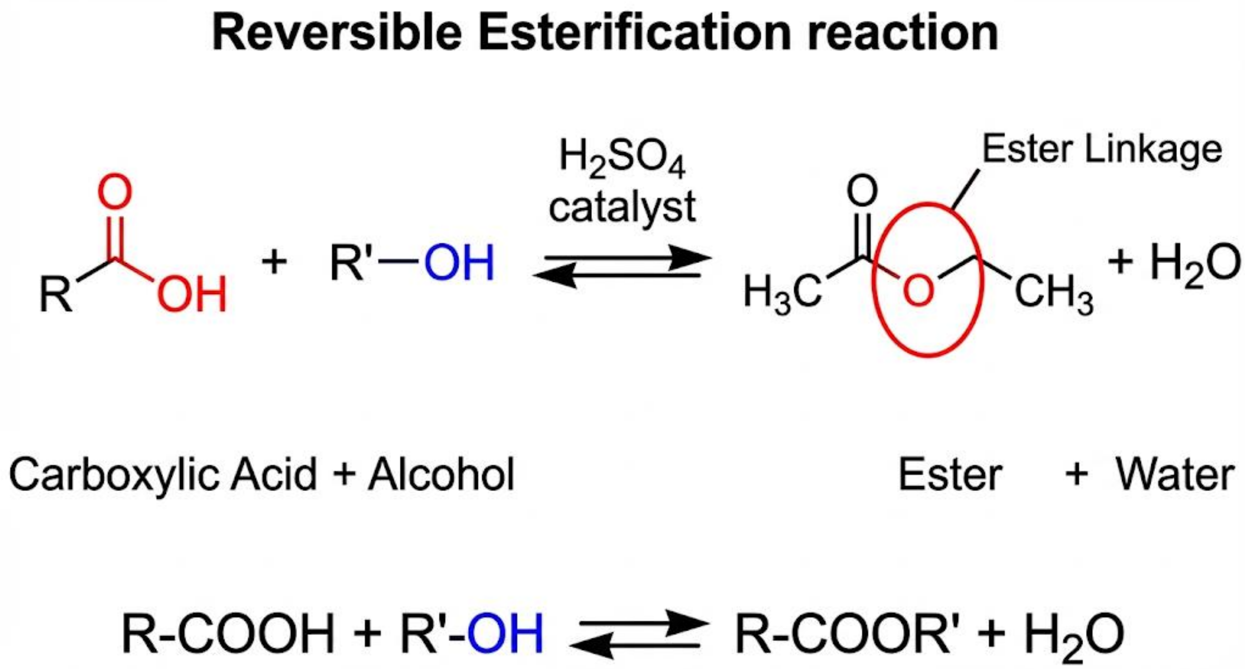
- Intermediate in synthesis of organic compounds.

Evaluation

1. Write the general formula and structure of aldehydes and ketones.
2. Give three methods of preparing aldehydes and three methods of preparing ketones with balanced equations.
3. Outline the principle of the 2,4-DNP test.
4. Explain, with equations, why aldehydes give a positive Tollen's test but ketones do not.
5. List four uses of aldehydes and ketones in industry and everyday life.

Assignment

1. Write balanced equations to show the conversion of:
   - (a) Ethanol to Ethanal
   - (b) Propan-2-ol to Propanone
   - (c) Calcium acetate to Acetone
2. Distinguish between ethanal and propanone using:
   - (a) Fehling's test
   - (b) Schiff's test
3. Draw a table comparing aldehydes and ketones under: Structure, Reactivity, Oxidation, and Uses.
4. State the industrial importance of formaldehyde and acetone.
5. Explain the **hydroformylation reaction**, giving an equation for the preparation of an aldehyde.



CARBOXYLIC ACIDS

Definition

Carboxylic acids are **organic compounds** that contain the **carboxyl group (–COOH)** as their functional group.
General formula:

$$R\text{–}COOH$$

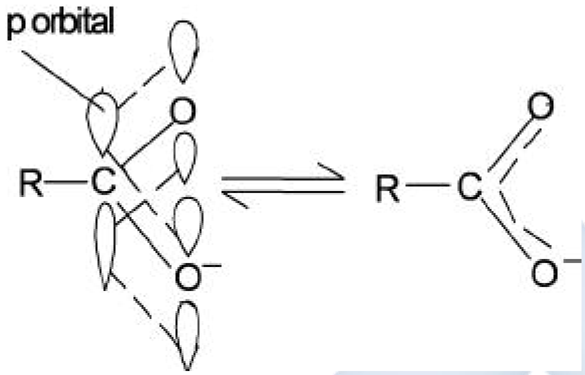
where **R** is an alkyl or aryl group.

Examples:

- Methanoic acid ($HCOOH$)
- Ethanoic acid (CH_3COOH)
- Benzoic acid (C_6H_5COOH)

Structure of the Carboxyl Group

- Consists of a **carbonyl group (C=O)** and a **hydroxyl group (–OH)** attached to the same carbon atom.
- Exhibits **resonance stabilization**, making the carboxyl group less reactive towards electrophiles but strongly acidic compared to alcohols.



(Resonance structures of –COOH showing delocalization of electrons between C=O and C–O bonds.)

Nomenclature

- **IUPAC naming:** Replace the final –e of the alkane with –oic acid.
  - CH_3COOH → ethanoic acid
  - HCOOH → methanoic acid
- **Common names:** Derived from sources (e.g., formic acid from ants, acetic acid from vinegar).

Physical Properties

1. **State:** Lower members (HCOOH, CH_3COOH) are colourless liquids with sharp smell; higher members are solids.
2. **Boiling Point:** Higher than alcohols due to **hydrogen bonding** between acid molecules (dimer formation).

3. **Solubility:** Lower members are soluble in water because of hydrogen bonding; solubility decreases as chain length increases.

4. **Taste:** Sour and corrosive in concentrated form.

Laboratory Preparation

1. **Oxidation of Primary Alcohols**:

$$CH_3CH_2OH + [O] \xrightarrow{K_2Cr_2O_7/H^+} CH_3COOH + H_2O$$

2. **Oxidation of Aldehydes:**

$$CH_3CHO + [O] \rightarrow CH_3COOH$$

Chemical Properties of Carboxylic Acids

1. **Acidic Character:**

    - Ionization in water:

    $$RCOOH \rightleftharpoons RCOO^- + H^+$$

    - Turns blue litmus paper red.
    - Stronger acids than alcohols because the carboxylate ion ($RCOO^-$) is stabilized by resonance.

2. **Reaction with Bases:**

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

3. **Reaction with Carbonates:**

$$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2$$

4. **Reaction with Active Metals:**

$$2CH_3COOH + 2Na \rightarrow 2CH_3COONa + H_2$$

5. **Esterification:**
   With alcohols in the presence of concentrated H_2SO_4:

$$CH_3COOH + CH_3OH \xrightarrow{H_2SO_4} CH_3COOCH_3 + H_2O$$

6. **Reduction:** Carboxylic acids can be reduced to primary alcohols using $LiAlH_4$.

Industrial Uses of Carboxylic Acids

- Ethanoic acid (vinegar) → food preservation, solvent, ester manufacture.
- Formic acid → leather tanning, textile dyeing, silage preservation.
- Benzoic acid → preservative in soft drinks and fruit juices.
- Dicarboxylic acids → raw materials for polymers (nylon, polyesters).

ESTERS

Definition

Esters are **organic compounds** derived when the hydrogen atom of the –COOH group in carboxylic acids is replaced with an alkyl group (–R').

General formula:

R–COOR'

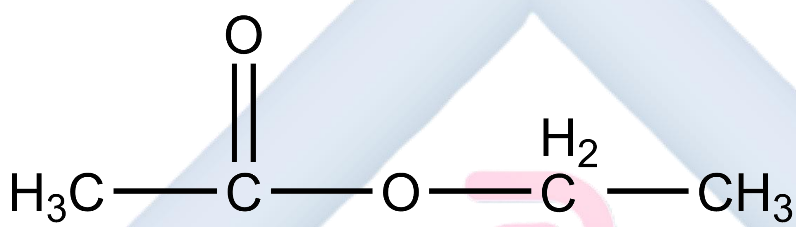
Examples:

- Methyl ethanoate (CH_3COOCH_3)

- Ethyl ethanoate ($CH_3COOCH_2CH_3$)

Structure and Nomenclature

- Formed by combining the **alkyl group from alcohol** and the **acid part from carboxylic acid**.
- Named as **alkyl alkanoate**.
    - CH_3COOCH_3 → methyl ethanoate
    - $C_2H_5COOCH_3$ → methyl propanoate



Structure of ethyl ethanoate with labelled acid and alcohol portions.

Physical Properties of Esters

1. Pleasant fruity smell → used in perfumes and flavourings.
2. Lower esters are liquids; higher esters are solids (waxy).
3. Insoluble in water (cannot form hydrogen bonds easily).
4. Low boiling points compared to acids of similar molar mass (absence of H-bonding between molecules).

Chemical Properties of Esters

1. **Hydrolysis (Reversible):** In acidic medium → alcohol + acid.

$$CH_3COOCH_3 + H_2O \xrightarrow{H^+} CH_3COOH + CH_3OH$$

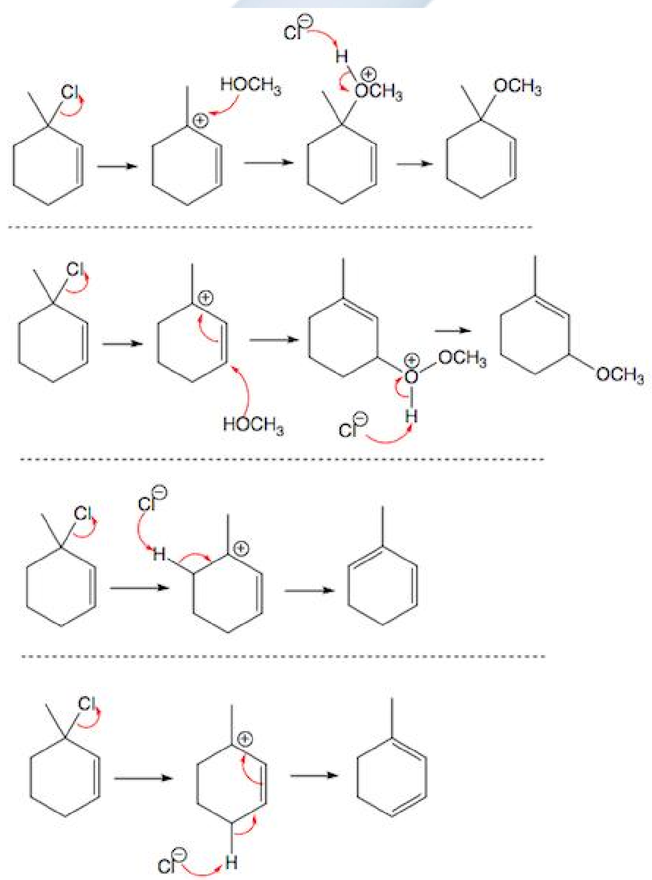
2. **Saponification (Irreversible):** In alkaline medium → salt + alcohol.

$$CH_3COOCH_3 + NaOH \rightarrow CH_3COONa + CH_3OH$$

3. **Combustion:** Burn with clean flame → CO_2 + H_2O.

Mechanism of Esterification (simplified)

1. Protonation of the carbonyl oxygen of the carboxylic acid.
2. Nucleophilic attack by alcohol on the carbonyl carbon.
3. Proton transfer and elimination of water.
4. Formation of ester.



Stepwise reaction mechanism showing intermediate structures

Industrial Uses of Esters

- Perfumes, flavourings (e.g., banana flavour: isoamyl acetate).
- Solvents for paints, varnishes (ethyl ethanoate).
- Manufacture of plastics and synthetic fibres.
- Soap making (hydrolysis of fats/oils → glycerol + fatty acid salts).
- Plasticizers (phthalate esters in PVC).

Relationship between Carboxylic Acids and Esters

- Carboxylic acids + alcohols ⇌ esters + water (esterification).
- Esters + water ⇌ acids + alcohols (hydrolysis).
- This **reversible interconversion** is key in organic synthesis.

Comparative Table

Property	**Carboxylic Acids (R–COOH)**	**Esters (R–COOR')**
Functional group	–COOH	–COO–
Smell	Pungent, sour	Pleasant, fruity
Solubility in water	Soluble (H-bonding)	Insoluble
Boiling point	High (H-bonding)	Low (no intermolecular H-bonds)
Reaction with base	Neutralized to form salts	Hydrolysed to acids + alcohols
Industrial importance	Preservatives, food, solvents	Perfumes, plastics, solvents

Evaluation

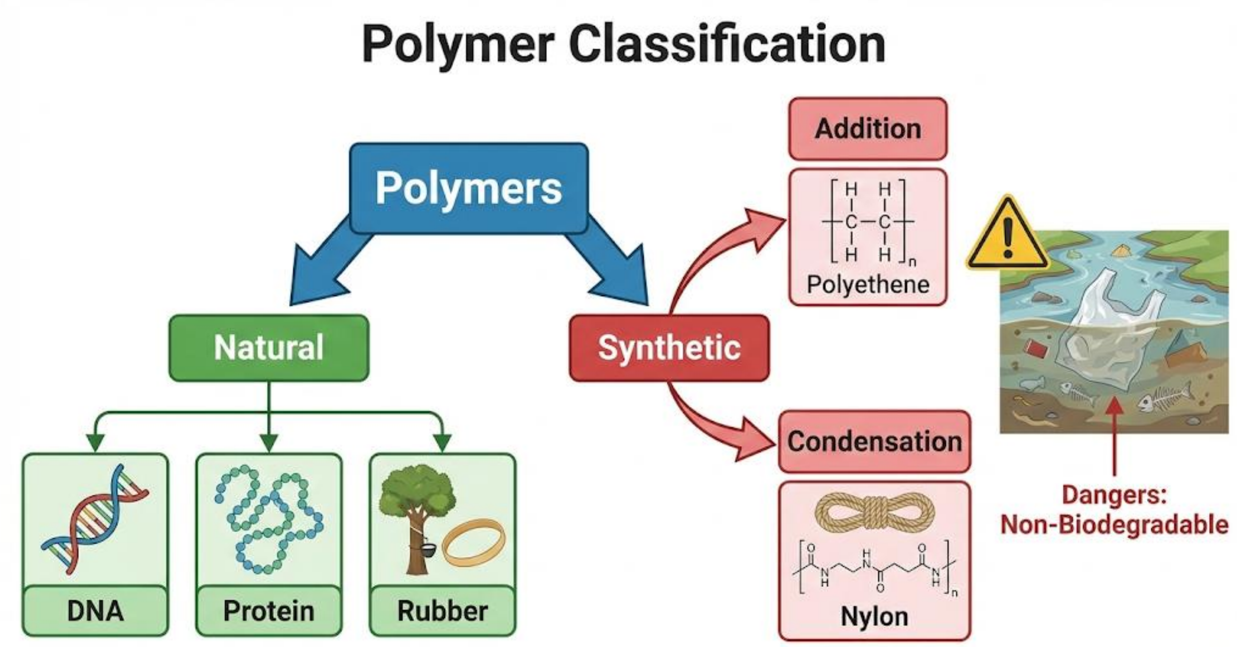
1. Define a carboxylic acid and give the general formula.
2. Explain why carboxylic acids have higher boiling points than esters of similar molar mass.
3. Write the structural formula of ethyl propanoate and state the alcohol and acid from which it is derived.
4. State two laboratory preparations of carboxylic acids with equations.
5. Describe the mechanism of esterification between ethanol and ethanoic acid.

Assignment

1. Write balanced equations for the reactions of ethanoic acid with:
   (a) sodium hydroxide,
   (b) sodium carbonate,
   (c) ethanol in the presence of conc. H_2SO_4.
2. Write the IUPAC names and structures of the esters formed from:
   (i) propanoic acid + ethanol,
   (ii) methanoic acid + propanol,
   (iii) benzoic acid + methanol.
3. Differentiate between **acid hydrolysis** and **alkaline hydrolysis** of esters with equations.
4. Discuss three industrial uses of esters and three of carboxylic acids, giving reasons for their importance.

WEEK 11: POLYMERS AND PLASTICS

Types, uses, dangers



Meaning of Polymers

- The word **polymer** is derived from Greek: *poly* = many, *meros* = parts.
- A polymer is a **macromolecule** made up of repeating structural units (monomers) linked together by **covalent bonds**.
- The **degree of polymerization (DP)** is the number of repeating units in a polymer chain.
- **Molar mass** of polymers is usually very high (up to millions).

Examples of Monomer → Polymer:

- Ethene → Polyethene
- Glucose → Starch/Cellulose
- Isoprene → Natural Rubber
- Amino acids → Proteins

Types of Polymers

Polymers are classified in different ways:

A. Based on Origin

1. **Natural Polymers** (biopolymers):
   - Found in living organisms.
   - Examples:
     - **Carbohydrates:** Starch, cellulose, glycogen (polyglucose).
     - **Proteins:** Chains of amino acids (enzymes, hemoglobin, keratin).
     - **Nucleic acids:** DNA and RNA (polymers of nucleotides).
     - **Natural rubber:** Polyisoprene (from latex of rubber tree).
2. **Synthetic Polymers** (man-made):
   - Prepared in industries.
   - Examples:
     - Polyethene, polystyrene, PVC, nylon, terylene, bakelite, silicone rubbers.

B. Based on Method of Polymerization

1. **Addition Polymers**
   - Formed when **unsaturated monomers (C=C double bonds)** combine without elimination of any by-product.
   - Requires heat, pressure, and catalyst.
   - Examples:
     - **Polyethene:**

$$nCH_2 = CH_2 \xrightarrow{catalyst} -(CH_2 - CH_2)-_n$$

- **Polystyrene:** from styrene ($C_6H_5CH=CH_2$).
- **Teflon (PTFE):** from tetrafluoroethene ($CF_2=CF_2$).

2. **Condensation Polymers**
   - Formed when **monomers with two functional groups** combine with elimination of small molecules like H_2O, HCl, NH_3.
   - Examples:
     - **Nylon-6,6:** hexamethylenediamine + adipic acid.
     - **Terylene (Dacron):** ethane-1,2-diol + terephthalic acid.
     - **Bakelite:** phenol + formaldehyde.

C. Based on Thermal Behavior

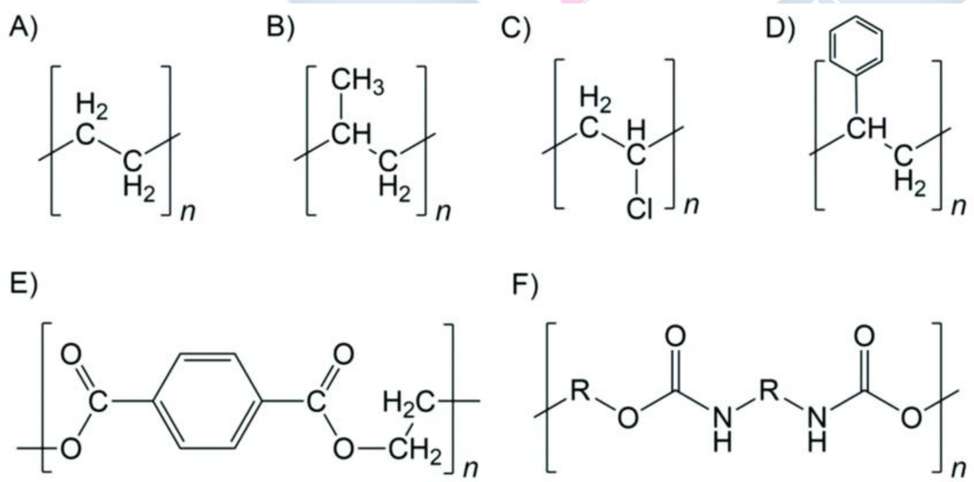
1. **Thermoplastics**
   - Soften on heating and can be remolded.
   - Consist of linear or branched chains without cross-links.
   - Examples: Polyethene, PVC, polystyrene.
2. **Thermosetting Plastics**
   - Once set, cannot be remolded; undergo permanent hardening.
   - Have extensive cross-linking between chains.
   - Examples: Bakelite, melamine, epoxy resins.

D. Based on Mechanical Properties

1. **Elastomers:** stretchable, regain shape (natural rubber, neoprene).
2. **Fibres:** thread-like polymers with high tensile strength (nylon, terylene, silk).
3. **Plastics:** rigid or flexible solids (polyethene, PVC, polystyrene).

Structures and Examples

1. **Polyethene (Polythene):** $–(CH_2–CH_2)–_n$
2. **Polyvinyl chloride (PVC):** $–(CH_2–CHCl)–_n$
3. **Polystyrene:** $–(CH_2–CHC_6H_5)–_n$
4. **Teflon (PTFE):** $–(CF_2–CF_2)–_n$
5. **Nylon-6,6:**
   $–NH–(CH_2)_6–NH–CO–(CH_2)_4–CO–_n$
6. **Terylene:**
   $–O–CH_2–CH_2–O–CO–C_6H_4–CO–_n$



Uses of Polymers and Plastics

- **Polyethene (PE):** Plastic bags, films, bottles, buckets, pipes.
- **PVC:** Electrical wire insulation, water pipes, vinyl flooring, roofing sheets.
- **Polystyrene:** Disposable cups, packaging foams, insulation materials.
- **Teflon (PTFE):** Non-stick cookware, electrical insulation, gaskets.
- **Nylon:** Textiles, ropes, nets, toothbrush bristles, parachutes.
- **Terylene:** Clothes, curtains, conveyor belts, upholstery.
- **Bakelite:** Electrical switches, plugs, handles of cooking utensils.
- **Rubber (natural & synthetic):** Tyres, footwear, waterproofing, shock absorbers.
- **Proteins & DNA:** Biological functions – enzymes, hormones, hereditary materials.

Dangers of Polymers and Plastics

1. **Environmental Pollution (Non-biodegradability):**
    - Plastics persist in soil and water for decades.
    - Block drainage systems → flooding.
    - Create unsightly refuse heaps.
2. **Toxicity on Burning:**
    - Burning PVC → releases **HCl gas**.
    - Incomplete combustion of plastics → **CO, dioxins** (carcinogenic).
3. **Marine Pollution:**
    - Plastics in oceans are ingested by fish, turtles, and birds → suffocation and death.
    - Microplastics enter the food chain, affecting humans.
4. **Health Hazards:**

- Plastic packaging may leach toxic additives (phthalates, BPA) into food and drinks → endocrine disruption.

5. **Resource Depletion:**
    - Synthetic polymers depend on petroleum; accelerates fossil fuel depletion.
6. **Fire Hazard:**
    - Many plastics are highly flammable, producing dense toxic smoke.

Control Measures and Solutions

- **Recycling and Reuse:** Collect, melt, and remold plastics.
- **Biodegradable Plastics:** Production of eco-friendly polymers from starch or polylactic acid (PLA).
- **Proper Waste Management:** Enforcing laws against indiscriminate disposal.
- **Substitution:** Use paper, cloth, or glass instead of plastics.
- **Public Education:** Awareness campaigns on reducing plastic use.
- **Legislation:** Bans or restrictions on single-use plastics (shopping bags, straws).

Evaluation

1. Define the terms (a) polymer, (b) monomer, (c) polymerization.
2. Differentiate between addition and condensation polymerization with equations.
3. List three differences between thermoplastics and thermosetting plastics.
4. Write the repeating unit of:
   (a) polyvinyl chloride,
   (b) nylon-6,6.
5. Explain two dangers associated with burning plastics.

Assignment

1. Write balanced polymerization equations for the following monomers:
   (a) Ethene,
   (b) Tetrafluoroethene,
   (c) Phenol + formaldehyde.
2. Give four differences between natural and synthetic polymers with examples.
3. Discuss **five dangers** of indiscriminate use of plastics and suggest practical remedies for each.
4. (a) Why is PVC suitable for electrical cable insulation?
   (b) Why is bakelite preferred for electrical switches and plugs?
5. Describe how **nylon-6,6** is formed, naming its monomers and showing the equation.